

Greedy Packing of Nested Rings: Placement Rules, a Golden Counterexample, and a Tribonacci Floor

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Abstract

We study packings of annuli (“rings”) of common width into a disk, where a ring may nest inside the hole of a strictly larger one – a circular, selection-oriented relative of the Recursive Circle Packing Problem. The two natural objectives, number of placed rings and total contact area, genuinely diverge: area is a superadditive function of the radius, cardinality is not. For *superincreasing* radii (each exceeding the sum of all smaller ones) we prove that the descending greedy maximizes every positive, strictly increasing, superadditive objective (in any placement variant), and – our main structural theorem – that the placement rule is irrelevant: any choice among feasible containers yields the lexicographically maximal feasible set, for containers of arbitrary shape and in every dimension. Both hypotheses are sharp: placement irrelevance holds unconditionally for at most three rings and fails at four, and *twin instances* rule out every placement rule that is a function of the observable state. Measuring the violation of superincreasingness by $\rho = \max_i(\sum_{j>i} r_j)/r_i$, the additive relaxation of the model has universal threshold exactly $\rho = 1$. The geometric model resists further: we prove, with no tangency idealization, that the four-ring rigid family of counterexamples has infimum exactly the Tribonacci constant $T \approx 1.83929$. Yet T is *not* the global threshold: an explicit golden family (pan $\varphi + 1$, radii $\{\varphi, 1, \varphi/2 + 2\varepsilon, \varphi/2 + \varepsilon\}$) breaks placement obliviousness at $\rho = \varphi + 3\varepsilon$ for every small $\varepsilon > 0$. This *proves* $\tau \leq \varphi < T$ for the geometric threshold τ and refutes the natural Tribonacci-threshold conjecture. The matching bound $\tau \geq \varphi$ we prove for pan exchanges with pair profiles – and with larger profiles outside an explicit heavy region – and conjecture in general: the golden value is thus a theorem in one direction and open only in the other, with T the exact floor of the rigid nested family and the conjectured universal floor of nested exchanges. Towards this we develop a program that proves nested-exchange floors above T for a hierarchy of templates – an exact positive-width curve with global minimum $13/7$, closed-form profile thresholds, and blocking walls for generic containers with metallic-mean floors – and complement it with a phase diagram for the divergence of the objectives and a decoupling of hardness into a geometric layer and a combinatorial (subset-sum) layer, of which superincreasingness eliminates exactly the latter.

1 Introduction

Drop a batch of squid rings into a frying pan and a natural optimization problem appears: place as many rings as possible flat on the pan – or, if one cares about searing, maximize the total

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ring-pan contact area – exploiting the fact that a sufficiently small ring fits inside the hole of a larger one. Abstracted, this is the problem of packing annuli of common width into a disk with recursive nesting, a circular-container relative of the *Recursive Circle Packing Problem* (RCPP) introduced by Pedroso, Cunha and Tavares [1] to model the “telescoping” of tubes in shipping containers, and solved exactly by Gleixner, Maher, Müller and Pedroso [2]. That literature is algorithmic: it packs a fixed set of rings into as few rectangular containers as possible. Here we ask structural questions instead: which objectives the natural greedy algorithm optimizes, when the choice of *where* to put each ring is irrelevant, and which conditions on the radii govern the answers.

Our starting point is a classical phenomenon in one dimension. For bin packing with *divisible* item sizes, Coffman, Garey and Johnson [3] proved that First Fit Decreasing is optimal. We are not aware of an analogue for geometric packing of disks, let alone with nesting. We prove one, with *superincreasing* radii (each radius exceeding the sum of all smaller ones) playing the role of divisibility, and we determine exactly how far the hypothesis can be relaxed. The answer is a surprise: the additive relaxation of the model has threshold exactly 1, while the geometric model resists up to (conjecturally, exactly) the *golden ratio* – an explicit golden family fails at $\rho = \varphi + 3\varepsilon$. The Tribonacci constant – the natural conjectured threshold, which our own program long targeted – survives as the exact floor of the rigid nested family, the proved floor of a hierarchy of nested-exchange templates, and the conjectured universal floor of nested exchanges.

Contributions: main theorems. (1) A *superadditivity dichotomy*: under superincreasing radii the selection greedy is optimal for every positive, strictly increasing, superadditive value function of the radius – contact area is one – while cardinality (not superadditive) admits counterexamples even under superincreasing radii (Section 3). The selection statement is best seen as a general fact about downward-closed set systems with superincreasing weights, in the lineage of greedy analyses on independence systems [13, 14] and of the superincreasing knapsack [15]; the geometric content of this paper begins with the next item. (2) A *placement-obliviousness theorem*: under superincreasing radii, *any* rule for choosing among feasible containers yields the lexicographically maximal feasible set; the proof is container-shape- and dimension-agnostic (Section 4). (3) *Sharpness*: placement obliviousness holds unconditionally for $n \leq 3$ rings and fails at $n = 4$; twin instances with an identical prefix rule out every state-based placement rule, deterministic or randomized (Section 5). (4) *Thresholds*: the additive model has universal threshold exactly $\rho = 1$; the geometric four-ring rigid subfamily has infimum exactly the Tribonacci constant T , proved with no idealization (Theorem 17, full proof in Appendix B); and an explicit *golden family* refutes the natural conjecture that the geometric threshold equals T : placement obliviousness fails at $\rho = \varphi + 3\varepsilon$ for every small $\varepsilon > 0$, which *proves* $\tau \leq \varphi < T$ (Theorem 19), and the family realizes every $\rho \in (\varphi, T)$ at a fixed width (Corollary 20). The matching bound $\tau \geq \varphi$ is proved for pan exchanges with pair profiles (Theorem 66), extended to larger profiles outside an explicit heavy region (Theorem 67), and conjectured in general (Conjecture 21): the golden value is a theorem in one direction and open only in the other (Section 6).

Contributions: the nested-exchange program (the Tribonacci floor). (5) *The width program*: the blocked-exchange infimum with relative width $\omega > 0$ is governed by the deformed Tribonacci cubic $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$; the full lower-bound curve is in closed form and its global minimum is exactly $13/7$, attained at a rational corner (Section 7); the combinatorial

profile threshold is in closed form for every profile size, with golden plateau and exact Tribonacci crossing $\omega_T = 1/T - 1/2$. (6) *Generic containers*: an exact corona criterion for wall-tangent packings (with two corrective counterexamples and a $k = 5$ “pentagram” certificate), and a hierarchy of blocking walls – occupant holes, “blocking costs the slack”, and the double pocket – yielding metallic-mean floors (silver ratio $1 + \sqrt{2}$, golden line $\varphi^2 - (\varphi/2)\omega$) that push every blocked exchange with extra occupants above T – throughout the width convention $\omega \in (0, 1)$ of the program: for widths $\omega < 0.9626$ with one extra occupant, up to 0.624 and 0.896 with two and three, and for all $\omega \in (0, 1)$ with four or more (Section 8). The remaining gaps of the program are stated explicitly there and in Open Problem 24. (7) A phase diagram for the divergence of the two objectives, and a decoupling of hardness into a geometric and a combinatorial layer, the latter with an explicit PARTITION reduction (Sections 9–10).

2 Model and preliminaries

Definition 1 (Rings and placements). Fix a width $w > 0$. A *ring* of outer radius $r > 0$ is an annulus with inner (hole) radius $\max(0, r - w)$; if $r \leq w$ it is a solid disk with no hole. Let $K \subset \mathbb{R}^d$ be a compact *container* (the *pan*; our motivating case is a disk of radius R in the plane, $d = 2$, but Theorems 6 and 9 hold for arbitrary K and d , with rings read as spherical shells). A *placement* of a subset S of the rings is a forest: each placed ring has as parent either the pan or a placed ring with $r_{\text{child}} \leq r_{\text{parent}} - w$ (it sits inside the parent’s hole); the children of a common parent must be packable as balls (by outer radius) with pairwise disjoint interiors inside the parent’s region — K itself, or the hole ball of radius $r - w$. Feasibility is a property of the assignment (siblings may be rearranged freely inside their container).

Definition 2 (Objectives; superincreasing radii). For radii $r_1 > r_2 > \dots > r_n$, the *contact area* of a placed ring is $a(r) = \pi(r^2 - \max(0, r - w)^2)$; the two objectives are $N = |S|$ and $A = \sum_{i \in S} a(r_i)$ over feasible S . The radii are *superincreasing* if $r_i > \sum_{j > i} r_j$ for all i ; the violation is measured by $\rho = \max_i (\sum_{j > i} r_j) / r_i$. We call $(\sum_{j > i} r_j) / r_i$ the *tail* (ratio) of ring i , so that ρ is the largest tail; “the tail of x ” always refers to this ratio. The *lex-max* set L is defined greedily: scanning $i = 1, \dots, n$, include i whenever $(L \cap \{1, \dots, i - 1\}) \cup \{i\}$ is feasible. Since feasibility is downward closed, L is the unique lexicographically maximal feasible set, and the *selection greedy* (add ring i if the resulting *set* remains feasible) computes it. Note that the selection greedy queries a *set-feasibility oracle*; it is an information bound, not an efficient algorithm – deciding sibling packability is itself hard (Section 10). The placement greedies below, by contrast, only ever attempt concrete insertions. A *descending greedy with placement rule* processes rings in decreasing radius and places each into *some* feasible container (the pan or the hole of an already placed ring), skipping it only if no container admits it.

Lemma 3 (Row lemma). *Balls with radii $x_1, \dots, x_k$ satisfying $\sum_j x_j \leq C$ pack inside a ball of radius C : place them tangent in a row along a diameter. Consequently, if a ball of radius m is removed from any packing, any collection of balls of total radius at most m can be inserted into the vacated ball without disturbing the rest.*

Proof. With centers on a diameter at coordinates $p_j = -C + 2\sum_{l < j} x_l + x_j$, each satisfies $|p_j| \leq C - x_j$. $\square$

3 The superadditivity dichotomy

Lemma 4. *The contact area a is increasing and superadditive on $(0, \infty)$: $a(x) + a(y) \leq a(x+y)$.*

Proof. Monotonicity is clear. For $x, y \geq w$: $\pi w(2x - w) + \pi w(2y - w) = \pi w(2(x+y) - 2w) \leq \pi w(2(x+y) - w)$. For $x, y \leq w$ with $x + y \leq w$: $x^2 + y^2 \leq (x+y)^2$. For $x, y \leq w < x+y$: $x^2 + y^2 \leq w(x+y) \leq 2w(x+y) - w^2$, using $x^2 \leq wx$, $y^2 \leq wy$ and $x+y \geq w$. For $x \leq w \leq y$: $\pi x^2 + \pi w(2y - w) \leq \pi w(2x + 2y - w)$ since $x^2 \leq 2wx$. $\square$

Lemma 5 (Lexicographic dominance). *Let the radii be superincreasing and let v be any positive, increasing, superadditive function of the radius. If S, T are feasible sets and the first index (in decreasing radius order) at which they differ belongs to S , then $\sum_S v(r_i) \geq \sum_T v(r_i)$, with equality only if $T \subseteq S$.*

Proof. Let i be the first differing index and $P = S \cap \{1, \dots, i-1\} = T \cap \{1, \dots, i-1\}$. If $T \setminus P = \emptyset$ then $T \subseteq S$ and monotonicity of the sum under inclusion suffices. Otherwise $T \setminus P \subseteq \{i+1, \dots, n\}$, so $\sum_{T \setminus P} r_j < r_i$ by superincreasingness, and by iterated superadditivity and monotonicity $\sum_{T \setminus P} v(r_j) \leq v(\sum_{T \setminus P} r_j) < v(r_i)$. Hence $\sum_T v = \sum_P v + \sum_{T \setminus P} v < \sum_P v + v(r_i) \leq \sum_S v$. $\square$

Theorem 6 (Selection greedy; arbitrary container, any dimension). *With superincreasing radii, the selection greedy computes the lex-max feasible set and therefore maximizes $\sum_{i \in S} v(r_i)$ for every positive, strictly increasing, superadditive v ; in particular it maximizes the contact area A .*

Proof. Feasibility is downward closed, so the selection greedy computes L (if it could add $i \notin L$, any maximal feasible extension would lexicographically beat L). Lemma 5 shows L dominates every feasible set. $\square$

Remark 7 (Positioning). Theorem 6 is at heart a statement about arbitrary downward-closed set systems with superincreasing weights, and as such it belongs to the classical circle of ideas around greedy algorithms on independence systems [13, 14] and the superincreasing knapsack, whose lexicographic structure was analysed by Gupte [15]. The hypotheses on v are all needed: positivity rules out $v \equiv -1$ (superadditive and non-decreasing, yet maximized by the empty set), and strict monotonicity is what makes the dominance in Lemma 5 strict. The geometric novelty of this paper lies in Theorem 9, where the exchange argument must respect the geometry of nested containers.

Proposition 8 (Cardinality is not rescued). *There are superincreasing instances on which every descending greedy is suboptimal for N . Example (verified): pan $R = 10$, $w = 4.8$, radii $\{9.95, 5.0, 4.3, 0.6\}$. Greedy places $\{9.95, 5.0\}$ ($N = 2$, optimal area), while $\{5.0, 4.3, 0.6\}$ packs in a row in the pan ($N = 3$). The obstruction is structural: N corresponds to $v \equiv 1$, which is not superadditive, and Lemma 5 fails for it.*

4 Placement obliviousness under superincreasing radii

Theorem 9 (Placement rule irrelevance). *Let the radii be superincreasing (weakly: $r_i \geq \sum_{j>i} r_j$ suffices) and let $K \subset \mathbb{R}^d$ be an arbitrary container. Every descending greedy, with an arbitrary rule for choosing among feasible containers, places exactly the lex-max feasible set. In particular best fit (place in the feasible container of smallest capacity), worst fit (largest capacity) and random placement are all optimal for every positive, strictly increasing, superadditive objective.*

Proof. Let F denote the greedy's placement just before processing ring i , with placed set G , and suppose $G \cup \{i\}$ is feasible via a witness placement P ; it suffices to show some container of F admits i (the converse direction is immediate, and then greedy's set equals L by induction as in Theorem 6). The available containers – the pan plus one hole per ring of G – coincide in F and P , and in P the smallest ring i parents nobody.

Let m be the largest ring of G assigned different containers by F and P (if none, P restricted to G equals F and i 's container in P witnesses admission). Let u be F 's container for m and v be P 's. Since the greedy processes rings in decreasing order, when F placed m the occupants of u were exactly the rings larger than m that F assigns to u ; by maximality of m these coincide with the rings larger than m that P assigns to u , and F certified that this set together with m packs in u . Build P' from P : (i) every ring smaller than m that P kept in u is moved into the ball of radius r_m vacated by m inside v – they fit in a row by Lemma 3, because their total radius is at most $\sum_{j: r_j < r_m} r_j \leq r_m$, and they touch nothing else in v ; (ii) m moves to u , joining exactly the larger-than- m rings whose compatibility F certified. Nesting constraints persist: u and v are the pan or holes of rings larger than m , which do not move, and every ring moved into v is smaller than r_m , which fitted in v . Thus P' is a feasible witness agreeing with F on all rings of radius $\geq r_m$: the first disagreement strictly shrinks. Iterating at most $|G|$ times yields a witness P^* agreeing with F on all of G and placing i in some container c^* ; since the occupants of c^* in P^* are exactly those in F , the greedy admits i there. $\square$

Remark 10. The proof uses only (a) downward closure of packings, (b) the row lemma applied *inside the vacated ball*, and (c) the total-radius bound from superincreasingness. Nothing refers to the shape of K or the dimension: the theorem covers rectangular pans (griddles) and spherical shells nested in arbitrary containers in $\mathbb{R}^d$ verbatim. Computational corroboration of the strongest consequence: on 100 random superincreasing instances, best-fit, worst-fit and random placement produced identical, optimal outcomes without exception.

5 Sharpness: the $n = 4$ transition and twin instances

Proposition 11 ($n \leq 3$, disk pan). *For a disk pan and arbitrary radii, every descending greedy on at most three rings places the lex-max set.*

Proof. The only branching point is ring r_2 , when both the pan ($r_1 + r_2 \leq R$; two circles fit in a disk iff their radii sum to at most its radius) and the hole of r_1 ($r_2 \leq r_1 - w$) admit it. Let G be the greedy's set after r_2 and suppose $G \cup \{r_3\}$ is feasible via a witness P ; we check that the greedy's placement F admits r_3 in each of the four combinations of choices. If F nested r_2 : should P nest it too, the containers of F and P have identical occupants and P 's home for r_3 works in F ; should P keep r_2 in the pan, then $r_1 + r_2 \leq R$, so $r_1 + r_3 < R$ and F 's pan (holding only r_1) admits r_3 . If F kept r_2 in the pan: should P do likewise, the placements agree; should P nest r_2 , then $r_2 \leq r_1 - w$, hence $r_3 < r_2 \leq r_1 - w$ and F 's *empty* hole of r_1 admits r_3 . In every case r_3 is admitted, and rings outside the lex-max set are never admissible, so the greedy's set is the lex-max set. $\square$

Lemma 12 (Cap confinement). *Circles of radii 10, 4.9, 4.8 do not pack in a disk of radius 15.*

Proof. Write c_A, c_X, c_Y for the centers, measured from the pan center. Then $|c_A| \leq 5$, $|c_X| \leq 10.1$, $|c_Y| \leq 10.2$, $|c_A - c_X| \geq 14.9$, $|c_A - c_Y| \geq 14.8$, $|c_X - c_Y| \geq 9.7$. From $|c_A| + |c_X| \geq 14.9$, $a := |c_A| \in [4.8, 5]$. Let $u = -c_A/|c_A|$. Then $c_X u = (|c_X - c_A|^2 - |c_X|^2 - a^2)/(2a) \geq (120 - a^2)/(2a) \geq$

9.5 (decreasing in a), so the transverse part of c_X has norm at most $\sqrt{10.1^2 - 9.5^2} \leq 3.43$; similarly $c_Y \cdot u \geq (115 - a^2)/(2a) \geq 9.0$ with transverse norm at most $\sqrt{10.2^2 - 9^2} = 4.8$. Hence $|c_X - c_Y|^2 \leq 10.1^2 + 10.2^2 - 2(9.5 \cdot 9.0 - 3.43 \cdot 4.8) \leq 67.98$, i.e. $|c_X - c_Y| \leq 8.25 < 9.7$, a contradiction. $\square$

Theorem 13 (Failure at $n = 4$). *Placement obliviousness fails for four rings: with $R = 15$, $w = 0.3$ and radii $\{10, 5, 4.9, 4.8\}$ (all four form the lex-max set, witnessed by the pan pair $\{10, 5\}$ at exact tangency and the hole pair $\{4.9, 4.8\}$ filling the hole 9.7 exactly), best fit nests the 5, forces the 4.9 into the pan, and blocks the 4.8 everywhere by Lemma 12 and the exact two-circle conditions; worst fit places all four (Figure 1).*

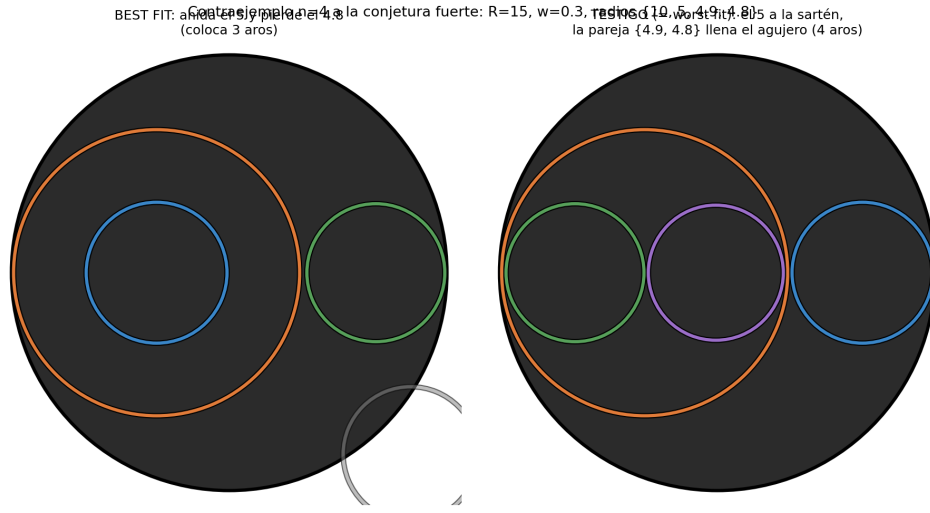


Figure 1: The $n = 4$ counterexample of Theorem 13: best fit (left) nests the 5 and loses the 4.8; the witness (right), realized by worst fit, places all four rings.

Theorem 14 (Twin instances; no state-based rule). *Let $R = 15$, $r_1 = 10$, $r_2 = 5$, $w = 0.505$ (hole 9.495), and*

$$I_1 = \{10, 5, 4.99, 4.50\}, \quad I_2 = \{10, 5, 4.76, 4.74\}.$$

In I_1 the correct placement of the 5 is the pan (best fit fails, worst fit succeeds); in I_2 it is the hole (worst fit fails, best fit succeeds). At the decisive step the observable state – containers, capacities, occupants, incoming ring, R and w – is identical in I_1 and I_2 . Consequently no deterministic placement rule that is a function of the state attains the lex-max set on all instances, and for every randomized rule some instance is failed with probability at least $1/2$.

Proof ingredients. Feasibilities are two-circle-exact except for two triples, both settled rigorously: $\{10, 4.99, 4.50\}$ does not pack in the disk of radius 15 (cap confinement as in Lemma 12 yields $|c_X - c_Y| \leq 8.80 < 9.49$), and the pan pair $\{10, 5\}$ is diametrically rigid, whence no third circle of radius ≥ 4.7 fits (confinement again: for $z = 4.76$, $|c_z - c_B| \leq \sqrt{204.86 - 20 \cdot 8.8} < 5.37 < 9.76$). Packability of $\{10, 4.76, 4.74\}$ in radius 15 is exhibited by an explicit boundary configuration (angles 180° , 30.8° , -32.0° ; all pairwise constraints verified). The key phenomenon enabling the twins is that triple-packability is *not monotone in the sum*: the unbalanced pair $\{4.99, 4.50\}$ (sum 9.49) fails beside the 10 while the balanced pair $\{4.76, 4.74\}$ (sum 9.50) succeeds. $\square$

Remark 15. The obstruction is informational, not existential: following any witness of the lex-max set is itself a legal greedy execution (a ring outside the lex-max set is never admissible), so some execution always succeeds; what cannot exist is a state-based rule that finds it. For rules with access to the full input the question becomes one of computational complexity and remains open (Section 12).

6 Thresholds: the Tribonacci floor and the golden counterexample

Theorem 16 (Additive model: threshold exactly 1). *Replace sibling feasibility by its additive surrogate (a set of siblings fits iff their radii sum to at most the container capacity; exact for at most two circles and sufficient in general by Lemma 3). Then the universal threshold for placement obliviousness is exactly 1: obliviousness holds for all instances with $\rho \leq 1$ (the proof of Theorem 9 applies verbatim), while for every $\varepsilon > 0$ some instance with $\rho \leq 1 + \varepsilon$ fails – individual instances with larger ρ may of course still succeed. Indeed, fix r_1 , set $s = r_1/2$ and $R = r_1 + s = \frac{3}{2}r_1$, pick any $w \in (\frac{r_1}{4}, \frac{r_1}{2} - \delta)$, and take $r_2 = s$ and the exact pair $r_3 = \frac{r_1}{4} + \frac{2\delta}{3}$, $r_4 = \frac{r_1}{4} + \frac{\delta}{3}$ (so $r_3 + r_4 = s + \delta$ and $r_4 < r_3 < r_2 < r_1$ strictly), with any $0 < \delta < \min(\frac{r_1}{4}, \frac{\varepsilon r_1}{2})$: the witness fills the pan with $\{r_1, r_2\}$ and the hole of r_1 with the pair ($r_3 + r_4 = s + \delta \leq r_1 - w$), while best fit nests r_2 ($r_2 = s \leq r_1 - w$ since $w \leq r_1/2$), sends r_3 to the pan, and blocks r_4 additively ($r_1 + r_3 + r_4 > R$ since $r_3 + r_4 > s$; $r_2 + r_4 > r_1 - w$ since $w > r_1/4$; $r_4 > r_2 - w$ likewise). The three tail ratios are $(2s + \delta)/r_1 = 1 + \delta/r_1$, $(s + \delta)/s = 1 + 2\delta/r_1$ and $r_4/r_3 < 1$, so $\rho = 1 + 2\delta/r_1 \leq 1 + \varepsilon$; verified instances reach $\rho = 1.03$ and 1.02 .*

The four-ring counterexample family carries a genuine floor at the Tribonacci constant. An earlier version of this statement relied on a tangency idealization ($r_3 \rightarrow r_2$, $w \rightarrow 0$); the following theorem removes it entirely: the rigid configuration is not assumed but *emerges* as the extremal case of a concavity argument.

Theorem 17 (Rigid-subfamily Tribonacci floor, no idealization). *Normalize $r_1 = 1$ and write $t = r_2$, $p = r_3$, $q = r_4$, $\omega = w > 0$. Let $\mathcal{F}$ be the family of four-ring instances with (F1) diametral pan $R = 1 + t$, (F2) $p + q \leq 1 - \omega$ (the witness pair fits the hole of r_1), and (F3) the triple $\{1, p, q\}$ does not pack in the disk of radius $1 + t$. Then every instance of $\mathcal{F}$ satisfies $\rho > T$, strictly, for every width $\omega > 0$ and all strict radii $q < p < t < 1$; moreover $\inf_{\mathcal{F}} \rho = T$ and the infimum is not attained. The family is nonempty only for $t < t^* = 1/T$, and it contains the $n = 4$ counterexample and the twin I_1 of Theorem 14.*

Proof sketch (complete proof in Appendix B). For two circles of radii a, b internally tangent to the wall of a disk of radius R , the minimal angular separation $\theta(a, b)$ obeys the half-angle identity $\sin^2(\theta/2) = f(a)f(b)$ with $f(x) = x/(R - x)$. Placing the triple $\{1, p, q\}$ wall-tangent with the unit ring in the middle packs it whenever $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$ (the two non-adjacent arguments are separated by monotonicity of θ), and this angular condition reduces *algebraically* to

$$\psi(p) + \psi(q) \geq \tau_t, \quad \psi(x) = \sqrt{\frac{t-x}{x}}, \quad \tau_t = \frac{t}{\sqrt{1+t}},$$

(an algebraic *sufficient* condition for packing, which is all the contrapositive needs: (F3) forces $\psi(p) + \psi(q) < \tau_t$). In the coordinates $\alpha = \psi(p)$, $\beta = \psi(q)$ the sum $p + q = U(\alpha) + U(\beta)$ with $U(z) = t/(1 + z^2)$, and U is concave on $[0, \tau_t]$ exactly when $\tau_t \leq 1/\sqrt{3}$, i.e. $t \leq (1 +$

$\sqrt{13}/6 = 0.7676\dots$, which covers the relevant range with room to spare; hence the infimum of $p + q$ over the blocked (open) region sits at the *corner* of its closure, $p = t$, $q = b(t)$, where $b(t) = t(1 + t)/(1 + t + t^2)$ is the Descartes pocket of the rigid pair. This gives the two pressures $t + b(t) < p + q \leq 1 - \omega < 1$ (the first one strict, whence the strict floor and the non-attainment), whose compatibility is precisely the cubic $t^3 + t^2 + t < 1$, and then $\rho \geq (p + q)/t > 1 + (1 + t)/(1 + t + t^2) =: L(t)$ with L strictly decreasing and $L(t^*) = T$. Exactness of the pocket at the rigid pair (needed only for the direction $\inf \leq T$) is a rigidity computation: the constraint left-hand side factors exactly as $4(t^2 + t + 1)(b(t) - q)$; and the approximating family lives genuinely inside $\mathcal{F}$ by a closure-and-monotonicity lemma for the packing region (compactness of witnesses), which shows the blocked interval in p is half-open with an attained endpoint. $\square$

Remark 18 (Slack pans). The hypothesis $R = r_1 + r_2$ can be relaxed to $R \geq r_1 + r_2$ at no cost: non-packability is antitone in the container (a packing in a smaller disk is a packing in a larger one after translation), so (F3) at radius R implies (F3) at radius $r_1 + r_2$ and Theorem 17 applies verbatim; the infimum remains T . Slack in the pan never helps the blocker — what it changes is the set of possible *occupants*, the subject of Section 8.

Define the *geometric threshold* as

$$\tau := \inf\{\rho(I) : \text{some greedy execution fails to place the lex-max set on } I\},$$

so that placement obliviousness holds for every instance with $\rho < \tau$ and fails at values of ρ arbitrarily close to τ (possibly at τ itself, if the infimum is attained); “floor” always refers to an infimum of ρ over a family of blocked exchanges. The natural conjecture — which we ourselves held until late in this project — is that $\tau = T$. It is *false*. The witness capacity that lifts the floor from φ to T in Theorem 17 exists only when the greedy’s container for the pivotal ring is a *hole*; when it is the pan itself, the floor drops to the first rung of the ladder, the Descartes pocket, and the golden ratio appears:

Theorem 19 (The golden family: the threshold is at most φ). *For every $\omega \in (1 - \varphi/2, \varphi - 1) = (0.1910, 0.6180)$ and every*

$$0 < \varepsilon < \varepsilon^* := \min\left(\frac{T - \varphi}{3}, \frac{\varphi/2 - \omega}{2}, \frac{4\sqrt{5} - 8 - \varphi/2}{2}\right)$$

(all three bounds positive on the window; the middle one is read as $(\varphi - \omega - \varphi/2)/2$ when it is the binding constraint on $s_1 \leq \varphi - \omega$), the instance

$$R = \varphi + 1, \quad w = \omega, \quad r = \left\{\varphi, 1, \frac{\varphi}{2} + 2\varepsilon, \frac{\varphi}{2} + \varepsilon\right\}$$

has $\rho = \varphi + 3\varepsilon < T$, its lex-max set is all four rings, and the descending greedy under worst fit places only three: placement obliviousness fails at $\rho = \varphi + 3\varepsilon$.

Proof. Write $s_1 = \frac{\varphi}{2} + 2\varepsilon$, $s_2 = \frac{\varphi}{2} + \varepsilon$. Every feasibility below is exact. *The value of ρ :* the tails are $(1 + s_1 + s_2)/\varphi = \varphi + 3\varepsilon/\varphi$ (by the golden identity $(1 + \varphi)/\varphi = \varphi$), $s_1 + s_2 = \varphi + 3\varepsilon$ (dominant) and $s_2/s_1 < 1$, and $\varphi + 3\varepsilon < T$ for $\varepsilon < (T - \varphi)/3$. *The witness:* place 1 in the hole of φ (capacity $\varphi - \omega \geq 1 \iff \omega \leq \varphi - 1$) and $\{\varphi, s_1, s_2\}$ as a corona in the pan; the corona exists by an *exact* certificate. By monotonicity of θ it suffices that the symmetric trio $\{\varphi, \sigma, \sigma\}$

with $\sigma = s_1$ has angular sum below 2π , i.e. $2A + B < \pi$ with $A = \theta(\varphi, \sigma)/2$, $B = \theta(\sigma, \sigma)/2$. Here $f(\varphi) = \varphi$, and when $2A > \pi/2$ (otherwise the claim is immediate from $B < \pi/2$),

$$2A + B < \pi \iff \sin B < \sin 2A \iff f(\sigma)^2 < 4\varphi f(\sigma)(1 - \varphi f(\sigma)) \iff f(\sigma)(1 + 4\varphi^2) < 4\varphi,$$

which with $f(\sigma) = \sigma/(\varphi^2 - \sigma)$, $8\varphi + 5 = (2 + \sqrt{5})^2$ and $\varphi^3 = 2 + \sqrt{5}$ reduces to $\sigma(8\varphi + 5) < 4\varphi^3$, i.e. $\sigma < 4(\sqrt{5} - 2) = 0.9442\dots$; this holds since $s_1 = \varphi/2 + 2\varepsilon < 4(\sqrt{5} - 2)$ by the third bound in ε^* . (At $\varepsilon = 0$ the certificate is the exact identity $4\varphi(\sqrt{5} - \varphi) = 4 > 1$.) So all four rings are placeable. *The greedy under worst fit*: $\varphi \rightarrow \text{pan}$; the ring 1 fits both the pan ($\varphi + 1 \leq R$, exact tangency, as in Theorem 13) and the hole of φ , and worst fit – the container of largest capacity, $R = \varphi + 1 > \varphi - \omega$ – chooses the *pan*. Now $\{\varphi, 1\}$ fills the pan diametrically: the pair is rigid and, by the exact necessity of Proposition 31 (rescaled), any third circle has radius at most $b_2(\varphi, 1) = \varphi/2$ – the golden pocket identity. Hence neither s_1 nor s_2 fits the pan; the hole of φ admits one of them but not both ($s_1 + s_2 = \varphi + 3\varepsilon > \varphi - \omega$, two-circle exact); the hole of m admits neither ($s_2 > 1 - \omega \iff \omega > 1 - \frac{\varphi}{2} - \varepsilon$); and s_2 cannot nest in s_1 ; and s_1 fits alone in the hole of φ ($s_1 \leq \varphi - \omega$, via ε^*), which is where the greedy puts it before s_2 is stranded. An exhaustive placement tree confirms the count: three rings with 1 in the pan, four with 1 in the hole (`code/aureo.py`). $\square$

Corollary 20 (The family realizes every $\rho \in (\varphi, T)$). *For $\rho_0 \in (\varphi, T)$ fix the width $\omega = 0.3$ (any value in the window works) and take $s_2 = \varphi/2 + \varepsilon$, $s_1 = s_2 + \eta$ with $2\varepsilon + \eta = \rho_0 - \varphi$ and $0 < \eta < \rho_0 - \varphi$ small. The dominant tail is $s_1 + s_2 = \rho_0$ exactly. All walls hold explicitly: $s_1 < 1$ and the corona certificate $s_1 < 4(\sqrt{5} - 2)$, since $s_1 = \varphi/2 + \frac{\rho_0 - \varphi}{2} + \frac{\eta}{2} < \varphi/2 + \frac{T - \varphi}{2} + \frac{\eta}{2} = 0.9195\dots + \frac{\eta}{2} < 0.9443$ for η small; $s_2 > \varphi/2 = 0.809 > 0.7 = 1 - \omega$ (H_m blocked); $s_1 + s_2 = \rho_0 > \varphi > \varphi - \omega$ (the hole of φ rejects the pair); $s_2 > s_1 - \omega$ (their gap is $\eta < 0.3 = \omega$, so no nesting); and $s_1 \leq \varphi - \omega = 1.318$ (s_1 fits alone in the hole, where the greedy strands it). $\square$*

The instance is golden four times over: the occupant is φ (the unique positive fixed point of $2b(A)A = 1 + 2b(A)$), the pocket is $b_2(\varphi, 1) = \varphi/2$, the tail of φ is $(1 + \varphi)/\varphi = \varphi$, and $\rho \rightarrow \varphi$. The family is not degenerate: the ω -window has width 0.427, ε ranges over an interval, and the diametral tangency survives pan slack up to $\delta^* = 0.0248$ (numerically, via the angular criterion). It escaped the earlier structured searches because it lives at a codimension-one corner (pan tangency) that random radii never visit.

Conjecture 21 (Golden threshold). *Placement obliviousness holds for every instance with $\rho < \varphi$; together with Corollary 20 (the family realizes every $\rho \in (\varphi, T)$) and the failures above T , this would make the geometric threshold exactly the golden ratio, $\tau = \varphi$: $\varphi - 1$ measures the protection that disk geometry grants the greedy over additive bookkeeping, and T is the exact floor of the rigid nested family (Theorem 17), the proved floor of every nested template of Sections 7–8, and the conjectured universal floor of nested exchanges (the remaining gaps are listed in Open Problem 24(b)). Evidence: the pan-exchange φ -floor is proved in full for pair profiles in Theorem 66 (Appendix D.6), for extra small rings by adjunction (Corollary 69), and for larger profiles outside a single heavy cell (Theorems 67 and 68) – one occupant at every width via the golden fixed point, the evacuation branch at every $\omega \in (0, 1)$ via $\Psi_B(1) = \varphi$, and the strict-leaf ladder with $\Psi_1(\frac{1}{2}) = \Psi_2(\varphi/2) = \varphi$ – every proved floor of the nested exchange exceeds $T > \varphi$ with margin, and every residual cell of the exchange assembly is closed computationally with duality-certificate backing (Appendix E; the profile and dust directions of the scaling are exact by the*

pocket- φ lemma, and the heavy pan cells, the entire nested template and the container part have been upgraded to written proofs by the insertion, compaction and trio lemmas, Theorems 72–74, the hole-corona closures and the geometric-tail lemma – the latter making every written shadow budget uniform in the occupant count), leaving as the numeric core the occupant-count direction of the computational scaling closures alone.

7 The width program: profile thresholds and the 13/7 corner

Towards Conjecture 21, the proof of Theorem 9 localizes all difficulty in a single *exchange step*: m is the largest ring that the greedy F and a witness P place in different containers $u = c_F(m)$, $v = c_P(m)$, and the set S of rings smaller than m that P keeps in u must be reinserted using the resources freed by moving m to u : the vacated disk D_m (radius r_m), the hole H_m (capacity $r_m - w$, travelling with m), nesting, and the geometry of v . Normalize $r_m = 1$ and $\omega = w/r_m$. *Width convention*: unless a result explicitly states otherwise, throughout the exchange-step program (this section, the next, and Appendices C and D) we assume $0 < \omega < 1$, so that the pivot m is a genuine ring with a nonempty hole H_m . The model does admit $\omega \geq 1$ (a solid-disk pivot); that regime is open except in the results that state “every $\omega > 0$ ” explicitly: Theorem 22, Corollary 41’s Tribonacci closure, Corollary 59, and cases (i)–(ii) of Theorem 66. Their proofs route the $\omega \geq 1$ regime through walls that do not use H_m (the trio frontier and the witness capacity) and so remain valid for a solid pivot. Everything ρ imposes on S is the pair of inequalities $\sum S \leq \rho$ and $\sum_{l>j} x_l \leq \rho x_j$; a lower bound for ρ over all *blocked* exchanges is therefore a lower bound for any failure of placement obliviousness. We call a *wall* any such necessary condition for blocking: the contrapositive of an explicit unblocking placement, which every blocked exchange must therefore satisfy. Two regimes must be distinguished. When u is the *hole* of a ring (the *nested* exchange), the witness placement supplies the capacity wall $\sum S \leq \text{cap}(u)$, and this section and the next prove floors that all exceed T for a hierarchy of nested templates: the Tribonacci constant is the exact floor of the rigid family and the conjectured universal floor of nested exchanges. When u is the *pan*, that wall is absent, the floor drops to the Descartes pocket, and the golden family of Theorem 19 realizes it: the *pan* exchange governs the global threshold of Conjecture 21. Complete proofs for every statement of this section are given in Appendix C.

The combinatorial layer is closed. For two-ring profiles the blocked threshold (a *floor* in the terminology of Section 6: an infimum over blocked exchanges) is exactly $\rho_2^*(\omega) = \max(1, 2(1 - \omega))$; in particular no two-ring blocking is compatible with $\rho < T$ for $\omega \leq 1 - T/2 = 0.0804$. For three rings the threshold is closed,

$$\rho_3^*(\omega) = \max\left(1, \min(2(1 - \omega), \max(\varphi, \frac{2}{1+2\omega}))\right),$$

with a purely additive proof and a *golden plateau* $\rho_3^* \equiv \varphi$ on $\omega \in [\frac{\sqrt{5}-2}{2}, 1 - \frac{\varphi}{2}]$; and a general additive tree argument shows $\rho_k^* = \rho_3^*$ for every $k \geq 3$. Consequently the combinatorial threshold crosses T exactly at

$$\omega_T = \frac{1}{T} - \frac{1}{2} = T^2 - T - \frac{3}{2} = 0.043689\dots :$$

below ω_T the exchange step never blocks with $\rho < T$, with no geometry at all (Propositions 34–37, Theorem 36, Corollary 38).

The canonical template with positive width. For $\omega > \omega_T$ geometry enters through the canonical template (v a diametral pan with one big neighbour α ; u the hole of α , capacity $\alpha - \omega$; $S = \{\sigma_1 \geq \sigma_2\}$ placed there by the witness). Blocking must defeat four resources, giving the walls $\sigma_2 > 1 - \omega$ (H_m), $\sigma_2 > \alpha - \omega - 1$ (σ_2 next to m inside u), $\sigma_1 + \sigma_2 \leq \alpha - \omega$ (the witness), and non-packability of the triple $\{\alpha, \sigma_1, \sigma_2\}$ in the pan. Two structural facts organize the optimum. First, on the blocking frontier of the triple one has the closed form $t(\sigma_1) + t(\sigma_2) = t(b(\alpha))$ with $t(s) = \sqrt{(1-s)/s}$ (the Descartes pocket read in the t -coordinate), and the frontier slope satisfies the α -independent identity

$$\kappa = -\frac{\partial \sigma_2}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \quad g(s) = s^3(1-s), \quad \kappa \geq 1 \text{ for all } \alpha > 1,$$

so the cheapest blocked pair sits at $\sigma_1 \rightarrow 1$, $\sigma_2 \rightarrow b(\alpha)$; moreover $1 + b(\alpha) \geq \alpha$ iff $\alpha \leq T$, so no blocking exists at all for $\alpha \leq T$. Second, the witness capacity deforms the cubic: with T_c the positive root of $\alpha^3 = c(\alpha^2 + \alpha + 1)$, the witness branch has infimum $\Phi(\omega) = T_{1+\omega} - \omega$, strictly increasing and concave with $\Phi'(\omega) = (2\alpha + 1)/(\alpha^2(\alpha^2 + 2\alpha + 3))$ at $\alpha = T_{1+\omega}$. The full curve of the blocked infimum $T_{\text{can}}(\omega)$ is obtained in closed form *as a proved lower bound* — the matching upper bound is exact on the witness branch and at the corner below, and inherits the angular-criterion caveat (the criterion is proved sufficient, not necessary, so the matching upper bound is conditional there) on the other two branches: $2(1 - \omega)$ on $(0, \omega_1]$, a sextic-algebraic middle branch on $[\omega_1, 1/7]$, and Φ on $[1/7, 0.30]$, where $\omega_1 = 0.0413570\dots$ is the root of $4\omega^3 - 20\omega^2 + 25\omega - 1$ in $(1/25, 1/14)$ (Lemma 39, Theorem 40, Corollary 41, Propositions 42 and 44).

Theorem 22 (The 13/7 corner). *For all $\omega > 0$ (including the solid-pivot regime $\omega \geq 1$: for $\omega \geq \frac{1}{7}$ the proof below uses only (B1) and (W), which do not use H_m and remain valid for a solid pivot),*

$$T_{\text{can}}(\omega) \geq \frac{13}{7} = T + 0.017856\dots,$$

with equality exactly in the limit at the rational corner $(\omega, \alpha, \sigma_2) = (\frac{1}{7}, 2, \frac{6}{7})$, where all four walls saturate simultaneously (note $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7} \cdot 7$). The bound is unconditional – in particular it does not depend on the exactness of any angular feasibility criterion – and the approximating family is genuine. Thus positive width only raises the Tribonacci floor, uniformly.

The fine structure is curious: the curve is *not* monotone on $(0, 1/7]$ – ω_1 is a local minimum, followed by a bump of height $1.1 \cdot 10^{-4}$ at $\omega_{\text{peak}} \approx 0.0445$ (a root of an explicit degree-8 polynomial) before descending to the corner.

8 Generic containers: coronas, walls, and metallic floors

The canonical template assumes v contains only the big neighbour and m . The remaining gap towards Conjecture 21 is genericity of the containers, and it is attacked by a toolkit of independent walls, each adversarially verified. Throughout, v is a pan of radius R with occupants $\{\alpha\} \cup O \cup \{m\}$, $O = \{\sigma_1 \geq \dots \geq \sigma_j\}$, $\sigma_i \geq m = 1$, u is the hole of α , $S = \{\sigma_1 \geq \sigma_2\}$, and holes may be occupied arbitrarily, at any nesting depth. Complete proofs for every statement of this section are given in Appendix D.

The universal trio frontier. For a trio $\{A, x, y\}$ wall-tangent in a disk of *arbitrary* radius R (interior domain, and under the necessary hypothesis $A \geq \min(x, y)$), the angular blocking condition is *linear* in the coordinates $T_c(x) = \sqrt{(c-x)/x}$, $c = R - A$: θ -infeasibility $\iff T_c(x) + T_c(y) \leq \tau_R = c/\sqrt{AR}$. The pocket generalizes to $b_R(A) = ARc/(AR + c^2)$, increasing in R , and the frontier slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$, $g_c(s) = s^3(c-s)$, is uniform in R . Without the hypothesis $A \geq \min(x, y)$ the equivalence is false (an explicit counterexample pins the failure region exactly), a fact found by adversarial verification (Lemma 46).

An exact corona criterion. Call a *corona* a packing of k circles all wall-tangent. For a fixed cyclic order the corona constraints form a linear feasibility system in the angular gaps, so coronas are decidable exactly; every subset of a corona yields a necessary *certificate* (its induced cyclic sum of pairwise angles is at most 2π). For $k = 4$ the certificates are exactly sufficient – for arbitrary symmetric separations in $(0, \pi]$, by a negative-cycle argument: a violating cycle with U ascents needs more than $2U$ edges, so on four nodes only $U = 1$ cycles can violate, and those are exactly the subset certificates – and the global criterion collapses to *two* inequalities: the top trio, plus the total of the *zigzag* order (a_1, a_3, a_2, a_4) , which minimizes the cyclic total by a convexity/majorization argument ($\theta = g(\log f(a) + \log f(b))$ with g convex). Two natural guesses are refuted by explicit counterexamples: the consecutive-arc condition alone is *not* sufficient for $k \geq 4$, and the sorted order is *not* optimal (the zigzag is: large circles must not be angular neighbours). For $k = 5$ sufficiency requires exactly one additional certificate, the *pentagram* (the five-diagonal star $\{5/2\}$, with combinatorially sharp example $\theta = \pi$ on the diagonals), conjecturally redundant for separations of geometric origin (Lemmas 47–48, Theorems 49–51).

Occupant walls and metallic floors. Each extra occupant's hole is itself a reinsertion resource, and blocking it is *taxed*. The basic count: with free holes, blocking forces every extra occupant to within one width of m ($o_i < \sigma_2 + \omega$), and the largest occupant's tail gives $\rho > (j+2)/(1+\omega)$ – each occupant pays $1/(1+\omega)$ – whence, with a refined branch $\rho > 4/(1+2\omega)$ for $\omega \geq \frac{1}{2}$, $\rho > T$ for $\omega < 2/T - \frac{1}{2} = 0.5874$ when $j = 1$ and for *all* ω when $j \geq 2$; adding occupants is strictly suboptimal for the blocker, since the canonical curve satisfies $\Phi(\omega) < 2$ identically (the polynomial identity $(2+\omega)^3 - (1+\omega)((2+\omega)^2 + (2+\omega) + 1) = 1$). For occupied holes the tax is quantified by the *opposite-disk lemma*: in a disk of capacity c with largest piece σ , if the remaining mass is at most $c - \sigma$ the whole family packs (the wall-tangent σ leaves a free disk of radius exactly $c - \sigma$ opposite, filled in a row); hence blocking a hole costs at least its slack: every *node* y – an extra occupant o_i , or any ring of radius $\geq r_m$ nested at any depth inside one – satisfies $y < \sigma_2 + \omega + X_y$, where X_y is the total radius of y 's hole content (α and m themselves are excluded: α 's hole is u , governed by its own walls, and excluding m is essential). Following the *minimal* such ring and balancing two tails yields, for arbitrary hole occupancy at any depth,

$$\rho > \Psi(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + 1},$$

the metallic mean of $2(1 - \omega)$: $\Psi(0) = 1 + \sqrt{2}$ is the silver ratio, $\Psi(1/4) = 2$ exactly, and $\Psi > T$ iff $\omega < (T - 1)^2/2 = 0.3522$. Allowing m itself to carry children splits by an evacuation dichotomy into the same bound and a second metallic mean, the root of $u^2 - (2 - \omega)u - 1$, which dominates automatically; its own crossing is at $(T - 1)^2$ exactly – twice the previous one, and the certifying identity is the Tribonacci polynomial: $(T - 1)^2 T - (2T - T^2 + 1) = T^3 - T^2 - T - 1$ (Lemmas 52 and 55, Proposition 53, Theorems 57 and 58).

The double-pocket wall and the golden line. The geometric wall needs no angular criterion at all: blocking implies the four rings $\{\alpha, o_1, \sigma_1, \sigma_2\}$ do not pack in the pan, hence not in the disk of radius $\bar{R} = \alpha + o_1$ (containment), where the pair $\{\alpha, o_1\}$ is diametrically rigid and leaves exactly *two* Descartes pockets $b_2(\alpha, o_1) = \alpha o_1(\alpha + o_1)/(\alpha^2 + \alpha o_1 + o_1^2)$, one on each side (their mirror centers are $4b_2$ apart: the exact identity $y_0 = 2b_2$). Thus blocking forces $\sigma_1 > b_2(\alpha, o_1)$. Feeding this wall into the program of the previous walls produces a *golden* optimum: the binding corner is $\alpha = 2$, $o_1 = \sqrt{5} - 1$ – the pair with unit pocket, $b_2(2, \sqrt{5} - 1) = 1$ exactly, self-dual in the sense $A_{\max}(\sqrt{5} - 1) = 2$, $A_{\max}(2) = \sqrt{5} - 1$ – and, for one extra occupant and *any* hole occupancy, both branches of the analysis meet the same line: $\rho > T$ unconditionally for all $\omega < \omega_A$, via

$$\rho > \varphi^2 - \frac{\varphi}{2}\omega, \quad \omega_A = 2 - 2(T - 1)(\varphi - 1) = 0.962585\dots$$

(the golden line itself is proved in all but one corner configuration – branch B with a nested node inside the occupant and $\alpha < o_1$, possible only for $\omega < \frac{1}{2}$ – where the proved bound is $\geq 2 > T$, via $\Psi_B(\frac{1}{2}) = 2$ exactly). The proof combines two exact facts – b_2 is concave in α ($\partial^2 b_2 / \partial \alpha^2 = -6\alpha o^3(\alpha + o)/D^3$) and $D^2 - \alpha o^2(o + 2\alpha) = (\alpha + o)(\alpha^3 + \alpha^2 o + o^3) > 0$ – with one-variable boundary certificates anchored at the identity $f_1 \equiv$ curve at $o_1 = \sqrt{5} - 1$. For $j \geq 2$ occupants the ladder $\Psi_j = (1 - \omega) + \sqrt{(1 - \omega)^2 + j}$ (crossing T at $1 - (T^2 - j)/(2T)$: 0.624 for $j = 2$, 0.896 for $j = 3$, all ω for $j \geq 4$) covers the corresponding ranges unconditionally, for arbitrary nested occupancy of the holes: a *leaf argument* closes the case where occupants' holes contain further rings of radius $\geq r_m$. Every occupant subtree contains a leaf (a node none of whose hole contents reach r_m); taking L the largest of the j disjoint leaves, its tail collects the other $j - 1$ leaves together with m , σ_1 , σ_2 and all small mass W , while the blocking wall at a leaf reads $L < \sigma_2 + \omega + W$; optimizing the resulting two-branch program returns exactly Ψ_j , with the crossing at $\sigma_2 = 1 - \omega$ – the same metallic-mean mechanism as Ψ . Small rings anywhere compatible with each theorem's template – and with S still a pair – are free for all the combinatorial walls, whose unblocking placements are local to D_m , H_m , the holes, and the slot next to m inside u (Lemma 60, Theorem 61, Proposition 62, Corollary 63).

Three-piece profiles: the zigzag pays. Larger reinsertion profiles are *cheaper* for the adversary in pure nest accounting ($\rho_3^* \leq \rho_2^*$), but in the canonical template the geometry reverses the sign: blocking with $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\}$ and arbitrary occupancy of H_m forces $\rho > \max(\Phi(\omega), \frac{13}{7})$ for *all* ω , with no exceptional branch. If $\sigma_3 \leq \sigma_1 - \omega$ then σ_3 (the *dust*) rides inside σ_1 and the pair program is inherited wall by wall (the 13/7 corner bound applies); if nothing nests and the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona, the witness chain gives $\sigma_1 + \sigma_2 \geq 1 + b(\alpha) \geq \Phi(\omega)$; and if the trio *does* pack, Lemma U_4 read backwards forces the *zigzag* certificate to fail, and the zigzag implies exactly $\sigma_2 > b(\alpha)$ – the Descartes pocket by a third route, via $\sin^2 A + \sin^2 B > 1$. Crossing that wall with the witness capacity lands on the *golden deformation* γ_ω , the root of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$ with $\gamma_0 = \varphi$, and the tail bounds meet at another rational corner: $\gamma_{2/7} = 2$ exactly ($8 = 4 + 2 + 2$) with value $\frac{17}{7}$ – the sister of the 13/7 corner (Theorem 64, Proposition 65).

Status. Together with Theorem 22, every blocked *nested* exchange in the templates above satisfies $\rho > T$: for all ω in the canonical template – with S a pair or a triple – for $\omega < 0.9626$ with one extra occupant under arbitrary hole occupancy, for $\omega < 0.624$ and $\omega < 0.896$ with two and three, and for all $\omega \in (0, 1)$ with four or more. All these floors exceed $T > \varphi$, so for

Conjecture 21 the decisive open front is the *pan* exchange (Theorem 66, Open Problem 24). Of the remaining nested gaps, the width tips at $j = 1$ ($\omega \geq 0.9626$, where the golden line still gives $\varphi^2 - \frac{\varphi}{2}\omega \geq \varphi^2 - \frac{\varphi}{2} = 1.809 > \varphi$) and at $j = 3$ ($\omega \geq 0.896$, where $\Psi_3(\omega) > \Psi_3(1) = \sqrt{3} > \varphi$) matter only for pinning the nested threshold at exactly T . Three nested gaps were also open *at the golden level*: the $j = 2$ tip for $\omega \in [\varphi/2, 1)$, where the proved floor $\Psi_2(\omega)$ crosses φ at $\Psi_2(\varphi/2) = \varphi$ exactly and no other wall applies; the quantitative “gap lemma” (small rings against the geometric walls; tiny σ_2), below which only the combinatorial floor $\rho_3^*(\omega)$ survives (above φ only for $\omega \leq 1 - \varphi/2$); and profiles of four or more pieces beyond the reduction branch (and $k \geq 3$ with extra occupants beyond it), which sit on the same combinatorial floor. All three now carry written proofs (Theorems 73 and 74: the entire nested template, every j), with duality-certificate backing and adversarial verification, in Appendix E. Each statement in this section is backed by a verification script and an adversarial verification report in the repository (Appendix A).

9 The two objectives diverge: a phase diagram

Contact area behaves like $2\pi w \sum r_i$ minus a per-ring penalty πw^2 ; cardinality counts rings. With uniform width the hole of a large ring is large, so nesting usually reconciles the two objectives, and the divergence survives only marginally: small rings that fit k times in the pan but $k - 1$ times in the large ring’s hole. The minimal instance is $R = 10$, $w = 1$, radii $\{9.0, 4.2, 4.2, 4.2\}$: the area optimum is the 9.0 with one nested 4.2 ($N = 2$, $A \approx 76.7$), the cardinality optimum the three 4.2’s ($N = 3$, $A \approx 69.7$). Figure 2 maps the divergence region for the family “one large ring b plus unlimited equal small rings of radius s ”: a staircase governed by the proved optimal thresholds for n equal circles in a disk [8, 9, 10, 11, 12]; its upper edge is exactly the three-circle threshold $0.4641R$.

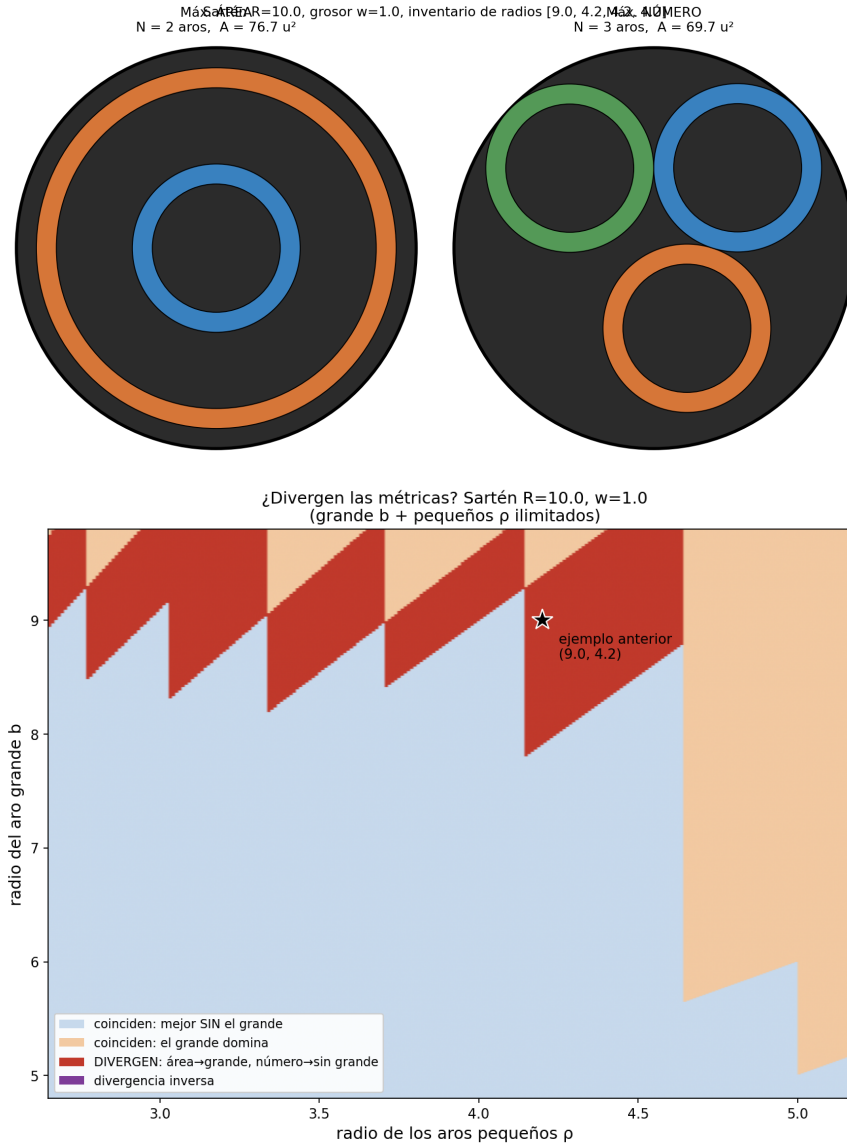


Figure 2: Top: the minimal divergence instance; area optimum (left) versus cardinality optimum (right). Bottom: phase diagram of the divergence band for uniform width.

10 Hardness decouples into two layers

If all radii lie within a band of width less than w , no nesting is possible and the decision problem “do all rings fit?” is exactly circle packing of the outer radii into the pan.

The geometric layer. Since Theorems 6 and 9 hold for arbitrary containers, our model contains, as the special case of a *square* pan with a no-nesting band, the problem of deciding whether given circles fit into a square, which is NP-hard by reduction from 3-PARTITION [4]; hence the geometric layer of the model is NP-hard. Membership in NP is itself delicate, because tight packings may require coordinates without short descriptions – a phenomenon made precise by the $\exists\mathbb{R}$ -completeness framework for two-dimensional packing [6]. For the disk pan specifically, NP-hardness of deciding whether given disks fit into a circular container is established in the journal version of Fekete–Keldenich–Scheffer [5, Theorem 5.1] (the conference version does not contain this result – we thank the referee for the precise pointer). Integer-programming formulations for circle packing in rectangular containers are studied in [16].

The combinatorial layer. Independently of the geometry, the *additive surrogate* (Theorem 16) is weakly NP-hard, and here the reduction can be made exact despite the per-ring penalty in the area.

Proposition 23 (The additive layer is weakly NP-hard). *Deciding whether an instance of the additive surrogate admits a feasible set of contact area at least a given threshold is weakly NP-hard, already for instances with a no-nesting radius band.*

Proof. Reduce from EQUAL-CARDINALITY PARTITION: given positive integers $a_1, \dots, a_n$, n even, with $\sum_i a_i = 2B$, decide whether some S with $|S| = n/2$ has $\sum_S a_i = B$. (This variant is NP-hard: from a PARTITION instance $b_1, \dots, b_m$ with $\sum_i b_i = 2B_0$, take $K > \sum_i b_i$ and the $2m$ numbers $b_1 + K, \dots, b_m + K, K, \dots, K$ (m copies of K); an m -subset with t items of the first kind has sum $mK + \sum_{\text{chosen}} b_i$, so m -subsets of sum $mK + B_0$ correspond exactly to subsets of the original instance of sum B_0 , of any size, padded to cardinality m with K -items.) Fix any width $w > 0$, set $M = 3B$, pick $\delta \in (\frac{w}{3B}, \frac{w}{2B})$, and take radii $r_i = \delta(a_i + M)$ with pan capacity $R^+ = \delta(\frac{n}{2}M + B)$. All radii lie in $[3B\delta, 5B\delta]$, so $r_i > w$ (each is a genuine ring, $a(r) = 2\pi wr - \pi w^2$) and $r_{\max} - r_{\min} = 2B\delta < w$ (no nesting): feasibility is exactly $\sum_S r_i \leq R^+$. Write $s = |S|$ and $t = \sum_S a_i \leq 2B$; the area is $A(S) = 2\pi w\delta(sM + t) - \pi w^2 s$, and we ask whether $A(S) \geq \Theta := 2\pi w\delta(\frac{n}{2}M + B) - \pi w^2 \frac{n}{2}$. If a balanced partition exists, taking S with $s = n/2$, $t = B$ attains Θ exactly. Conversely: $s > n/2$ is infeasible, since it would force $(s - \frac{n}{2})M \leq B - t \leq B < M$. For $s = \frac{n}{2} - q$ with $q \geq 1$, the requirement $A(S) \geq \Theta$ rearranges to $wq \geq 2\delta(qM + B - t)$, but

$$2\delta(qM + B - t) \geq 2\delta(qM - B) \geq 2\delta q(M - B) = 4\delta qB > wq$$

using $t \leq 2B$, $q \geq 1$, $M = 3B$ and $\delta > \frac{w}{4B}$ (implied by $\delta > \frac{w}{3B}$): impossible. Finally $s = n/2$ forces $t \geq B$ from $A \geq \Theta$ and $t \leq B$ from feasibility, i.e. $t = B$: a balanced partition. $\square$

Superincreasing radii eliminate exactly this combinatorial layer – precisely as in the superincreasing knapsack [15] – leaving only the geometric one: by Theorem 6, the area optimum then reduces to n calls to a sibling-packing feasibility oracle. Useful blackboxes for the remaining geometry include the critical-density theorem for disks in a disk (any family of total area at most half the container’s packs, and $1/2$ is tight) [5] and Split Packing [7].

11 Generalizations

The model invites several generalizations, all of which we record here both as context for the theorems (two of them hold with no extra work) and as a research program.

(a) *Container shape and dimension.* Theorems 6 and 9 are stated and proved for arbitrary containers in $\mathbb{R}^d$: rectangular griddles, and tubes/spherical shells in three dimensions – the original RCPP setting – are covered verbatim, because the exchange argument acts inside the vacated ball of the moved ring only. The sharpness results (Section 5) are disk-specific; their analogues for squares and in $\mathbb{R}^3$ are open.

(b) *Variable width.* With per-ring widths w_i , a large ring may have a small hole, which destroys the “nesting rescue” and should widen the divergence band of Section 9 dramatically; the superadditivity lemma must be re-examined since a depends on both r and w .

(c) *Flexible rings.* Real squid rings bend: a ring partially resting on another still touches the pan outside the overlap, up to a lifted ramp of width δ around each overlap. The idealization $\delta = 0$ (contact area equals the annulus minus the overlapped region) defines a continuous relaxation in which partial placements trade contact for cardinality; $\delta > 0$ penalizes overlaps by a dead band proportional to the overlap perimeter.

(d) *Infinite inventories.* Unlimited supplies (already used in the phase diagram) suggest degenerate limits: maximal nested chains $r, r-w, r-2w, \dots$, density questions, and Apollonian-type limit configurations.

(e) *Objectives.* Theorem 6 covers every positive, strictly increasing, superadditive value; weighted counts and concave values fall outside and, by Proposition 8, genuinely so.

12 Open problems

Open Problem 24. Prove Conjecture 21. Two halves. (a) *The pan exchange has floor φ :* proved in full for pair profiles in Theorem 66 (adversarially verified): blocking the pan exchange forces $\rho > \varphi$ for one occupant at every width $\omega > 0$ (both reinserted rings exceed the pocket $b(o_1)$ by containment and rigidity, and the two tails meet at the golden fixed point $2b(A)A = 1 + 2b(A)$, whose crossing factors as $(A^2 - A - 1)(2A + 1)$); for the evacuation branch at every $\omega \in (0, 1)$ (a *strict leaf* yields the Ψ_B program and $\Psi_B(1) = \varphi$ exactly); and for the remaining branch through the strict-leaf ladder ($\Psi_1(\frac{1}{2}) = \varphi$, $\Psi_2(\varphi/2) = \varphi$, $\Psi_3(1) = \sqrt{3} > \varphi$), with a U_4 -based wall ($o_2 < 1 + 1/o_1$) and a pincer on the smallest node above o_2 closing the rest. Larger profiles are closed in the light, nested, heavy-large and heavy- Ψ_B regimes of Theorem 67, and $p = 3$ with one occupant at every width. After Theorem 68, extra small rings adjoin to the profile (Corollary 69, a complete proof), and the single heavy cell $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ and the solid-pivot regime $\omega \geq 1, j \geq 3$ now carry *written proofs* (Theorem 72, via the insertion lemma, with a single certified maximization – the same standard as Theorem 68). (b) *Assemble the universal reinsertion lemma* for the nested exchange from the proved floors. The $j = 1$ and $j = 3$ width tips matter only for pinning the nested threshold at exactly T ; the three nested gaps that were open at the golden level itself, the $j = 2$ tip $\omega \in [\varphi/2, 1)$, the profiles of four or more pieces beyond the reduction branch, and the quantitative gap lemma now ALL carry written proofs (Theorems 73 and 74: the entire nested template, every j), and the exchange assembly – the case partition over the containers of m , the container port, and the $\alpha \notin v$ campaign – is carried out there with its facts at theorem level; the container port itself is now written (the pan-destination case is a corollary of the two nested theorems via the tower root, and the hole-

destination case assembles from exact pincers); the shared-root cell carries a certified-corner closure ($\pi + 4 \arcsin(1/\sqrt{3}) < 2\pi$) and the pocket- φ lemma closes the profile and dust directions of the scaling exactly. The two hole-corona branches of the port are closed by running the nested template inside the hole (automatic regime for $k \geq 3$ occupants, bounded family with the trio bound for $k \leq 2$, and a strong breathing lemma $X_{>m} + \omega > \varphi$ exact after discounting dust). A geometric-tail lemma makes the written theorems’ shadow budgets uniform in the occupant count (suffix masses decay geometrically and the leader’s own cascade forbids a tight pair with a full tail), so their sweep caps drop out. What remains for a full proof of Conjecture 21 is exactly: the occupant-count direction j of the *computational* scaling closures (numeric, swept to $j \leq 9$ with exact geometric cascade growth), and written proofs replacing the remaining adversarially verified computational closures.

Open Problem 25. Can a placement rule with access to the full input attain the lex-max set with polynomially many calls to a sibling-packing oracle, or is there a complexity-theoretic obstruction?

Open Problem 26. Characterize the divergence band of Section 9 exactly; determine the approximability of the cardinality objective; decide whether the additive surrogate admits a pseudopolynomial algorithm for integer radii; and settle membership in NP (versus $\exists\mathbb{R}$ -completeness) of disk-into-disk packability, whose NP-hardness is [5, Theorem 5.1].

Acknowledgements

The results of Sections 6–8 were developed with extensive computer assistance (Anthropic’s Claude) under a draft–verify workflow: every result was independently rederived from its bare statement by an adversarial verification pass before being accepted, and several statements were corrected or sharpened by that pass; the reports are part of the repository. To be explicit about epistemic status: the mathematical guarantee for every claim is the written proof and, where a proof delegates an identity, an exact symbolic computation; the verification workflow and the numerical sweeps are quality control and evidence, never a substitute for proof.

A Verification map

Every theorem of Sections 6–8 is backed in the accompanying repository by (i) a self-contained draft with complete proofs, (ii) a verification script whose blocks separate exact symbolic identities (**sympy**) from numerical sweeps, all green, and (iii) an *adversarial verification report*: an independent rederivation of each result from its statement alone, followed by an audit and directed attacks. The correspondence, as “verified claims (script blocks)”, is: Theorem 17 – **suelo_rigido**, 10/10 claims (7 blocks); slack pans and the universal trio frontier – **universal** (5/5 blocks); the κ identity – **h1**, 6/6 claims (5/5 blocks); the width curve and Theorem 22 – **grosor** 8/8 claims, **esquina** 6/6 claims (5/5 blocks); profile thresholds – **perfil_tres**, **cuatro** (**cuatrok** 5/5 blocks); the corona criterion – **corona** (5/5 blocks); occupant walls, the opposite-disk lemma and the metallic floors – **ocupantes** (5/5), **bloqueadores** (6/6); the double-pocket wall and the golden line – **bolsillo** (6/6); three-piece profiles and the golden deformation – **stripe** (5/5); the golden counterexample family, Theorem 19 – **aureo** (5/5); the pan-exchange golden floor – **batalla2** (6/6), **microcelda** (5/5) and, for larger profiles, **perfilp** (5/5) and **rstar** (6/6); the corona-versus-tails campaign of Appendix E – **coronacolas** (5/5),

coronanidada (5/5), the duality certificates **zigzag** (5/5), and the exchange-assembly book-keeping **ensamblaje** (6/6), **puertocii** (7/7) and the scaling layer **escala** (5/5); the written-proof layer – **compactacion** (5/5), **insercion** (7/7), **insercionanidada** (6/6), **gaplemma** (5/5), **coronaagujero** (5/5), **colageometrica** (5/5), **optimizacion** (5/5), **repack** (5/5), **r2bcert** (5/5), **arcolp** (5/5), **bolsillos** (5/5), **f3cierre** (5/5, the conditional duality lemma; its adversarial round refuted a stronger closure claim and extended the arc-LP validation to $k = 6$ against an independent LP solver), **r2bmulti** (5/5, the multipiece certificates; its adversarial round found no soundness gap and corrected the domain claim to the sweep box) and **areduccion** (5/5, the golden-threshold reduction lemma and the heavy branches; its adversarial round confirmed the algebra by hand, found no soundness gap, and contributed the $4t_0 = \varphi - 1$ incompatibility), **espxy** (5/5, the diametral-pocket geometry of the $X_Y > 0$ mirrored corona; its adversarial round verified the algebra by hand and contributed $p(\varphi, 1; \varphi + 1) = \varphi/2$) and **espvals** (5/5, the vacuity of the diametral-pocket sliver under the global tail wall and the floor rigidity; its adversarial round refuted a proposed vacated-ball witness and supplied the missing wall, correcting the **espxy** report’s legality claim – the errata is recorded) **auditcolas** (5/5, the global-tail audit of every existence claim in the campaign; its adversarial round refuted a partial-tail first version and contributed the forbidden-triple mechanism) and **f3vacio** (5/5, the forced-sub-pocket lemma closing the near-equal-tops vacuity; its adversarial round corrected the declared generator domain and verified the confinement attack on the lower bound). Several statements were corrected or sharpened by the adversarial rounds (a missing hypothesis in the trio frontier, a refuted “leak” family, the $\alpha \geq o_1$ case split, a refuted induction in the duality layer, a repaired extension lemma and case split in the assembly); the reports record each incident.

A fourth layer is machine-checked. The repository contains a Lean 4 development (**lean/**, core only, no **mathlib**) that formalizes the *exact-certificate layer* on which the proofs rest: arithmetic in $\mathbb{Q}$ and $\mathbb{Q}[\sqrt{5}]$ with a decidable real order, and polynomial identities over $\mathbb{Z}[A]$ and $(\mathbb{Q}[e])[d]$. Its forty-one theorems cover the golden arithmetic of Theorem 19 ($\varphi^2 = \varphi + 1$, $b_2(\varphi, 1) = \varphi/2$, the fixed point and its factorization, the witness certificate at $s^* = 4(\sqrt{5} - 2)$, the ε_0 margin and the width window), the metallic-mean certificates of Theorem 66 (including $\Psi(1/2) = \Psi_B(1) = \Psi_2(\varphi/2) = \varphi$ and the monotonicity certificate for g), the Tribonacci floor ($P(\varphi) = -1$, the bracket $1.8392 < T < 1.8393$ and a certificate that P increases strictly on $[1, \infty)$), the 13/7 corner, the additive threshold family, the pincer and boundary-corner certificates of Theorem 68, and the duality-certificate identities of Appendix E (the vanishing Descartes discriminant and pocket $\varphi/2$ of the golden pair, the critical margin identity, the (N) -heritage golden line, the mirror corner $b_2(2, \sqrt{5} - 1) = 1$, the strong-breathing chain $\varphi(3 - \varphi) = 2\varphi - 1 = \sqrt{5}$, $\sqrt{5} - (\varphi - 1) = \varphi$, the golden π -trio identity behind $\theta(\varphi, 1/\varphi) + \theta(1/\varphi, 1) + \theta(1, \varphi) = \pi$ at $R = 2\varphi$, the reduction-lemma thresholds $t_0 = (\varphi - 1)/4 = 1/(4\varphi)$ with $5t_0 = \varphi - \beta^*$ and $4t_0 = \varphi - 1$ exact at $\beta^* = (9 - \sqrt{5})/8$, and the diametral-pocket identities $\varphi^2 + \varphi + 1 = 2\varphi^2$, $x^*(\varphi) = \varphi/2$ and $\varphi(4 - \varphi) = 3\varphi - 1$). There are no **sorrys** and no new axioms (**#print axioms** reports only **propext**, **Classical.choice** and **Quot.sound**); **native_decide** is not used, so every proof is checked by the Lean kernel. The geometric layer proper – corona rigidity, angular criteria, placement trees, evacuations – is *not* formalized and remains covered by the scripts and reports above.

B Complete proof of Theorem 17

Throughout, work in the normalization of Theorem 17: $r_1 = 1$, $t = r_2$, $p = r_3$, $q = r_4$, width $\omega > 0$, pan radius $R = 1 + t$, and recall $b(t) = t(1+t)/(1+t+t^2)$, $T^3 = T^2 + T + 1$, $t^* = 1/T$ (equivalently $t^{*3} + t^{*2} + t^* = 1$), $\varphi = (1 + \sqrt{5})/2$. All exact algebraic identities used below are additionally verified in `code/rigido.py` (block V5).

Lemma 27 (Half-angle identity). *Let two circles of radii a, b be internally tangent to the wall of a disk of radius R (centers at distances $R - a$, $R - b$ from the center). If $a + b \leq R$, the minimal angular separation $\theta(a, b)$ guaranteeing disjoint interiors is well defined, lies in $[0, \pi]$, satisfies*

$$\sin^2 \frac{\theta(a, b)}{2} = f(a)f(b), \quad f(x) = \frac{x}{R - x},$$

and is increasing in each argument; two wall-tangent circles at angular separation γ have disjoint interiors iff $\gamma \geq \theta(a, b)$.

Proof. By the law of cosines the center distance $D(\gamma)^2 = (R - a)^2 + (R - b)^2 - 2(R - a)(R - b) \cos \gamma$ is increasing on $[0, \pi]$, and disjointness is $D \geq a + b$. At the threshold, $1 - \cos \theta = [(a + b)^2 - (a - b)^2] / (2(R - a)(R - b)) = 2ab / ((R - a)(R - b))$, i.e. $\sin^2(\theta/2) = f(a)f(b)$; and $f(a)f(b) \leq 1$ is exactly $a + b \leq R$. Monotonicity is that of f . $\square$

Lemma 28 (Construction). *Let $q \leq p \leq t < 1$ and $R = 1 + t$. If $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$, then $\{1, p, q\}$ packs in the disk of radius R .*

Proof. All three angles are defined ($1 + p \leq R \iff p \leq t$, and $p + q \leq 2t < R$). Place the three circles wall-tangent with the unit circle in the middle: centers at angles 0 , $+\theta(1, p)$, $-\theta(1, q)$. Each circle is contained in the disk (internal wall tangency). The pairs $(1, p)$ and $(1, q)$ are separated by exactly their thresholds. For (p, q) , the angular difference is $\Delta = \theta(1, p) + \theta(1, q)$ and the separation is $\gamma = \min(\Delta, 2\pi - \Delta)$. If $\gamma = \Delta$: by monotonicity (Lemma 27, $1 \geq p$), $\theta(1, q) \geq \theta(p, q)$, so $\Delta \geq \theta(p, q)$. If $\gamma = 2\pi - \Delta$: the hypothesis gives $2\pi - \Delta \geq \theta(p, q)$. $\square$

Lemma 29 (Algebraic reduction). *Define $\psi(x) = \sqrt{(t - x)/x}$ (decreasing, $\psi(t) = 0$) and $\tau_t = t/\sqrt{1 + t}$; note the exact identities $\psi(b(t)) = \tau_t$ and $t/(1 + \tau_t^2) = b(t)$. If $q \leq p \leq t < 1$ and $\psi(p) + \psi(q) \geq \tau_t$, then $\theta(1, p) + \theta(1, q) + \theta(p, q) \leq 2\pi$, and hence $\{1, p, q\}$ packs in the disk $R = 1 + t$ by Lemma 28.*

Proof. Let $A = \sqrt{f(1)f(p)}$, $B = \sqrt{f(1)f(q)}$, $C = \sqrt{f(p)f(q)}$ (the sines of the half-angles); the goal is $\arcsin A + \arcsin B + \arcsin C \leq \pi$. If $\arcsin A + \arcsin B \leq \pi/2$ there is nothing to prove, since $\arcsin C \leq \pi/2$. Otherwise the goal is equivalent, both sides lying in $[0, \pi/2]$, to $C \leq \sin(\arcsin A + \arcsin B) = A\sqrt{1 - B^2} + B\sqrt{1 - A^2}$. Dividing by $C > 0$ and writing $G(x) = f(1)/f(x) - f(1)^2$,

$$\frac{A\sqrt{1 - B^2}}{C} = \sqrt{G(q)}, \quad \frac{B\sqrt{1 - A^2}}{C} = \sqrt{G(p)}, \quad G(x) = \frac{(1 + t)(t - x)}{t^2 x},$$

the last identity by direct computation with $f(1) = 1/t$, $f(x) = x/(1 + t - x)$; the radicals are real because $f(1)f(x) \leq 1 \iff x \leq t$. Thus $\sqrt{G(x)} = (\sqrt{1 + t}/t)\psi(x)$ and, by hypothesis, $\sqrt{G(p)} + \sqrt{G(q)} = (\sqrt{1 + t}/t)(\psi(p) + \psi(q)) \geq (\sqrt{1 + t}/t)\tau_t = 1$, which is the required inequality. $\square$

The contrapositive used below: hypothesis (F3) implies $\psi(p) + \psi(q) < \tau_t$. This is the only direction ever used – the *constructive* one – so no exactness of any angular criterion is assumed anywhere in this appendix.

Lemma 30 (Corner minimization). *Let $0 < q \leq p \leq t < 1$ with $\psi(p) + \psi(q) < \tau_t$. Then: (1) $p, q > b(t)$; (2) if $t \geq 1/\varphi$ then $p + q > 1$; (3) if $t < 1/\varphi$ then $p + q > t + b(t)$.*

Proof. (1) Each summand is $< \tau_t$, ψ is decreasing and $\psi(b(t)) = \tau_t$. (2) $p + q > 2b(t)$ and $2b(t) - 1 = (t^2 + t - 1)/(t^2 + t + 1) \geq 0$ exactly when $t \geq 1/\varphi$. (3) Substitute $\alpha = \psi(p)$, $\beta = \psi(q)$, so $p = U(\alpha)$, $q = U(\beta)$ with $U(z) = t/(1 + z^2)$ strictly decreasing; the region is $\alpha, \beta \geq 0$, $\alpha + \beta < \tau_t$. Increasing α to $\tau_t - \beta$ decreases p , so $p + q > U(\tau_t - \beta) + U(\beta) =: W(\tau_t - \beta)$. Now $U''(z) = 2t(3z^2 - 1)/(1 + z^2)^3 \leq 0$ for $z \leq 1/\sqrt{3}$, and $\tau_t \leq 1/\sqrt{3} \iff 3t^2 - t - 1 \leq 0 \iff t \leq (1 + \sqrt{13})/6 = 0.7676\dots$, which covers $t < 1/\varphi = 0.618\dots$ with room to spare. Hence U is concave on $[0, \tau_t]$, W is concave, and its minimum over $[0, \tau_t]$ is at the endpoints: $W \geq W(0) = U(0) + U(\tau_t) = t + t/(1 + \tau_t^2) = t + b(t)$, by the identity $t/(1 + \tau_t^2) = b(t)$. $\square$

Proof of the strict floor in Theorem 17. Let $I \in \mathcal{F}$, normalized. By (F3) and the contrapositive of Lemmas 28–29, $\psi(p) + \psi(q) < \tau_t$. If $t \geq 1/\varphi$, Lemma 30(2) gives $p + q > 1$, contradicting (F2) ($p + q \leq 1 - \omega < 1$); so $t < 1/\varphi$ and Lemma 30(3) gives the blocking pressure $p + q > t + b(t)$. Chaining with the witness pressure (F2),

$$t + b(t) < p + q \leq 1 - \omega < 1,$$

and $t + b(t) < 1 \iff t(1 + t) < (1 - t)(1 + t + t^2) = 1 - t^3 \iff t^3 + t^2 + t < 1 \iff t < t^*$ (which also proves that $\mathcal{F}$ forces $t < t^*$). Finally

$$\rho \geq \rho_2 = \frac{p + q}{t} > \frac{t + b(t)}{t} = 1 + \frac{1 + t}{1 + t + t^2} =: L(t),$$

and L is strictly decreasing ($L' = -(t^2 + 2t)/(1 + t + t^2)^2 < 0$) with $L(t^*) = 1/t^* = T$ (an identity modulo $t^{*3} + t^{*2} + t^* = 1$). Since $t < t^*$: $\rho > L(t) > L(t^*) = T$. $\square$

It remains to prove exactness of the infimum. Two ingredients: the rigid pocket is exact, and infeasibility survives small perturbations.

Proposition 31 (Exact rigid pocket). *For $0 < q \leq t < 1$, the triple $\{1, t, q\}$ packs in the disk of radius $R = 1 + t$ if and only if $q \leq b(t)$.*

Proof. Rigidity. Centers satisfy $|c_1| \leq t$, $|c_t| \leq 1$ and $|c_1 - c_t| \geq 1 + t$; the triangle inequality forces equality throughout, so up to rotation $c_1 = (-t, 0)$, $c_t = (1, 0)$. *Sufficiency.* The pocket circle tangent to the wall and to both has radius $b(t)$; any $q \leq b(t)$ placed concentrically inside it works. *Necessity.* Let X be the center of q , $d = |X| > 0$, and γ the angle between X and $(1, 0)$. Disjointness reads $d^2 + t^2 + 2dt \cos \gamma \geq (1 + q)^2$ and $d^2 + 1 - 2d \cos \gamma \geq (t + q)^2$, which bound $\cos \gamma$ from below and above; compatibility of the two bounds requires $(1 + t)d^2 \geq (1 + q)^2 + t(t + q)^2 - t(1 + t)$, and with $d \leq R - q = 1 + t - q$:

$$(1 + t)(1 + t - q)^2 - (1 + q)^2 - t(t + q)^2 + t(1 + t) \geq 0.$$

The left-hand side factors *exactly* as $4(t^2 + t + 1)(b(t) - q)$ (symbolic identity, V5), whence $q \leq b(t)$. $\square$

Lemma 32 (Closure and monotonicity of packability). *Let $D = \{(t, p, q) : 0 < q \leq p \leq t < 1\}$ and say $z = (t, p, q)$ packs if there are centers with $|c_1| \leq t$, $|c_2| \leq 1 + t - p$, $|c_3| \leq 1 + t - q$, $|c_1 - c_2| \geq 1 + p$, $|c_1 - c_3| \geq 1 + q$, $|c_2 - c_3| \geq p + q$. Let $E \subseteq D$ be the packable set. Then: (1) E is downward closed in (p, q) ; (2) E is closed in D ; (3) for fixed t, q , the set $\{p \in [q, t] : (t, p, q) \in E\}$ is empty or a closed interval $[q, p_{\max}]$ with the maximum attained; in particular, if $(t, t, q) \notin E$, then $(t, p, q) \notin E$ for all $p \in (p_{\max}, t]$ – the infeasibility of the rigid corner propagates to $\delta := t - p \in [0, \delta_0)$, $\delta_0 = t - p_{\max} > 0$.*

Proof. (1) The same centers witness (all six constraints relax). (2) Witness centers live in a fixed compact ball; take a convergent subsequence of witnesses along $z_k \rightarrow z$ and pass to the limit in the six non-strict continuous constraints. (3) By (1) the set is an initial segment, by (2) it is closed. $\square$

Proposition 33 (The infimum is T and is not attained). $\inf\{\rho(I) : I \in \mathcal{F}\} = T$.

Proof. $\geq$ and non-attainment: the strict floor above. $\leq$: fix $t < t^*$, set $\varepsilon_t = (1 - t - b(t))/2 > 0$ and $q_t = b(t) + \min(\varepsilon_t, (t - b(t))/2)$. By Proposition 31, (t, t, q_t) is infeasible; by Lemma 32(3) there is $\delta \in (0, \min(\varepsilon_t, t - q_t))$ with $(t, t - \delta, q_t)$ infeasible. The instance $I_t = (1, t, t - \delta, q_t)$ with $\omega = 1 - (t - \delta + q_t) > 0$ satisfies (F1)–(F3) and the strict orderings, so $I_t \in \mathcal{F}$, and $\rho(I_t) = \rho_2 \leq (t + b(t) + \varepsilon_t)/t = L(t) + \varepsilon_t/t \rightarrow L(t^*) = T$ as $t \rightarrow t^*$ (the other tails are smaller: $\rho_1 = t + p + q < 3t < T$ and $\rho_3 = q/p < 1$). $\square$

Proof of Remark 18. A packing in a disk of radius $R' \leq R$ is a packing in the disk of radius R (translate to concentric position; containment is preserved, pairwise constraints are unchanged). Hence non-packability at radius R implies non-packability at radius $r_1 + r_2 \leq R$, so an instance with a slack pan satisfying (F2) and (F3) at its own radius satisfies (F2)–(F3) at radius $r_1 + r_2$, and the floor applies verbatim; the approximating family of Proposition 33 has $R = r_1 + r_2$, which is admissible, so the infimum is unchanged. $\square$

C Complete proofs for the width program

This appendix proves every statement of Section 7. Throughout, normalize $r_m = 1$ and $\omega = w/r_m \in (0, 1)$, and write $\beta := 1 - \omega$. The reinsertion resources of the exchange step are the two root nests $A = N(1)$ (the vacated disk D_m) and $B = N(\beta)$ (the hole H_m), plus recursive nesting inside S itself: a ring x fits in the hole of a ring y iff $x \leq y - \omega$. For siblings in a disk of capacity c we use: one circle, $s \leq c$; two circles, $s + s' \leq c$, which is *exact* (sufficiency by tangent placement on a diameter; necessity from $s + s' \leq |c_1 - c_2| \leq |c_1| + |c_2| \leq (c - s) + (c - s')$); and for three or more only *constructive* certificates – the row of Lemma 3 and explicit placements – so that every lower bound below is unconditional. For a profile $S = \{s_1 \geq \dots \geq s_k\} \subset (0, 1)$,

$$\rho_{\text{needed}}(S) = \max\left(\sum S, \max_j (\sum_{l>j} s_l) / s_j\right),$$

and $\rho_k^*(\omega) = \inf\{\rho_{\text{needed}}(S) : S \text{ not reinsertable in } A \uplus B\}$. Two elementary facts are used without mention: ρ_{needed} is monotone under adding smaller rings, and if a prefix $\{s_1, \dots, s_j\}$ is already non-reinsertable then so is S (every placement restricts).

C.1 Profile thresholds

Proposition 34 (Two rings). $S = \{\sigma_1 \geq \sigma_2\}$ is non-reinsertable iff $\sigma_1 + \sigma_2 > 1$ and $\sigma_2 > 1 - \omega$. Hence $\rho_2^*(\omega) = \max(1, 2(1 - \omega))$, approached by $\sigma_1 = \sigma_2 \rightarrow \max(\frac{1}{2}, 1 - \omega)^+$.

Proof. The possible placements are: both in A (iff $\sigma_1 + \sigma_2 \leq 1$, two-circle exact); one in each nest (the one in B measures $\geq \sigma_2$, so it requires $\sigma_2 \leq \beta$); σ_2 nested in σ_1 (requires $\sigma_2 \leq \sigma_1 - \omega \leq \beta$); both in B (stronger than one in each). Both stated conditions kill all four; failure of either condition enables the first or second placement. The infimum: blocking forces $\rho \geq \sigma_1 + \sigma_2 > \max(1, 2\beta)$, and the twin family attains the bound in the limit (its tail ratios tend to 1 and 0). $\square$

Proposition 35 (Three rings: characterization). Let $s_1 \geq s_2 \geq s_3$ in $(0, 1)$. S is non-reinsertable iff exactly one of the following holds (the cases are disjoint by the position of s_1, s_2 relative to β):

- (i) $s_2 > \beta$ and $s_1 + s_2 > 1$ (then s_3 is arbitrary);
- (ii) $s_2 > \beta$, $s_1 + s_2 \leq 1$, $s_3 > \beta$, and the trio does not pack in the unit disk (possible only for $\omega > \frac{1}{2}$);
- (iii) $s_2 \leq \beta < s_1$, $s_1 + s_3 > 1$, $s_2 + s_3 > \beta$, and $s_3 > s_1 - \omega$;
- (iv) $s_1 \leq \beta$, $s_2 + s_3 > 1$, and $s_3 > s_1 - \omega$.

Proof. The nesting forests on three rings are finite: no edge; one edge (s_3 in s_1 , s_3 in s_2 , s_2 in s_1); the chain $s_3 \subset s_2 \subset s_1$; the star $\{s_2, s_3\}$ in the hole of s_1 . Top-level rings are distributed among A and B under the sibling rules above; S is reinsertable iff some combination is feasible.

(i). Neither s_1 nor s_2 fits in B ($> \beta$) and s_2 cannot nest in s_1 ($s_2 \leq s_1 - \omega < \beta < s_2$ is absurd), so both must go to A , and $s_1 + s_2 > 1$ forbids it: blocked for every s_3 . Conversely, with $s_2 > \beta$ and $s_1 + s_2 \leq 1$: if $s_3 \leq \beta$, place $\{s_1, s_2\}$ in a row in A and s_3 in B ; if $s_3 > \beta$, all three live in the band $(\beta, 1)$, B is useless, nobody nests ($s_i - \omega < \beta < s_3$), so all three must go to A and reinsertability is exactly packability of the trio in the unit disk – case (ii), whose condition is also sufficient by the same argument. Habitability of (ii): $s_1 + s_2 > 2\beta$ and $s_1 + s_2 \leq 1$ force $\omega > \frac{1}{2}$.

(iii). ($\Leftarrow$) $s_1 > \beta$ has no parent and does not fit in B : $s_1 \in A$. No edge survives: $s_3 > s_1 - \omega \geq s_2 - \omega$ kills all three. A companion of s_1 in A needs $s_1 + s_i \geq s_1 + s_3 > 1$: impossible. So s_2, s_3 must both go to B , and $s_2 + s_3 > \beta$ forbids it. ($\Rightarrow$) If $s_1 + s_3 \leq 1$: $\{s_1, s_3\}$ in A , s_2 in B . If $s_2 + s_3 \leq \beta$: $\{s_2, s_3\}$ in B , s_1 in A . If $s_3 \leq s_1 - \omega$: nest s_3 in s_1 , $s_1 \rightarrow A$, $s_2 \rightarrow B$.

(iv). ($\Leftarrow$) $s_2 + s_3 > 1$ makes every pair infeasible in any root ($s_i + s_j \geq s_2 + s_3 > 1 > \beta$), so each root hosts at most one top-level ring; two roots cannot host three rings unless someone nests, and $s_3 > s_1 - \omega$ kills all edges as in (iii). ($\Rightarrow$) If $s_2 + s_3 \leq 1$: $\{s_2, s_3\}$ in A , s_1 in B ($s_1 \leq \beta$). If $s_3 \leq s_1 - \omega$: nest s_3 in s_1 , $s_1 \rightarrow A$, $s_2 \rightarrow B$. $\square$

Theorem 36 (Three rings: closed formula). For all $\omega \in (0, 1)$,

$$\rho_3^*(\omega) = \max\left(1, \min(2(1 - \omega), \max(\varphi, \frac{2}{1+2\omega}))\right),$$

with extremal families (as $\varepsilon \rightarrow 0^+$): $\{\frac{1}{2} + \omega, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ on the hyperbola, $\{1/\varphi, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ on the golden plateau $\omega \in [\frac{\sqrt{5}-2}{2}, 1 - \frac{\varphi}{2}]$, and $\{\beta + \varepsilon, \beta + \varepsilon, \delta\}$ beyond. The proof is purely additive: neither case (ii) nor any three-circle geometry enters.

Proof. We bound the infimum of ρ_{needed} over each case of Proposition 35 and take the minimum.

Case (i). $\rho \geq s_1 + s_2 > \max(1, 2\beta)$, and the family $s_1 = s_2 = \max(\beta, \frac{1}{2}) + \varepsilon$, $s_3 = \delta \rightarrow 0$ is blocked with $\rho_{\text{needed}} \rightarrow \max(1, 2\beta) = \rho_2^*$. (This “pair plus dust” argument also shows $\rho_{k+1}^* \leq \rho_k^*$ for every k .)

Case (ii). Any blocked profile has $\sum S > 1$ (otherwise the row of Lemma 3 places the trio in A), so its infimum is ≥ 1 , the formula’s value on $\omega \geq \frac{1}{2}$.

Case (iii). From $s_3 > \max(1 - s_1, s_1 - \omega)$ and $s_2 \geq s_3$: $\sum S \geq g(s_1) := s_1 + 2\max(1 - s_1, s_1 - \omega)$, which decreases until $s_1 = (1 + \omega)/2$ and increases after, with minimum $(3 - \omega)/2$; and $s_1 > \beta$. If $\omega \leq \frac{1}{3}$ then $\beta \geq (1 + \omega)/2$ and $\sum S > g(\beta) = 3 - 5\omega \geq 2 - 2\omega$; if $\omega \in (\frac{1}{3}, \frac{1}{2})$, $\sum S > (3 - \omega)/2 \geq 2 - 2\omega$; if $\omega \geq \frac{1}{2}$, $\sum S \geq s_1 + s_3 > 1$. In all ranges case (iii) is dominated by the formula’s value: it never furnishes the minimum.

Case (iv). Set $q = s_2 + s_3$. Blocking gives $q > 1$ and $q \geq 2s_3 > 2(s_1 - \omega)$, with $s_1 \leq \beta$, while $\rho \geq \max(s_1 + q, q/s_1)$. For fixed s_1 both bounds grow with q , so $q \downarrow Q(s_1) := \max(1, 2(s_1 - \omega))$. On the branch $s_1 \geq \frac{1}{2} + \omega$ ($Q = 2(s_1 - \omega)$) both $3s_1 - 2\omega$ and $2 - 2\omega/s_1$ increase in s_1 : the branch minimum is $\max(\frac{3}{2} + \omega, \frac{2}{1+2\omega})$ at the left endpoint. On the branch $s_1 \leq \frac{1}{2} + \omega$ ($Q = 1$), $h(s_1) = \max(s_1 + 1, 1/s_1)$ decreases until the crossing $s_1^2 + s_1 = 1$, i.e. $s_1 = 1/\varphi$ where $h = \varphi$, and increases after; the branch minimum sits at $s_1^* = \min(\beta, \frac{1}{2} + \omega, 1/\varphi)$. Comparing (exact crossings): for $\omega \leq \omega_1 := \frac{\sqrt{5}-2}{2}$ the infimum is $\frac{2}{1+2\omega}$ (the crossing $\frac{2}{1+2\omega} = \frac{3}{2} + \omega$ is $2\omega^2 + 4\omega - \frac{1}{2} = 0$, positive root ω_1); for $\omega_1 \leq \omega \leq 1 - 1/\varphi$ it is φ ; for $1 - 1/\varphi \leq \omega < \frac{1}{2}$ it is $\max(2 - \omega, 1/\beta) = 1/\beta \geq 2\beta$: dominated by case (i). For $\omega \geq \frac{1}{2}$ case (iv) is empty ($s_2 + s_3 \leq 2\beta < 1$). The lower bounds are approached by the stated families ($s_2 = s_3 = q/2$ with s_1 at each branch minimum satisfies all blocking conditions with margin ε).

Global minimum. $\frac{2}{1+2\omega} < 2(1 - \omega)$ on $(0, \frac{1}{2})$ (equivalent to $\omega(1 - 2\omega) > 0$) and $\varphi \leq 2(1 - \omega) \iff \omega \leq 1 - \varphi/2$: the case-wise minimum is the stated formula. $\square$

Proposition 37 (All profile sizes). $\rho_k^*(\omega) = \rho_3^*(\omega)$ for every $k \geq 3$ and every $\omega \in (0, 1)$.

Proof. ($\leq$) Dust: append $k - 3$ rings $\delta \rightarrow 0$ to any blocked three-ring witness. ($\geq$) Let S be non-reinsertable with k rings and $\rho := \rho_{\text{needed}}(S)$.

Step 0. The row of all of S in A fails: $\sum S > 1$, so $\rho > 1$; this closes $\omega \geq \frac{1}{2}$. Assume $\omega < \frac{1}{2}$ and $\rho < 2\beta$ (otherwise done).

Step 1. Two rings $> \beta$ would give $\sum S > 2\beta \geq \rho \geq \sum S$. If only $s_1 > \beta$: the placement “ $s_1 \rightarrow A$ alone, the rest in a row in B ” fails, and since each $s_i \leq \beta$ fits alone the failure is the sum: $\sum S - s_1 > \beta$, whence $\sum S > 2\beta$: impossible. So $s_1 \leq \beta$.

Step 2. “ $s_1 \rightarrow B$, the rest in a row in A ” fails: $Q := \sum S - s_1 > 1$. From $\rho \geq s_1 + Q$ and $\rho \geq Q/s_1$: $\rho \geq \min_{s \leq \beta} \max(1 + s, 1/s)$, which equals φ if $\beta \geq 1/\varphi$ and $1/\beta$ otherwise; in either case $\rho \geq \rho_3^*$ for all $\omega \geq \frac{\sqrt{5}-2}{2}$ (on the plateau $\rho_3^* \leq \varphi$; beyond it $\rho_3^* = 2\beta \leq \varphi$ resp. $2\beta \leq 1/\beta$ for $\beta \leq 1/\sqrt{2}$). Assume then $\omega < \frac{\sqrt{5}-2}{2}$, where $\rho_3^* = \frac{2}{1+2\omega}$, and towards a contradiction $\rho < \frac{2}{1+2\omega}$. From $\rho \geq Q/s_1 > 1/s_1$ we get $s_1 > \frac{1}{2} + \omega$, in particular $s_1 - \omega > \frac{1}{2}$.

Step 3. Let $p := \#\{i : s_i > s_1 - \omega\}$; all such rings exceed $\frac{1}{2}$ and $\sum S \leq \rho < 2$ allows at most three, so $p \in \{1, 2, 3\}$.

$p \geq 3$: $s_1, s_2, s_3 > s_1 - \omega > \frac{1}{2}$. No pair fits together in a root (sums $> 1 > \beta$, two-circle exact) and nobody nests ($s_j \leq s_i - \omega \leq s_1 - \omega < s_j$ is absurd): the prefix $\{s_1, s_2, s_3\}$ is already non-reinsertable, so $\rho \geq \rho_{\text{needed}}(\{s_1, s_2, s_3\}) \geq \rho_3^*$.

$p = 2$: $s_3, \dots, s_k \leq s_1 - \omega$. The placement “ $\{s_3, \dots, s_k\}$ in a row inside the hole of s_1 (capacity $s_1 - \omega$), $s_1 \rightarrow A$, $s_2 \rightarrow B$ ” fails, and the only piece that can fail is the row: $R := \sum_{i \geq 3} s_i > s_1 - \omega$.

With $s_2 > s_1 - \omega$, the tail $Q = s_2 + R > 2(s_1 - \omega)$ and $Q > 1$ (Step 2): exactly the case-(iv) program of Theorem 36, whose value is $\geq \rho_3^*$. (This is the only biting branch: the dust of $(\leq)$ lives here.)

$p = 1$: $s_2 \leq s_1 - \omega$. The placement “ s_2 in the hole of s_1 , $s_1 \rightarrow B$, $\{s_3, \dots, s_k\} \rightarrow A$ in a row” fails: $R' := \sum_{i \geq 3} s_i > 1$. Then $\sum S = s_1 + s_2 + R' > (s_2 + \omega) + s_2 + 1 = 1 + 2s_2 + \omega$ and the tail of s_2 gives $\rho \geq R'/s_2 > 1/s_2$, so

$$\rho \geq \min_s \max(1 + 2s + \omega, 1/s) = \frac{(1+\omega) + \sqrt{(1+\omega)^2 + 8}}{2} \geq 2,$$

contradicting $\rho < 2\beta < 2$. $\square$

Corollary 38 (Exact combinatorial threshold). *For $\omega < \omega_T = \frac{1}{T} - \frac{1}{2}$ no profile of any size blocks the exchange step with $\rho < T$; for $\omega > \omega_T$ the witness $\{\frac{1}{2} + \omega, \frac{1}{2} + \varepsilon, \frac{1}{2} + \varepsilon\}$ blocks with $\rho = \frac{2}{1+2\omega} < T$.* $\square$

C.2 The blocking frontier of the canonical trio

Work in the canonical template: the trio $\{\alpha, \sigma_1, \sigma_2\}$ with $0 < \sigma_2 \leq \sigma_1 \leq 1 < \alpha$ must pack in the pan of radius $R = \alpha + 1$. With the half-angle machinery of Appendix B (Lemma 27), set $f(x) = x/(R - x)$, $\theta(a, b) = 2 \arcsin \sqrt{f(a)f(b)}$ and $F(\sigma_1, \sigma_2) = \theta(\alpha, \sigma_1) + \theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)$; by the constructive direction (Lemma 28), $F \leq 2\pi$ implies the trio packs, so non-packability implies $F \geq 2\pi$ on the closure. Each θ is strictly increasing in both arguments, so $F = 2\pi$ defines the *blocking frontier* $\sigma_2 = h(\alpha, \sigma_1)$ where it meets the band. Two evaluations used repeatedly: $f(\alpha) = \alpha$ (since $R - \alpha = 1$) and $1 - \alpha f(\sigma) = R(1 - \sigma)/(R - \sigma)$.

Lemma 39 (Closed frontier). *Let $\alpha > 1$ and $\sigma := 1/\sqrt{\alpha(\alpha + 1)}$. In the band $0 < \sigma_2 \leq \sigma_1 \leq 1$,*

$$F(\sigma_1, \sigma_2) = 2\pi \iff t(\sigma_1) + t(\sigma_2) = \sigma, \quad t(s) = \sqrt{\frac{1-s}{s}},$$

and $\sigma = t(b(\alpha))$. Consequently $h(\alpha, \sigma_1) = t^{-1}(\sigma - t(\sigma_1))$ with $t^{-1}(u) = 1/(1 + u^2)$, defined and strictly decreasing exactly for $\sigma_1 \in [s^, 1]$ where $s^* = 4\alpha(\alpha + 1)/(2\alpha + 1)^2$ is the diagonal point ($t(s^*) = \sigma/2$); the range of h is $[b(\alpha), s^*]$, with $h(\alpha, 1) = b(\alpha)$.*

Proof. On a frontier point all three angles are interior: for the pairs (α, σ_i) , $P = \alpha f(\sigma_i) < 1$ since $\sigma_i < 1$ (and $\alpha f(1) = 1$); for (σ_1, σ_2) , $\sigma_1 + \sigma_2 \leq 2 < R$. Write $A = \theta(\alpha, \sigma_1)/2$, $B = \theta(\alpha, \sigma_2)/2 \in (0, \pi/2)$, so that $\sin^2 A = \alpha f(\sigma_1)$, $\cos^2 A = R(1 - \sigma_1)/(R - \sigma_1)$ (same for B with σ_2), and on the frontier $\theta(\sigma_1, \sigma_2)/2 = \pi - A - B \in (0, \pi)$.

($\Rightarrow$) Taking sines, $\sin(\theta(\sigma_1, \sigma_2)/2) = \sin(A + B) = \sin A \cos B + \cos A \sin B$. The cross terms factor:

$$(\sin A \cos B)^2 = \alpha f(\sigma_1) \cdot \frac{R(1 - \sigma_2)}{R - \sigma_2} = \alpha R t(\sigma_2)^2 f(\sigma_1) f(\sigma_2),$$

because $(1 - \sigma_2)/(R - \sigma_2) = t(\sigma_2)^2 f(\sigma_2)$; symmetrically with $t(\sigma_1)$. Since $\sin(\theta(\sigma_1, \sigma_2)/2) = \sqrt{f(\sigma_1)f(\sigma_2)} \neq 0$, dividing gives $1 = \sqrt{\alpha R} (t(\sigma_1) + t(\sigma_2))$, the stated equation with $\sigma = 1/\sqrt{\alpha R}$. The identity $(1 - b)/b = 1/(\alpha(\alpha + 1))$ for $b = b(\alpha)$ gives $t(b(\alpha)) = \sigma$.

($\Leftarrow$) For fixed σ_1 , both $F(\sigma_1, \cdot)$ (increasing) and $t(\sigma_1) + t(\cdot)$ (decreasing) are strictly monotone in σ_2 ; by ($\Rightarrow$) their level sets coincide where the frontier meets the band, and by monotonicity each has a unique solution. Existence: on the diagonal, $F(s, s)$ increases from 0^+ to $F(1, 1) > 2\pi$, so there is a unique s° with $F = 2\pi$, and ($\Rightarrow$) forces $t(s^\circ) = \sigma/2$, i.e. $s^\circ = s^*$. For $\sigma_1 \in [s^*, 1]$, $F(\sigma_1, 0^+) \leq \pi < 2\pi \leq F(\sigma_1, \sigma_1)$: the frontier crosses, and $t(\sigma_1) \leq \sigma/2$ keeps $h(\sigma_1) \leq \sigma_1$. For $\sigma_1 < s^*$ it would need $\sigma_2 > \sigma_1$. Endpoints: $h(s^*) = s^*$ and $h(1) = t^{-1}(\sigma) = b(\alpha)$. $\square$

Theorem 40 (The κ identity). *On the frontier, for every $\alpha > 1$ and $\sigma_1 < 1$,*

$$\kappa := -\frac{\partial h}{\partial \sigma_1} = \sqrt{\frac{g(\sigma_2)}{g(\sigma_1)}}, \quad g(s) = s^3(1-s), \quad \text{and} \quad \kappa \geq 1,$$

with equality iff $\sigma_1 = \sigma_2 = s^$.*

Proof. The identity: $t'(s) = -1/(2\sqrt{g(s)})$, and differentiating $t(\sigma_1) + t(h(\sigma_1)) = \sigma$ gives $\kappa = t'(\sigma_1)/t'(\sigma_2) = \sqrt{g(\sigma_2)/g(\sigma_1)}$.

The inequality, in the t -coordinate: since t reverses order, $\sigma_2 \leq \sigma_1 \iff t_2 \geq t_1$ on the segment $t_1 + t_2 = \sigma$, and the claim is $G(t_2) \geq G(t_1)$ for

$$G(t) := g(1/(1+t^2)) = \frac{t^2}{(1+t^2)^4}, \quad G'(t) = \frac{2t(1-3t^2)}{(1+t^2)^5} :$$

G increases on $[0, 1/\sqrt{3}]$ and decreases after. Note $\alpha > 1 \iff \alpha(\alpha+1) > 2 \iff \sigma < 1/\sqrt{2}$. If $\sigma \leq 1/\sqrt{3}$, both $t_i \leq \sigma$ lie in the increasing range and $t_2 \geq t_1$ gives the claim, with equality only at $t_1 = t_2$. If $\sigma > 1/\sqrt{3}$ and $t_2 > 1/\sqrt{3}$ (otherwise the previous case applies), then $t_1 = \sigma - t_2 < 1/\sqrt{2} - 1/\sqrt{3}$, so $G(t_1) \leq G(1/\sqrt{2} - 1/\sqrt{3}) = 0.01574\dots$, while $t_2 \in (1/\sqrt{3}, \sigma)$ lies in the decreasing range, so $G(t_2) > G(\sigma) > G(1/\sqrt{2}) = \frac{8}{81}$: here $\kappa^2 > \frac{8/81}{0.0158} > 6$, far from tight. $\square$

Corollary 41 (Minimum of $\sigma_1 + \sigma_2$; Tribonacci closure). *For every $\alpha > 1$, the minimum of $\sigma_1 + \sigma_2$ over the non-packable region $\{F \geq 2\pi, 0 < \sigma_2 \leq \sigma_1 \leq 1\}$ is $1 + b(\alpha)$, attained at $(\sigma_1, \sigma_2) = (1, b(\alpha))$. Consequently a blocking satisfying the witness wall $\sigma_1 + \sigma_2 \leq \alpha - \omega$ forces $1 + b(\alpha) \leq \alpha - \omega$, i.e. $\alpha \geq T_{1+\omega}$ (the root of $\alpha^3 = (1+\omega)(\alpha^2 + \alpha + 1)$); in particular no blocking exists for $\alpha \leq T$, at any width.*

Proof. The region is non-empty ($F(1, 1) > 2\pi$) and a minimum cannot have $F > 2\pi$ (decrease σ_2 : the sum drops, F stays $\geq 2\pi$ by continuity until the frontier is crossed, and $F(\sigma_1, 0^+) < 2\pi$ guarantees it is crossed). On the frontier, $d(\sigma_1 + h)/d\sigma_1 = 1 - \kappa \leq 0$ (Theorem 40): the minimum sits at $\sigma_1 = 1$ and equals $1 + b(\alpha)$. The sign of $1 + b(\alpha) - (\alpha - \omega)$ is governed by the polynomial identity $1 + b(\alpha) - \alpha = -(\alpha^3 - \alpha^2 - \alpha - 1)/(\alpha^2 + \alpha + 1)$ and its deformation $b(\alpha) - (\alpha - 1 - \omega) = -[\alpha^3 - (1+\omega)(\alpha^2 + \alpha + 1)]/(\alpha^2 + \alpha + 1)$, whose numerator has the single positive root $T_{1+\omega}$ (Descartes' rule); $T_1 = T$. $\square$

C.3 Positive width: the exact curve and the 13/7 corner

The full blocking program of the canonical template adds the remaining resources to the trio wall. With capacities $c_2 := 1 - \omega$ (the hole H_m) and $c_4 := \alpha - \omega - 1$ (σ_2 back in u next to m , two-circle exact), a realizable blocking satisfies (closures for infima)

$$(B1) \text{ trio non-packable, } (B2) \sigma_2 \geq c_2, \quad (B4) \sigma_2 \geq c_4, \quad (W) \sigma_1 + \sigma_2 \leq \alpha - \omega,$$

plus the band $\sigma_2 \leq \sigma_1 \leq 1$ and $\alpha - \omega \geq 1$, and $T_{\text{can}}(\omega)$ is the infimum of $\rho = \max(\sigma_1 + \sigma_2, (1 + \sigma_1 + \sigma_2)/\alpha)$ over blockings. (At $\omega = 0$ the program is empty – (B2) would force $\sigma_2 \geq 1$ – which is why the limit value T uses only (B1)+(W).)

Proposition 42 (Witness branch). *The infimum of ρ over $(B1)+(W)$ alone is $\Phi(\omega) = T_{1+\omega} - \omega$, attained in the limit $\alpha = T_{1+\omega}$, $\sigma_1 \rightarrow 1$, $\sigma_2 \rightarrow b(\alpha) = \alpha - 1 - \omega$. Moreover Φ is strictly increasing and concave, with*

$$\Phi'(\omega) = \frac{2\alpha + 1}{\alpha^2(\alpha^2 + 2\alpha + 3)} \Big|_{\alpha=T_{1+\omega}} > 0, \quad \Phi'(0) = \frac{2T + 1}{7T^2 + 4T + 3} = 0.1374\dots,$$

and on $[0, \frac{1}{7}]$ the chord-tangent bounds $T + (13 - 7T)\omega \leq \Phi(\omega) \leq T + \Phi'(0)\omega$ hold, the chord endpoint being exact: $T_{8/7} = 2$ and $\Phi(\frac{1}{7}) = \frac{13}{7}$.

Proof. By Corollary 41, blocking at a given α forces $\rho \geq \sigma_1 + \sigma_2 \geq 1 + b(\alpha)$ and (W) forces $\alpha \geq T_{1+\omega}$; b is strictly increasing ($b' = (2\alpha + 1)/(\alpha^2 + \alpha + 1)^2$), so the infimum over admissible α is $1 + b(T_{1+\omega}) = T_{1+\omega} - \omega$ (the cubic identity with equality), approached by $\sigma_1 = 1 - \varepsilon$, $\sigma_2 = b(\alpha) + \varepsilon'$. The derivative: implicit differentiation of $\alpha^3 = (1 + \omega)(\alpha^2 + \alpha + 1)$ and substitution of $1 + \omega = \alpha^3/(\alpha^2 + \alpha + 1)$ reduce the numerator of $d\alpha/d\omega - 1$ to $2\alpha + 1$ (symbolic identity). Concavity: $\frac{d}{d\alpha} \log \Phi' = \frac{2}{2\alpha+1} - \frac{2}{\alpha} - \frac{2\alpha+2}{\alpha^2+2\alpha+3} < 0$ and α increases with ω . $T_{8/7} = 2$ because $2^3 = 8 = \frac{8}{7} \cdot 7$. $\square$

Adding (B2) gives the second branch: every full blocking has $\rho \geq \sigma_1 + \sigma_2 \geq 2(1 - \omega)$. The maximum of the two branches is minimized at their crossing $T_{1+\omega} = 2 - \omega$, i.e. at the root $\omega_\times = 0.0754315\dots$ of $2\omega^3 - 10\omega^2 + 14\omega - 1$, giving the uniform floor $T_{\text{can}}(\omega) \geq 2(\alpha_\times - 1) = T + 0.00985\dots$ for $\omega \in (0, 0.30]$ – already enough for “width only raises the floor”. The exact curve is finer. For fixed α , pass to the coordinate $t_i = t(\sigma_i)$, where (B1) becomes *linear* (Lemma 39): $t_1 + t_2 \leq \Theta(\alpha) := t(b(\alpha))$, and the walls read $t_2 \leq \tau_c := t(c)$, $c := \max(c_2, c_4)$. If $\alpha > 2 + \omega$ there is no blocking at all ((B4) and the band give $\sigma_1 + \sigma_2 \geq 2c_4 > \alpha - \omega$, violating (W)); so $\alpha - \omega \in [1, 2]$ effectively.

Lemma 43 (Conditional minimum). *Let $\alpha \in (1, 2 + \omega)$ and $c = \max(1 - \omega, \alpha - \omega - 1) < 1$. The infimum of $S_0 = \sigma_1 + \sigma_2$ over non-packable trios with $\sigma_2 \geq c$ and $\sigma_2 \leq \sigma_1 \leq 1$ is*

$$(a) \ 1 + b(\alpha) \text{ if } c \leq b(\alpha); \quad (b) \ c + t^{-1}(\Theta(\alpha) - t(c)) \text{ if } b(\alpha) < c \leq s^*(\alpha); \quad (c) \ 2c \text{ if } c > s^*(\alpha),$$

where $t^{-1}(u) = 1/(1 + u^2)$ as in Lemma 39. Moreover each case is non-decreasing in α on its domain.

Proof. $t^{-1}(\cdot)$ is decreasing, so S_0 drops when either t_i rises. In the truncated box $\{t_1 \leq t_2 \leq \tau_c, t_1 + t_2 \leq \Theta\}$: if $2\tau_c \leq \Theta$ (case (c)) the box corner (τ_c, τ_c) is admissible and optimal, $S_0 = 2t^{-1}(\tau_c) = 2c$. Otherwise the minimum lives on the segment $t_1 + t_2 = \Theta$, where

$$\frac{dS_0}{dt_2} = -(t^{-1})'(\Theta - t_2) + (t^{-1})'(t_2) = 2[\sqrt{G(t_1)} - \sqrt{G(t_2)}] \leq 0$$

by the G -lemma inside Theorem 40 ($((t^{-1})'(x) = -2x/(1 + x^2)^2, ((t^{-1})'/2)^2 = G)$: push t_2 to its cap. If $\tau_c \geq \Theta$ (case (a)): $t_2 = \Theta$, $t_1 = 0$, i.e. $(\sigma_1, \sigma_2) = (1, b(\alpha))$. If $\tau_c < \Theta$ (case (b)): $t_2 = \tau_c$, $t_1 = \Theta - \tau_c$. Monotonicity in α : (a) is $1 + b(\alpha)$ with b increasing; (b) with $c = c_2$ fixed has $dS_0/d\alpha = (t^{-1})'(\Theta - \tau_c) \Theta'(\alpha) \geq 0$ ($(t^{-1})' < 0$, $\Theta' < 0$); (c) with the moving cap $c = c_4$ ($\alpha \geq 2$, $\tau_4 = t(c_4)$) has $dS_0/d\alpha = \tau_4'(-2)[\sqrt{G(\tau_4)} - \sqrt{G(t_1)}] + (t^{-1})'(t_1) \Theta' \geq 0$, since $\tau_4' < 0$, $G(\tau_4) \geq G(t_1)$ ($\tau_4 \geq t_1$ on the segment, same G -lemma) and the last two factors are both negative. $\square$

Proposition 44 (The exact curve). *For $\omega \in (0, 0.30]$, the proved lower bound for T_{can} is*

$$T_{\text{can}}(\omega) \geq \begin{cases} 2(1 - \omega) & \omega \in (0, \omega_1], \\ \alpha_m(\omega) - \omega & \omega \in [\omega_1, \frac{1}{7}], \\ \Phi(\omega) & \omega \in [\frac{1}{7}, 0.30], \end{cases}$$

where $\omega_1 = 0.0413570\dots$ is the root in $(\frac{1}{25}, \frac{1}{14})$ of $4\omega^3 - 20\omega^2 + 25\omega - 1$ and $\alpha_m(\omega) \in (1, 2]$ is the unique root of $t(\alpha - 1) + t(1 - \omega) = \Theta(\alpha)$, an algebraic function of degree 6 (its polynomial $P(\alpha, \omega)$, of bidegree $(6, 2)$, is irreducible over $\mathbb{Q}[\alpha, \omega]$). Equality holds on the witness branch and at the corner; on the other two branches the matching upper bound inherits the angular-criterion caveat.

Proof. Fix ω and minimize the value of Lemma 43 over α subject to (W).

H_m branch ($\omega \leq \omega_1$). Case (c) gives $S = 2(1 - \omega)$, constant in α , admissible iff (W): $\alpha \geq 2 - \omega$, and iff $c_2 > s^*(\alpha)$, i.e. $\alpha \leq \bar{\alpha}(\omega)$. The window is non-empty iff $s^*(2 - \omega) \leq 1 - \omega$, and $(1 - \omega) - s^*(2 - \omega)$ has numerator $-(4\omega^3 - 20\omega^2 + 25\omega - 1)$; the cubic is strictly increasing on $(0, \frac{5}{6})$ (derivative $(5 - 2\omega)(5 - 6\omega)$), so the window exists exactly for $\omega \leq \omega_1$. The other cases do not undercut $2(1 - \omega)$ (monotonicity in Lemma 43 plus continuity across case junctions).

Mixed branch ($\omega_1 < \omega \leq \frac{1}{7}$). For $\alpha \leq 2$, case (c) is empty (it would need $1 - \omega \geq s^*(\alpha) \geq s^*(2 - \omega)$, against the cubic) and case (a) needs $b(\alpha) \geq 1 - \omega$, impossible ($b(2) = \frac{6}{7} \leq 1 - \omega$). Case (b) with $c = c_2$ has value $S_0^b(\alpha) = 1 - \omega + t^{-1}(\Theta - t(1 - \omega))$, increasing in α , and (W) is equivalent to $\Xi(\alpha) := \Theta(\alpha) - t(\alpha - 1) \geq t(1 - \omega)$ with Ξ strictly increasing on $(1, 2]$, $\Xi(1^+) = -\infty$ and $\Xi(2) = t(\frac{6}{7}) \geq t(1 - \omega) \iff \omega \leq \frac{1}{7}$: the admissible α form $[\alpha_m, 2]$ and the minimum is $S_0^b(\alpha_m) = \alpha_m - \omega$ ((W) with equality). For $\alpha \in (2, 2 + \omega)$: if $b(\alpha) \leq 1 - \omega$, relax (B4) to (B2) and use monotonicity, $S_0 \geq S_0^b(\alpha) > S_0^b(\alpha_m)$; if $b(\alpha) > 1 - \omega$, $S_0 \geq 1 + b(\alpha) > 2 - \omega > \alpha_m - \omega$. Squaring the defining equation twice yields the polynomial P (necessary condition; the right branch is pinned by $\alpha_m \in (1, 2]$), with the junction identities $P(2 - \omega, \omega) = (\omega - 1)^3(4\omega^3 - 20\omega^2 + 25\omega - 1)$ (so $\alpha_m(\omega_1) = 2 - \omega_1$) and $\alpha_m(\frac{1}{7}) = 2$.

Witness branch ($\omega \geq \frac{1}{7}$). Now $b(T_{1+\omega}) = T_{1+\omega} - 1 - \omega \geq 1 - \omega \iff T_{1+\omega} \geq 2 \iff \omega \geq \frac{1}{7}$: the optimum of case (a) satisfies (B2), (W) forces $\alpha \geq T_{1+\omega}$ (Corollary 41) and the minimum is $\Phi(\omega)$; larger α and the cases (b)/(c₄) are increasing and start at $\Phi(\omega)$; smaller α admit no blocking (for $\alpha \leq 2$, $\Xi(\alpha) \leq \Xi(2) = t(\frac{6}{7}) \leq t(1 - \omega)$ with equality only at the corner; case (c) would need $\omega \leq \omega_1$). $\square$

Proof of Theorem 22. Lower bound. Take any blocking at width ω . If $\alpha \geq 2$: by Corollary 41 alone (no (B2), (B4), (W) needed), $\rho \geq 1 + b(\alpha) \geq 1 + b(2) = \frac{13}{7}$. If $\alpha < 2$: for $\omega \geq \frac{1}{7}$, by Corollary 41 and (W) blocking forces $\alpha \geq T_{1+\omega} \geq T_{8/7} = 2$, contradicting $\alpha < 2$: empty. For $\omega \leq \omega_1$: $\rho \geq 2(1 - \omega) \geq 2(1 - \omega_1) > \frac{13}{7}$ (since $\omega_1 < \frac{1}{14}$). For $\omega_1 < \omega < \frac{1}{7}$: $\rho \geq V(\omega) := \alpha_m(\omega) - \omega$, and $V > \frac{13}{7}$ there: if $V(\omega') = \frac{13}{7}$ then $(\alpha, \omega) = (\frac{13}{7} + \omega', \omega')$ satisfies $P = 0$, but

$$P(\frac{13}{7} + \omega, \omega) = \frac{4(7\omega - 1)Q_5(\omega)}{7^6},$$

$$Q_5 = 33614\omega^5 + 235298\omega^4 + 620830\omega^3 + 766066\omega^2 + 397831\omega - 289,$$

and Q_5 has no root in $[\frac{1}{25}, \frac{1}{7}]$ (exact root isolation; its only positive root is ≈ 0.0007), so the only root of the product there is $\omega = \frac{1}{7}$; since $V(\omega_1) - \frac{13}{7} = \frac{1}{7} - 2\omega_1 > 0$ and V is continuous, $V > \frac{13}{7}$ on all of $[\omega_1, \frac{1}{7}]$. Uniqueness of the infimum configuration: equality in the branch $\alpha \geq 2$ forces $\alpha = 2$, $\sigma_1 = 1$, $\sigma_2 = \frac{6}{7}$; then (B2) gives $\omega \geq \frac{1}{7}$ and (W) gives $\omega \leq \frac{1}{7}$.

The approximating family is genuine. At exactly $\alpha = 2$, $\omega = \frac{1}{7}$ there is no family: $\kappa > 1$ gives $\sigma_1 + h(2, \sigma_1) > \frac{13}{7} = \alpha - \omega$ for $\sigma_1 < 1$, violating (W); the corner is approached by opening (B4). Fix $\omega = \frac{1}{7}$ and let $\delta \downarrow 0$:

$$\alpha = 2 + \delta, \quad \varepsilon = \delta^2/4, \quad \sigma_1 = 1 - \varepsilon, \quad \sigma_2 = (\alpha - \omega - 1) + \varepsilon/2.$$

All walls hold ((B4) with slack $\varepsilon/2$; (B2) since $\alpha - \omega - 1 \geq 1 - \omega$ for $\alpha \geq 2$; (W) since $\sigma_1 + \sigma_2 = \alpha - \omega - \varepsilon/2$). Genuineness: since $\alpha > T_{8/7} = 2$, Corollary 41 gives $\sigma_2 > b(\alpha)$ strictly, so the trio $\{\alpha, 1, \sigma_2\}$ is non-packable by the *rigidity* of Proposition 31 (the disk $R = \alpha + 1$ is diametrically full; rescale by $t = 1/\alpha$) – not merely by the angular criterion – and the infeasibility propagates from $\sigma_1 = 1$ to $\sigma_1 = 1 - \varepsilon$ by Lemma 32(3) for all small ε . Then $\rho = \alpha - \omega - \varepsilon/2 \rightarrow \frac{13}{7}$. $\square$

Remark 45 (Fine structure). The curve is not monotone on $(0, \frac{1}{7}]$: differentiating the defining equation of α_m at the junction and using the exact coincidence of the leading terms $((\alpha - 1)^3(2 - \alpha) = (1 - \omega_1)^3\omega_1$ at $\alpha = 2 - \omega_1$) gives $V'(\omega_1^+) = c_0/(r' - c_0) > 0$ (numerically $+0.0721$): ω_1 is a *local minimum* of T_{can} , followed by a local maximum at $\omega_{\text{peak}} = 0.0444700$ (the unique root in $(\omega_1, \frac{1}{7})$ – by Sturm – of an explicit degree-8 factor R_8 of the resultant of P and $P_\alpha + P_\omega$), a bump of height $1.1 \cdot 10^{-4}$, before descending to the corner. T_{can} has two local minima, ω_1 and $\frac{1}{7}$, and the global one is the corner.

D Complete proofs for generic containers

This appendix proves every statement of Section 8; its final subsection (D.6) proves the golden floor of the pan exchange for pair profiles, in full. Normalize $r_m = 1$, $\omega = w/r_m$. The template: v is the pan, of radius R , containing $\{\alpha\} \cup O \cup \{m\}$ with $O = \{o_1 \geq \dots \geq o_j\}$, $j \geq 0$, $o_i \geq m = 1$ (rings larger than m agree in F and P); u is the hole of α , capacity $\alpha - \omega \geq 1$; the witness placed $S = \{\sigma_1 \geq \sigma_2\}$ in u , $\sigma_2 \leq \sigma_1 \leq 1$. Holes may be occupied arbitrarily: contents of a hole at any depth are parametrized by $X_y := \sum \text{children}(y)$. The exchange builds a placement P' that preserves the container assignments of all rings $\geq m$; *positions* inside each container are existential (a placement is a nesting forest whose sibling groups pack). A *node* is any ring of radius ≥ 1 that is an occupant of v other than α and m , or is nested (at any depth) inside such an occupant; nodes and their descendants live outside m . Unless stated otherwise, blocked-exchange walls are closures of strict inequalities, as in Appendix C.

D.1 The universal frontier and the corona criterion

Lemma 46 (Universal frontier). *Let $\{A, x, y\}$ be wall-tangent in a disk of arbitrary radius R , in the interior domain $A + x, A + y, x + y < R$. Set $c := R - A$, $T_c(x) := \sqrt{(c - x)/x}$ and $\tau_R := c/\sqrt{AR}$. If $A \geq \min(x, y)$, then*

$$F = \theta(A, x) + \theta(A, y) + \theta(x, y) \geq 2\pi \iff T_c(x) + T_c(y) \leq \tau_R.$$

The direction ($\Rightarrow$) is unconditional; the converse fails without the hypothesis ($R = 1$, $A = 0.01$, $x = y = 0.45$ is a counterexample).

Proof. The cross term factors for arbitrary R : with $\sin^2(\theta_{Az}/2) = f(A)f(z)$, $f(z) = z/(R - z)$, one has the identity

$$f(A)f(x)(1 - f(A)f(y)) = \frac{AR}{(R - A)^2} \cdot \frac{c - y}{y} \cdot f(x)f(y),$$

so $\sin(\theta_{Ax}/2)\cos(\theta_{Ay}/2) = \frac{\sqrt{AR}}{c}T_c(y)\sqrt{f(x)f(y)}$ and, adding the symmetric term, $\sin s = \frac{\sqrt{AR}}{c}(T_c(x)+T_c(y))\sin w$ with $s = (\theta_{Ax}+\theta_{Ay})/2$, $w = \theta_{xy}/2$. The linear form is thus $\sin s \leq \sin w$, while $F \geq 2\pi$ is $s+w \geq \pi$. Since $w \leq \pi/2$: $s+w \geq \pi \Rightarrow \sin s \leq \sin w$ always; the converse needs $s > w$, which the hypothesis provides: if $A \geq y$, monotonicity gives $\theta(A, x) \geq \theta(y, x) = 2w$, so $s > w$. $\square$

Uniform companions (same computations as in Appendix C, now for general R): the general pocket $b_R(A) = ARc/(AR + c^2)$, equal to Descartes' b_2 when $R = A + B$, increasing in R and always $\leq A$; and the slope identity $\kappa = \sqrt{g_c(y)/g_c(x)}$ with $g_c(s) = s^3(c - s)$, since T_c is a primitive of $-c/(2\sqrt{g_c})$.

Lemma 47 (Gap system). *A corona of $\{a_1, \dots, a_k\}$ in the disk R (all k circles wall-tangent) with cyclic order π exists iff the linear system*

$$\theta_{ij} \leq S_j - S_i \leq 2\pi - \theta_{ij} \quad (1 \leq i < j \leq k, S_1 = 0)$$

is feasible, where θ_{ij} is the angle of the pair in positions i, j of π .

Proof. The angular separation of positions $i < j$ is $\min(S_j - S_i, 2\pi - (S_j - S_i))$, and $\min(\Delta, 2\pi - \Delta) \geq \theta \iff \theta \leq \Delta \leq 2\pi - \theta$ because $\theta \leq \pi$; wall-tangent circles are contained in the disk and pairwise disjointness is exactly the angular condition. $\square$

Lemma 48 (Subset certificates). *If $\{a_1, \dots, a_k\}$ has a corona with cyclic order π , then every subset T , $|T| \geq 3$, satisfies $\sum_{\text{consecutive pairs of } T} \theta \leq 2\pi$ in the induced cyclic order. Consequently, if some T exceeds 2π in all its cyclic orders, no corona exists in any order – the unconditional direction used by all lower bounds.* $\square$

The naive criterion (“some order with consecutive sum $\leq 2\pi$ implies a corona”) is *false* for $k \geq 4$: the conditional triangle inequality $\theta(a, c) \leq \theta(a, b) + \theta(b, c)$ holds when $b \geq \min(a, c)$ but fails badly for small b ($R = 1$, $a = c = 0.45$, $b \rightarrow 0$), and $\{0.47, 0.47, 0.47, 0.02\}$ has a small cyclic sum while its top trio is already infeasible. The correct statement:

Theorem 49 (Exact criterion per order, $k = 4$). *For arbitrary symmetric $\theta_{ij} \in (0, \pi]$, the system of Lemma 47 for the order $(1, 2, 3, 4)$ is feasible iff the four trio certificates and the total $\theta_{12} + \theta_{23} + \theta_{34} + \theta_{14} \leq 2\pi$ hold.*

Proof. Necessity: Lemma 48. Sufficiency, by difference-constraint duality: each upper bound is an up-edge $i \rightarrow j$ of weight $2\pi - \theta_{ij}$, each lower bound a down-edge $j \rightarrow i$ of weight $-\theta_{ij}$; the system is feasible iff the digraph has no negative directed cycle, and it suffices to exclude *simple* cycles. A simple cycle Γ with U up-edges has weight $w(\Gamma) = 2\pi U - \sum_{e \in \Gamma} \theta_e$, so $w < 0$ requires $\#\text{edges}(\Gamma) > 2U$ (as $\theta \leq \pi$). With $k = 4$, a simple cycle has ≤ 4 edges, so only $U = 1$ can go negative; a $U = 1$ cycle is an up-edge $i \rightarrow j$ plus a decreasing chain from j to i , and its negativity condition $\theta_{ij} + \sum_{\text{chain}} \theta > 2\pi$ is exactly the subset certificate of $\{i, \text{chain}, j\}$: enumerating, sizes 3 give the four trios and size 4 gives the total. $\square$

Proposition 50 (The zigzag minimizes the total). *For $a_1 \geq a_2 \geq a_3 \geq a_4$, the three cyclic orders have totals $\sum_{i < j} \theta_{ij}$ minus one perfect matching, and the maximum-sum matching is $\theta_{12} + \theta_{34}$; hence $\min_{\pi} \text{TOT}(\pi) = \sum_{i < j} \theta_{ij} - \theta_{12} - \theta_{34}$, achieved by the zigzag order (a_1, a_3, a_2, a_4) .*

Proof. With $\sigma_i := \log f(a_i)$ (decreasing) one has $\theta(a_i, a_j) = g(\sigma_i + \sigma_j)$ for $g(s) = 2 \arcsin(e^{s/2})$ on $s \leq 0$, with $g''(s) = e^{-s}/(2(e^{-s} - 1)^{3/2}) > 0$: g is convex and increasing (continuity at $s = 0$ by passage to the limit). The three matchings have pair-sums $\{\sigma_1 + \sigma_2, \sigma_3 + \sigma_4\}$, $\{\sigma_1 + \sigma_3, \sigma_2 + \sigma_4\}$, $\{\sigma_1 + \sigma_4, \sigma_2 + \sigma_3\}$ with equal totals, and the first majorizes the other two; convexity turns majorization into the claimed maximality. $\square$

Theorem 51 (Lemma U_4 : global criterion, two inequalities). $\{a_1 \geq a_2 \geq a_3 \geq a_4\}$ admits a corona in the disk R (all pairs admissible) iff

$$\theta_{12} + \theta_{13} + \theta_{23} \leq 2\pi \quad \text{and} \quad \sum_{i < j} \theta_{ij} - \theta_{12} - \theta_{34} \leq 2\pi.$$

Proof. Trio certificates do not depend on the order, so a corona exists iff all trios pass and the minimal total does (Theorem 49 + Proposition 50). By monotonicity of θ , every trio is dominated pairwise by the top trio $\{a_1, a_2, a_3\}$, so one trio inequality suffices. Neither inequality is redundant (both are active on the frontier). The top-trio condition linearizes by Lemma 46 with automatic hypothesis ($A = a_1$ is the global maximum): it reads $T_c(a_2) + T_c(a_3) \geq \tau_R$ with $c = R - a_1$. $\square$

For $k = 5$ the subset-only criterion fails for exactly one pattern, the *pentagram*: of the 84 simple cycles, all with $U \geq 2$ are count-dominated or LP-dominated except the cycle through the five diagonals, and adding its certificate $\sum_D \theta \leq 4\pi$ restores an exact criterion for arbitrary θ ; for *geometric* θ (rank-one matrices $\sin^2(\theta_{ij}/2) = f_i f_j$) the identity $\prod_D f_i f_j = \prod_{\text{sides}} f_i f_j = (\prod f_i)^2$ obstructs the pattern and the best value found is 7.556 against the $4\pi = 12.57$ needed; its redundancy for geometric θ is stated as a conjecture in the repository and is *not used*: lower bounds only ever use the unconditional direction of Lemma 48. For $k \geq 5$ the subset quantifier does not commute with the order quantifier (explicit five-ring counterexamples in the repository), so the per-order criterion or the LP is the correct tool.

D.2 Occupant walls and metallic floors

Lemma 52 (Walls, free-hole template). *Suppose the holes of m , σ_1 and all o_i are free. If the exchange is blocked, then*

$$(B2) \ \sigma_2 > 1 - \omega; \quad (B3) \ \sigma_2 > \sigma_1 - \omega; \quad (B4) \ \sigma_2 > \alpha - \omega - 1; \quad (Bo) \ \sigma_2 > o_k - \omega \ \forall k;$$

$$(D) \ \sigma_1 + \sigma_2 > 1; \quad (W) \ \sigma_1 + \sigma_2 \leq \alpha - \omega.$$

In particular $o_k < \sigma_2 + \omega \leq 1 + \omega$ for all k : every extra occupant stays within one width of m .

Proof. Each line is the contrapositive of an explicit placement whose two resources are disjoint and whose feasibility criteria (pair, nesting, row) are exact: $\sigma_2 \rightarrow H_m$ with $\sigma_1 \rightarrow D_m$; $\sigma_2 \subset \sigma_1 \rightarrow D_m$; σ_2 next to m in u ($1 + \sigma_2 \leq \alpha - \omega$, pair-exact) with $\sigma_1 \rightarrow D_m$; $\sigma_2 \subset o_k$ with $\sigma_1 \rightarrow D_m$; the whole pair into D_m (a free unit disk). (W) is the legality of the witness's placement. $\square$

Proposition 53 (Occupant price). *Blocking in the free-hole template with $j \geq 1$ extra occupants implies*

$$\rho > 2 + \frac{j - 2\omega}{o_1} \geq \frac{j + 2}{1 + \omega} \quad (\omega \leq j/2), \quad \text{and for } j = 1, \ \omega \geq \frac{1}{2}: \quad \rho > \frac{4}{1 + 2\omega}.$$

Proof. The tail of o_1 contains $\{o_2, \dots, o_j, m, \sigma_1, \sigma_2\}$ (ties among equal radii are broken by index, so the tail of a ring collects all later copies: the *first-copy convention*). By (Bo), $\sigma_1 \geq \sigma_2 > o_1 - \omega$, and each $o_i \geq 1$: $\rho \geq ((j-1) + 1 + \sigma_1 + \sigma_2)/o_1 > (j + 2(o_1 - \omega))/o_1 = 2 + (j - 2\omega)/o_1$. If $\omega \leq j/2$ the expression decreases in o_1 and $o_1 < 1 + \omega$ (Lemma 52). For $j = 1 < 2\omega$, add (D): $\sigma_1 + \sigma_2 > \max(1, 2(o_1 - \omega))$, so $\rho > (1 + \max(1, 2(o_1 - \omega)))/o_1$, whose infimum over o_1 is $4/(1 + 2\omega)$ at $o_1 = \frac{1}{2} + \omega$. $\square$

Corollary 54 (Crossings; canonical optimality). *For $j = 1$: $\rho > 3/(1 + \omega) > T$ for $\omega < 3/T - 1 = 3T^2 - 3T - 4 = 0.6310\dots$, and combining with the fine branch, $\rho > T$ for all $\omega < 2/T - \frac{1}{2} = 2T^2 - 2T - \frac{5}{2} = 0.5873\dots$, with $\rho \geq \frac{13}{7}$ for $\omega \leq \frac{15}{26}$. For $j \geq 2$: $\rho > 4/(1 + \omega) > 2 > T$ for all $\omega < 1$. Moreover every extra-occupant blocking has $\rho > 2$ for $\omega < \frac{1}{2}$, while the entire canonical curve lies strictly below 2 – the witness branch by the exact identity $(2 + \omega)^3 - (1 + \omega)((2 + \omega)^2 + (2 + \omega) + 1) = 1 > 0$, i.e. $T_{1+\omega} < 2 + \omega$, hence $\Phi(\omega) < 2$ – so adding occupants is strictly suboptimal for the adversary: the canonical template is the optimal v in this class.* $\square$

D.3 Occupied holes: the opposite-disk lemma and the silver floor

Lemma 55 (Opposite disk). *Let a disk have capacity c , let $\sigma \leq c$ be the largest piece and C a multiset of pieces $\leq \sigma$. If $\sum C \leq c - \sigma$, then $\{\sigma\} \cup C$ packs in the disk.*

Proof. Place σ wall-tangent with center $(-(c - \sigma), 0)$. The disk of radius $r = c - \sigma$ centered at $(c - r, 0)$ is wall-tangent and exactly tangent to σ (center distance $(c - \sigma) + (c - r) = c = \sigma + r$): a free disk of radius $c - \sigma$, in which C fits in a row (Lemma 3). $\square$

Lemma 56 (General blocking wall Bo''). *Blocking implies, for every node y : $y < \sigma_2 + \omega + X_y$.*

Proof. Otherwise $\sum \text{children}(y) \leq (y - \omega) - \sigma_2$ and Lemma 55 packs $\{\sigma_2\} \cup \text{children}(y)$ in the hole of y (if some child exceeds σ_2 , apply the lemma with that child as the largest piece; the inequality is the same), with $\sigma_1 \rightarrow D_m$: unblocked. Positions are re-eligible, and inserting σ_2 (a ring $< m$) only changes the container of σ_2 : legal. $\square$

Theorem 57 (Silver floor; m without children). *Blocking with $j \geq 1$, holes occupied arbitrarily at any depth, m childless, implies $\rho > \Psi(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + 1}$, the positive root of $u^2 - 2(1 - \omega)u - 1$. In particular $\Psi(0) = 1 + \sqrt{2}$ (the silver ratio), $\Psi(\frac{1}{4}) = 2$, and $\Psi > T \iff \omega < (T - 1)^2/2 = 0.35220\dots$*

Proof. Let y^* be a node of minimal radius; its children are all < 1 (a child ≥ 1 would be a smaller node), so $X := X_{y^*}$ is a sum of rings smaller than m , living in the tail of m . With $\sigma := \sigma_2 > 1 - \omega$ (wall (B2), valid since H_m is free): the tail of m gives $\rho \geq \sigma_1 + \sigma_2 + X \geq 2\sigma + X$, and the tail of y^* with Lemma 56 gives $\rho \geq (1 + \sigma_1 + \sigma_2 + X)/y^* > (1 + 2\sigma + X)/(\sigma + \omega + X)$. Minimize $F(\sigma, X) = \max(2\sigma + X, (1 + 2\sigma + X)/(\sigma + \omega + X))$ over $\sigma \geq 1 - \omega$, $X \geq 0$: for fixed σ the first term grows in X and the second decreases, so $F(\sigma, \cdot)$ is bounded below by the crossing value $u = 2\sigma + X$ with $u(u + \omega - \sigma) = 1 + u$, i.e. the positive root of $u^2 - (1 + \sigma - \omega)u - 1 = 0$ (when the crossing falls at $X < 0$, the same algebra shows $F(\sigma, 0) = 2\sigma$ already exceeds the root). The root increases in σ , so the program minimum is at $\sigma = 1 - \omega$, giving $\Psi(\omega)$. $\square$

Theorem 58 (Evacuation dichotomy; m with children). *The conclusion of Theorem 57 holds without the hypothesis that m is childless.*

Proof. Evacuation to D_m : place σ_1 and all children of m in a row inside D_m (possible if $\sigma_1 + M \leq 1$, $M := \sum \text{children}(m)$; children of σ_1 travel inside σ_1) and σ_2 in the emptied H_m (possible if $\sigma_2 \leq 1 - \omega$). Contrapositive: blocking implies $\sigma_2 > 1 - \omega$ or $\sigma_1 + M > 1$. In branch A the proof of Theorem 57 applies verbatim (it used only $\sigma_2 > 1 - \omega$, Bo'' and the two tails; the mass $M \geq 0$ only fattens them). In branch B, let y^* be the minimal node, $s := \sigma_2 + X > 1 - \omega$ (Bo'' and $y^* \geq 1$) and $A := \sigma_1 + \sigma_2 + M + X > 1 + s$; the two tails give $\rho \geq A > 1 + s$ and $\rho \geq (1+A)/y^* > (2+s)/(s+\omega)$, whose optimization yields the metallic mean $\Psi_B(\omega)$, the positive root of $u^2 - (2-\omega)u - 1$. Since the positive root of $u^2 - bu - 1$ increases in b and $2-\omega \geq 2(1-\omega)$, $\Psi_B \geq \Psi$: branch B is dominated. (The crossing $\Psi_B = T$ happens at $\omega = (T-1)^2 -$ twice the branch-A threshold – via the *polynomial* identity $(T-1)^2 T - (2T-T^2+1) = T^3 - T^2 - T - 1$.) $\square$

Corollary 59. *In the canonical template ($j = 0$) with m arbitrary: branch A gives the full curve $T_{\text{can}}(\omega)$ (its program uses (B2) only as an inequality), branch B gives $\rho \geq \Phi(\omega)$ (Proposition 42 uses only (B1) and (W)); in both, $\rho > \Phi(\omega) > T$ for all $\omega > 0$.* $\square$

D.4 The double-pocket wall, the golden line, and Ψ_j

Throughout this subsection $j \geq 1$ and the walls above are in force; write $X := X_{o_1}$, $M := \sum \text{children}(m)$, $X_\sigma := \sum \text{children}(\sigma_1)$, and add the tariffed-nesting wall (B3'): $\sigma_2 + X_\sigma > \sigma_1 - \omega$ (Lemma 55 applied to the hole of σ_1).

Lemma 60 (Double pocket). *Blocking (template $j = 1$, arbitrary occupancy) implies $\sigma_1 > b_2(\alpha, o_1)$, where $b_2(A, B) = AB(A+B)/(A^2 + AB + B^2)$.*

Proof. Blocking implies $\{\alpha, o_1, \sigma_1, \sigma_2\}$ does not pack in the pan (full repacking is a legal resource: children travel inside their parents, positions are existential). Since $\alpha + o_1 \leq R$, non-packability is inherited by containment in the disk $\bar{R} = \alpha + o_1$, where the pair $\{\alpha, o_1\}$ is diametrically rigid: $|c_\alpha| \leq o_1$, $|c_{o_1}| \leq \alpha$, $|c_\alpha - c_{o_1}| \geq \alpha + o_1$ force equality throughout – the rescaled rigidity of Proposition 31. By the exact necessity of that proposition (rescaled), any third circle disjoint from both has radius $\leq b_2(\alpha, o_1)$, and there are *two* pockets, one on each side of the diameter, mutually disjoint with slack: their centers are at $(x_0, \pm y_0)$ with $y_0^2 - b_2^2 = 3b_2^2$, i.e. $y_0 = 2b_2$ exactly, so the mirror circles are $4b_2$ apart. If $\sigma_1 \leq b_2$ then also $\sigma_2 \leq b_2$, and placing each σ concentric in a pocket packs the quadruple in $\bar{R} \subseteq$ the pan: contradiction. $\square$

Theorem 61 (Golden line, both branches). *Blocking in the template ($j = 1$, arbitrary occupancy) implies $\rho > \varphi^2 - \frac{\varphi}{2}\omega$ for all ω in every case except the corner $\{\text{branch B}, \alpha < o_1, o_1 > \tilde{o} = 1.2956, \text{node-child}, \omega < \frac{1}{2}\}$, where the proved bound is $\geq 2 > T$. In particular $\rho > T$ for all $\omega < \omega_A = 2(\varphi^2 - T)(\varphi - 1) = 2 - 2(T-1)(\varphi - 1) = 0.962585\dots$*

Proof. Write $g := \sqrt{5} - 1$. *Tail chain.* The tail of o_1 contains $\{m, \sigma_1, \sigma_2\} \cup \text{children}(o_1) \cup \text{children}(m) \cup \text{children}(\sigma_1)$, so $\rho o_1 \geq 1 + \sigma_1 + \sigma_2 + X + M + X_\sigma$; eliminating X with Bo'' ($X > o_1 - \sigma_2 - \omega$):

$$(*) \quad \rho > 1 + \frac{1 + \sigma_1 - \omega + M + X_\sigma}{o_1}.$$

Branch A ($\sigma_2 \geq 1 - \omega$). From (W), $\alpha \geq \sigma_1 + \sigma_2 + \omega \geq 1 + \sigma_1$. Lemma 60 with b_2 increasing in α gives $\sigma_1 > b_2(1 + \sigma_1, o_1)$, i.e. $o_1 < N(\sigma_1) := (1 + \sigma_1)(\sqrt{1 + 4\sigma_1} - 1)/2$ (the condition $b_2(1 + \sigma_1, N) = \sigma_1$ is the quadratic $N^2 + (1 + \sigma_1)N - \sigma_1(1 + \sigma_1)^2 = 0$). With (*) and $M, X_\sigma \geq 0$: $\rho > 1 + (1 + \sigma_1 - \omega)/N(\sigma_1) =: 1 + h(\sigma_1)$, and h is decreasing in σ_1 : the comparison of derivatives

reduces to the polynomial identity $(1 + 4\sigma_1) - (1 + 2\sigma_1 - 2\sigma_1^2)^2 = 4\sigma_1^3(2 - \sigma_1) > 0$. The infimum is at $\sigma_1 = 1$: $N(1) = g$, $1/g = \varphi/2$, and $\rho > 1 + (2 - \omega)\varphi/2 = \varphi^2 - \frac{\varphi}{2}\omega$.

Branch B ($\sigma_1 + M > 1$). Using $\sigma_1 + M > 1$, Bo'', (B3'), (W) and Lemma 60:

- (I) $\rho o_1 \geq 1 + \sigma_1 + \sigma_2 + M + X + X_\sigma > 2 + o_1 - \omega + 2b_2(\alpha, o_1) - \alpha$,
- (II) $\rho \alpha \geq o_1 + 1 + \sigma_1 + \sigma_2 + M + X + X_\sigma > 2o_1 + 2 - \omega + 2b_2(\alpha, o_1) - \alpha$ [if $\alpha \geq o_1$],

(in (I): $X > o_1 - \omega - \sigma_2$, $X_\sigma > \sigma_1 - \omega - \sigma_2$, $-\sigma_2 \geq \sigma_1 + \omega - \alpha$ by (W), $\sigma_1 > b_2$; (II) is the tail of α , which contains o_1 and its children only when $\alpha \geq o_1$). Set $f_1 := 1 + (2 - \omega + 2b_2 - \alpha)/o_1$ and $f_2 := (2o_1 + 2 - \omega + 2b_2 - \alpha)/\alpha$. If $o_1 \leq g$, the short chain (only $\sigma_1 + M > 1$ and Bo'', valid in any order) gives $\rho > 1 + (2 - \omega)/o_1 \geq 1 + (2 - \omega)/g = \varphi^2 - \frac{\varphi}{2}\omega$. If $o_1 > g$, the admissible α -region is $[1 + \omega, A_{\max}(o_1)]$ where $b_2(A_{\max}, o_1) = 1$, and two exact facts drive the optimization: b_2 is concave in α ($\partial^2 b_2 / \partial \alpha^2 = -6\alpha o_1^3(\alpha + o_1)/D^3 < 0$, D the denominator), so f_1 is concave in α and minimized at the endpoints; and $D^2 - \alpha o_1^2(o_1 + 2\alpha) = (\alpha + o_1)(\alpha^3 + \alpha^2 o_1 + o_1^3) > 0$ gives $\alpha \partial b_2 / \partial \alpha < o_1$, so f_2 is strictly decreasing in α . For $o_1 \in (g, \tilde{o}]$ ($\tilde{o} = 1.29558\dots$; any order of α, o_1 , only (I) used): at $\alpha = A_{\max}$, f_1 - curve $= c_{10}(o_1) + \omega(1/g - 1/o_1)$ with the ω -coefficient ≥ 0 and $c_{10} \geq 0$ on $[g, o^* = 1.5958]$, with the exact identity $f_1 \equiv$ curve at $o_1 = g$ (the golden corner, for all ω); at $\alpha = 1 + \omega$, $f_1 \geq$ curve on $[g, o^*] \times [0, 1]$ with contact only at $(g, \omega \rightarrow 1)$. For $o_1 > \tilde{o}$ with $\alpha \geq o_1$ (forcing $o_1 \leq \frac{3}{2}$, since $A_{\max}$ is decreasing with $A_{\max}(\frac{3}{2}) = \frac{3}{2}$ exactly; also $A_{\max}(g) = 2$, $A_{\max}(2) = g$ - the corner is self-dual): $f_2 \geq f_2(A_{\max}) = c_{20}(o_1) + \omega c_{21}(o_1)$ with $c_{21} \geq 0$ for $o_1 \leq 2$ and $c_{20} \geq 0$ on $[\tilde{o}, \frac{3}{2}]$. For $o_1 > \tilde{o}$ with $\alpha < o_1$ (a region requiring $\omega < \frac{1}{2}$: it needs $1 + \omega \leq \alpha < o_1 < N_1(\omega)$, and $N_1(\frac{1}{2}) = \frac{3}{2}$ exactly with N_1 decreasing, where $b_2(1 + \omega, N_1) = 1$): if the children of o_1 are all < 1 , the two α -free bounds $\rho > 1 + o_1 - \omega$ (tail of m : $\sigma_1 + M > 1$ and Bo'') and $\rho > 1 + (2 - \omega + 2b_2(1 + \omega, o_1))/o_1$ (tail of o_1 , which now contains α ; Bo'', (B3'), (W) cancel α , and Lemma 60 applies with $\alpha \geq 1 + \omega$) have $\max \geq$ curve $+ 0.307$ throughout the region (certified on a grid with symbolic endpoints); if o_1 has a node child we are in branch B with $\omega < \frac{1}{2}$, and Theorem 58 gives $\rho > \Psi_B(\omega) > \Psi_B(\frac{1}{2}) = 2 > T$ ($\Psi_B(\frac{1}{2}) = 2$ exact). $\square$

Proposition 62 (The Ψ_j ladder, unconditional). *Blocking with $j \geq 1$ extra occupants and arbitrary nested occupancy implies*

$$\rho > \Psi_j(\omega) = (1 - \omega) + \sqrt{(1 - \omega)^2 + j} \quad (\text{root of } u^2 - 2(1 - \omega)u - j),$$

with $\Psi_j > T \iff \omega < 1 - (T^2 - j)/(2T)$: 0.3522 ($j = 1$), 0.6240 ($j = 2$), 0.8959 ($j = 3$), and all $\omega \in (0, 1)$ for $j \geq 4$ ($\Psi_j(\omega) > \Psi_j(1) = \sqrt{j} \geq 2$ there; for $\omega > 1$ the $(1 - \omega)$ term is negative and Ψ_j drops below $\sqrt{j}$).

Proof (leaf argument). Call a node a *leaf* if it has no node children. The node subtree of each occupant o_i is finite and non-empty, hence contains a leaf ℓ_i ; the subtrees are disjoint, so there are j distinct leaves $\ell_1, \dots, \ell_j \geq 1$. Let L be the largest, $X_L = \sum \text{children}(L)$ (all < 1 , as L is a leaf), $M = \sum \text{children}(m)$, and W the total mass of rings < 1 other than σ_1, σ_2 (so $X_L \leq W$, $M \leq W$, and X_L, M are disjoint). Three facts: (1) *wall at the leaf* (Lemma 56): $L < \sigma_2 + \omega + X_L \leq \sigma_2 + \omega + W$; (2) *tail of L*: the other $j - 1$ leaves, m, σ_1, σ_2 and all mass < 1 are $\leq L$ (first-copy convention), so $\rho L \geq (j - 1) + 1 + \sigma_1 + \sigma_2 + W \geq j + 2\sigma_2 + W$; (3) *tail of m*: $\rho \geq \sigma_1 + \sigma_2 + W \geq 2\sigma_2 + W$. In branch A of the evacuation dichotomy ($\sigma_2 \geq 1 - \omega$), facts (1)–(3) give $\rho > \max(2\sigma + W, (j + 2\sigma + W)/(\sigma + \omega + W))$ at $\sigma = \sigma_2$; minimizing over $\sigma \geq 1 - \omega$, $W \geq 0$, the crossing $u^2 + u(\omega - \sigma - 1) - j = 0$ ($u = 2\sigma + W$) increases in σ , with minimum at

$\sigma = 1 - \omega$: exactly $\Psi_j(\omega)$. (If the crossing falls at $W < 0$ – possible for $\omega > \frac{1}{2}$, $j = 1$, $\sigma \rightarrow 1 -$ the value 2σ at $W = 0$ satisfies $(2\sigma)^2 - 2(1 - \omega)(2\sigma) - j \geq j > 0$, so $2\sigma > \Psi_j$ anyway.) In branch B ($\sigma_1 + M > 1$), set $s := \sigma_2 + X_L$: by fact (1) and $L \geq 1$, $s > 1 - \omega$; the tail of L also contains M , disjoint from X_L , so $\rho L \geq j + \sigma_1 + M + \sigma_2 + X_L > j + 1 + s$ with $L < s + \omega$, and the tail of m gives $\rho \geq \sigma_1 + M + \sigma_2 + X_L > 1 + s$; minimizing $\max(1 + s, (j + 1 + s)/(s + \omega))$ yields, with $u = 1 + s$, the metallic mean $u^2 - (2 - \omega)u - j = 0$, whose root dominates Ψ_j ($2 - \omega \geq 2(1 - \omega)$, and the root increases in the linear coefficient). $\square$

The leaf argument is what closes the case caught by the adversarial verification (a node child inside the largest occupant): towers of nested nodes inflate X_{o_1} without paying the tail of m , but they always leave a leaf at the bottom of each subtree, and the largest leaf carries both the j -bonus in its tail and the leaf ceiling $\sigma_2 + \omega + W$.

Corollary 63 (Small rings are free for the combinatorial walls). *Proposition 53, Theorems 57 and 58 and Proposition 62 hold unchanged if the instance contains additional rings smaller than m anywhere compatible with each theorem's template (with S still the pair $\{\sigma_1, \sigma_2\}$).*

Proof. Every unblocking placement used by those walls is *local*: it lives in D_m (freed by the exchange), in H_m , in the holes of nodes or of σ_1 , or next to m inside u – never in the free space of v . An extra ring smaller than m , wherever it sits, intersects none of these resources (hole contents are already counted in the X 's; rings in v proper stay put, which is legal since no other position changes), and tails can only grow. (The geometric wall of Lemma 60 is *not* covered: it uses the global repacking of the pan, which does see small rings in v ; this is the “gap lemma” left open.) $\square$

D.5 Three-piece profiles in the canonical template

Here v is the canonical pan of radius $R = \alpha + 1$ with $\{\alpha, m\}$ diametral, u is the hole of α , and the witness placed $S = \{\sigma_1 \geq \sigma_2 \geq \sigma_3\} \subset (\omega, 1)$ in u ; the hole of m may carry mass $M \geq 0$. The reinsertion resources are D_m (rows of sum ≤ 1 , with α in place), H_m (capacity $1 - \omega$; its content M can be evacuated in a row to D_m), the slot next to m in u (row $1 + \Sigma \leq \alpha - \omega$), nesting inside S , and the repack of v : placing $\{\alpha\} \cup (\text{roots of } S)$ as a corona in the pan – incompatible with using D_m or the evacuation, both of which pin α . The witness gives (W) $\sigma_1 + \sigma_2 \leq \alpha - \omega$.

Theorem 64 (Three pieces; the zigzag pays). *Blocking with S a triple and arbitrary M implies $\rho > \max(\Phi(\omega), \frac{13}{7})$ for all $\omega \in (0, 1)$; in the zigzag branch below, $\rho > \frac{17}{7}$.*

Proof. Branch 1 ($\sigma_3 \leq \sigma_1 - \omega$): the dust rides the pair. Nest σ_3 in σ_1 inside each pair placement; their failures give five walls: (B1) “ $\sigma_3 \subset \sigma_1$; corona $\{\alpha, \sigma_1, \sigma_2\}$ in v ” fails, so the trio has no corona and $F(\sigma_1, \sigma_2) > 2\pi$ (corona $\iff F \leq 2\pi$ is exact for three circles); (B2/dichotomy) “ $\sigma_1(\sigma_3) + M$ in a row in D_m ; $\sigma_2 \rightarrow$ emptied H_m ” fails, so $\sigma_2 > 1 - \omega$ or $\sigma_1 + M > 1$; (B4) “ $\sigma_2 \rightarrow u$ next to m ; corona $\{\alpha, \sigma_1(\sigma_3)\}$ ” fails, so $\sigma_2 > \alpha - \omega - 1$; (BH) “ $\sigma_1(\sigma_3) \rightarrow D_m$; $\sigma_2 + M$ in a row in H_m ” fails, so $\sigma_2 + M > 1 - \omega$; and (W). From (B1)+(W), by Corollary 41, in both branches of the dichotomy $\sigma_1 + \sigma_2 \geq 1 + b(\alpha)$, $\alpha \geq T_{1+\omega}$ and $\rho \geq \sigma_1 + \sigma_2 \geq \Phi(\omega) > T$. Moreover: if $\sigma_2 > 1 - \omega$, the program (B1)+(B2)+(B4)+(W) is exactly that of Theorem 22, whose lower bound gives $\sigma_1 + \sigma_2 > \frac{13}{7}$; if instead $\sigma_1 + M > 1$ and $\sigma_3 > \omega$, then (BH) gives $\rho \geq \sigma_1 + \sigma_2 + \sigma_3 + M > \sigma_1 + (1 - \omega) + \omega = 1 + \sigma_1 \geq 1 + \Phi(\omega)/2 \geq 1 + T/2 = 1.9196 > \frac{13}{7}$.

Branch 2 ($\sigma_3 > \sigma_1 - \omega$): nothing nests (also $\sigma_2, \sigma_3 > \sigma_2 - \omega$ and the star needs $\sigma_2 + \sigma_3 \leq \sigma_1 - \omega$, impossible). Split by the trio.

2A: the trio $\{\alpha, \sigma_1, \sigma_2\}$ has no corona. Then $F > 2\pi$ and, as above, $\rho \geq \sum S > \sigma_1 + \sigma_2 \geq \Phi(\omega)$. Furthermore $\sigma_1 \geq (\sigma_1 + \sigma_2)/2 \geq \Phi/2$, so $\sigma_3 > \sigma_1 - \omega \geq \Phi/2 - \omega$ and $\rho > \frac{3}{2}\Phi - \omega$ when $\omega \leq \Phi/2$; by the chord bound $\Phi \geq T + (13 - 7T)\omega$ this exceeds 2.64 on $(0, \frac{1}{7}]$, while for $\omega \geq \frac{1}{7}$, $\Phi \geq \frac{13}{7}$: in all cases $\rho \geq \frac{13}{7}$.

2B: the trio has a corona. Three mixed placements fail: “ $\sigma_3(+M)$ in a row in H_m + trio corona” gives (m1) $\sigma_3 + M > 1 - \omega$; “ $\sigma_3 \rightarrow u$ next to m + trio corona” gives (m2) $\alpha < 1 + \omega + \sigma_3$; and “corona of the quadruple” fails, so by Theorem 51 – the top trio passing its certificate, by Lemma 48 – the zigzag fails: $\theta(\alpha, \sigma_2) + \theta(\alpha, \sigma_3) + \theta(\sigma_1, \sigma_2) + \theta(\sigma_1, \sigma_3) > 2\pi$. By monotonicity, $2[\theta(\alpha, \sigma_2) + \theta(\sigma_1, \sigma_2)] > 2\pi$, so $A + B > \pi/2$ with $A = \theta(\alpha, \sigma_2)/2$, $B = \theta(\sigma_1, \sigma_2)/2$, both $< \pi/2$ ($A \geq \pi/2$ would force $\sigma_2 \geq 1$; $\sigma_1 + \sigma_2 < R$). Hence $\sin A > \cos B$, i.e. $\sin^2 A + \sin^2 B > 1$: $f(\sigma_2)[f(\alpha) + f(\sigma_1)] > 1$ with $f(\alpha) = \alpha$ and $f(\sigma_1) \leq 1/\alpha$, so $f(\sigma_2)(\alpha + 1/\alpha) > 1$, which is exactly $\sigma_2 > b(\alpha)$. With (W): $2b(\alpha) < \sigma_1 + \sigma_2 \leq \alpha - \omega$, and since $\alpha \mapsto \alpha - \omega - 2b(\alpha)$ is strictly increasing ($1 - 2b' > 0$ for $\alpha \geq 1$), $\alpha > \gamma_\omega$, the root of $2b(\alpha) = \alpha - \omega$ – equivalently of $\alpha^3 = \alpha^2 + \alpha + \omega(\alpha^2 + \alpha + 1)$, with $\gamma_0 = \varphi$. The tails: by (m1), $\rho \geq \sum S + M > 2b(\alpha) + (1 - \omega) \geq \gamma_\omega + 1 - 2\omega$, and by (m2), $\rho > 2b(\alpha) + \alpha - 1 - \omega \geq 2\gamma_\omega - 1 - 2\omega$. The first bound decreases in ω and the second increases ($1 < \gamma' < 2$), and they cross exactly at $\gamma = 2$, $\omega = \frac{2}{7}$ ($8 = 4 + 2 + 2$ in the cubic), where both equal $3 - \frac{4}{7} = \frac{17}{7}$: $\rho > \frac{17}{7}$ throughout branch 2B. $\square$

Proposition 65 (Reduction branch with occupants). *In the templates with $j \geq 1$ extra occupants and S of $k \geq 3$ pieces with $\sum_{i \geq 3} \sigma_i \leq \sigma_1 - \omega$, all pair walls and bounds (Proposition 53, Theorems 57 and 58, Proposition 62) hold unchanged: blocking implies $\rho > \Psi_j(\omega)$.*

Proof. Nest $\sigma_3, \dots, \sigma_k$ in a row inside the hole of σ_1 ; every unblocking placement of those walls moves σ_1 (with its cargo travelling inside) and σ_2 exactly as in the pair case, and the tails only grow. $\square$

D.6 Golden floor for pan exchanges

Here u is the pan itself: F places m at top level, the witness P keeps m in the hole of a ring y ($1 + X_y^{\text{rest}} \leq y - \omega$), the pan holds occupants $O = \{o_1 \geq \dots \geq o_j\}$ (> 1 , top level, shared by F and P , arbitrarily occupied) and, per P , the pair $S = \{\sigma_1 \geq \sigma_2\}$. Resources: D_m (free unit disk in y 's hole), H_m , hole tariffs (Lemma 55), nesting, and the pan repack. F 's pan gives the pair necessities $o_i + 1 \leq R$ and $o_i + o_\ell \leq R$ for all $i \neq \ell$. Write $S_0 := \sigma_1 + \sigma_2$; the row placement of the pair in D_m fails, so (D) $S_0 > 1$; and the joint placement of the pair in y 's hole beside its remaining content X_y^{rest} fails, which by Lemma 55 gives the wall

$$\sigma_1 + \sigma_2 + X_y^{\text{rest}} > y - \omega. \quad (\text{Ry})$$

Strict leaf: a leaf whose hole is not y 's; with $j \geq 2$ at least $j - 1$ exist (j if y is not a leaf).

Theorem 66 (Golden floor, pan exchange, pair profiles). *Blocking implies $\rho > \varphi$ in each of the following cases, which together exhaust the template: (i) $j = 1$, every width $\omega > 0$ (including the solid-pivot regime $\omega \geq 1$: the proof uses no width) and every occupancy; (ii) $j = 2$, likewise every $\omega > 0$ and occupancy; (iii) the evacuation branch $\sigma_1 + M > 1$, every $j \geq 2$ and every $\omega \in (0, 1)$; (iv) $j \geq 3$, every $\omega \in (0, 1)$. (Note $j \geq 1$ always: the carrier y has $y \geq 1 + \omega > 1$, so y is an occupant or lies inside one.) The infimum is exactly φ , realized by the golden family of Theorem 19 inside case (i).*

Proof. $j = 1$. The placement “ $\sigma_i \rightarrow \text{pan}$, the other to D_m ” fails, so $\{o_1, m, \sigma_i\}$ does not pack in R , hence (antitone containment, $R \geq o_1 + 1$) not in the disk $o_1 + 1$, where $\{o_1, 1\}$ is rigid: $\sigma_i > b(o_1)$ by Proposition 31. The tails give $\rho > \max(2b(o_1), (1 + 2b(o_1))/o_1)$; b is increasing, $g(A) = (1 + 2b)/A$ decreasing (the certificate $(3A^2 + 3A + 1)(A^2 + A + 1) - 2A(2A + 1)$ has positive coefficients), and the crossing $2b(A)A = 1 + 2b(A)$ factors as $(A^2 - A - 1)(2A + 1)$: unique positive root $A = \varphi$, value φ ($2b(\varphi) = \varphi$, $g(\varphi) = (1 + \varphi)/\varphi = \varphi$).

$j = 2$: *the mirror-pocket wall*. The same placement fails, so $\{o_1, o_2, m, \sigma_2\}$ does not pack in R , hence not in the disk $o_1 + o_2$, where $\{o_1, o_2\}$ is rigid and the quadruple packs iff $m \leq b_2(o_1, o_2)$ and $\sigma_2 \leq b_2$ (necessity by Proposition 31 per circle; sufficiency by placing them concentric in the two mirror pockets, disjoint since $y_0 = 2b_2$, as in Lemma 60). Thus $b_2(o_1, o_2) < 1$, i.e. $o_2 < \bar{A}(o_1)$, the decreasing root of $b_2(o_1, \cdot) = 1$ with $\bar{A}(2) = \sqrt{5} - 1$ and $\bar{A}(\frac{3}{2}) = \frac{3}{2}$. The tails of o_1 and o_2 , with (D) $\sigma_1 + \sigma_2 > 1$ (row in D_m), give $\rho > \max((o_2 + 2)/o_1, 2/o_2)$ on $o_2 \in (1, \bar{A}(o_1))$: for $o_1 \leq \frac{3}{2}$, $3/o_1 \geq 2$; for $o_1 \in (\frac{3}{2}, 2)$ the crossing $o_2^* = \sqrt{1 + 2o_1} - 1$ is interior ($b_2(o_1, o_2^*) < 1$ there, with exact equality at $o_1 = 2$) and its value $2/o_2^* > 2/(\sqrt{5} - 1) = \varphi$; for $o_1 \geq 2$, $2/o_2 > 2/\bar{A}(2) = \varphi$. The crossing at $o_1 = 2$ is the golden corner $b_2(2, \sqrt{5} - 1) = 1$ once more.

Evacuation branch, $j \geq 2$. A strict leaf L' with $q := \sigma_2 + X_{L'} > 1 - \omega$ satisfies (Lemma 55) $L' < q + \omega$, and its tail collects $\{m, \sigma_1, \sigma_2, M\} \cup \text{children}(L')$: with $\sigma_1 + M > 1$, exactly the branch-B program of Theorem 58, so $\rho > \Psi_B(\omega)$, and $\Psi_B(1) = \varphi$ *exactly* with Ψ_B decreasing: $\rho > \varphi$ for all $\omega < 1$.

$j \geq 3$, *remaining branch*. The strict-leaf ladder of Proposition 62 needs only k *distinct* strict leaves, each a node with the wall (Bo) at it (not one per occupant): it gives $\rho > \Psi_k(\omega)$, and here $k \geq j - 1$ always ($k \geq j$ if y is not a leaf). The golden crossings are exact: $\Psi_1(\frac{1}{2}) = \varphi$, $\Psi_2(\varphi/2) = \varphi$, $\Psi_3(1) = \sqrt{3} > \varphi$ – covering $j \geq 4$ outright and $j = 3$ up to $\varphi/2$ (always if y is not a leaf). The rest of $j = 3$ is a case tree: $o_2 < 3/\varphi$ (tail of o_2 exceeds $3/o_2$), or $o_1 < 3$ (tail of o_1 exceeds $(3/\varphi + 3)/o_1$, and $(3/\varphi + 3)/\varphi = 3$ exactly), or $o_1 \geq 3$ with the node/dust dichotomy on o_1 ’s hole content: a node makes y ’s subtree or o_1 ’s contribute a third strict leaf (Ψ_3), dust lands in m ’s tail. In detail, for $o_1 \geq 3$: if $y = o_1$, (Ry) gives $X_1 \geq o_1 - \omega - S_0 > 0$ (as $S_0 < 2 \leq o_1 - \omega$); a node inside makes y a non-leaf, whence $jj = 3$ and Ψ_3 applies, while pure dust lives in m ’s tail and $\rho \geq S_0 + X_1 \geq o_1 - \omega \geq 2$. If y lies in another occupant’s subtree and is not a leaf, its subtree supplies the third strict leaf ($jj = 3$); if y is a leaf and $\omega < \varphi/2$, $\Psi_2 > \varphi$; and if y is a leaf, $\omega \geq \varphi/2$ and $\sigma_2 > \omega$, then $\rho \geq \sigma_1 + \sigma_2 > 2\omega \geq \varphi$. What survives is y a leaf outside o_1 ’s subtree, closed by the pincer below.

The pincer (closes y a leaf outside o_1 ’s subtree). Here $o_2 \geq 3/\varphi$, and no width, solid-disk or o_1 hypothesis is needed: it suffices that $s := \sigma_2 + \omega < 2$, which holds since $\sigma_2 \leq 1$ (the profile consists of rings smaller than m) and $\omega < 1$. Suppose $\rho \leq \varphi$. Write D for the total radius of the rings < 1 *other than* σ_1, σ_2 , and for a node v let X_v be its hole content and $T_v \geq X_v$ its whole subtree. (a) The tail of m collects σ_1, σ_2 and all of D , whence $\rho > 1 + D$ and so $D < \varphi - 1$. (b) If some node of $\{o_1\} \cup o_1$ ’s subtree had two node children, their subtrees are disjoint and each contains a leaf, and the occupant of $\{o_2, o_3\}$ not containing y contributes a third; all three are strict (their holes are not y ’s, as y lies outside o_1 ’s subtree), so $\rho > \Psi_3(\omega) > \Psi_3(1) = \sqrt{3} > \varphi$ in branch A (branch B is case (iii)). (c) Hence $\{o_1\} \cup o_1$ ’s subtree is a chain. Let V be the set of its nodes whose tail contains both o_2 and o_3 (so $o_1 \in V$, by the first-copy convention even if $o_1 = o_2$), and let $v^* = \min V$. The pieces $o_2, o_3, m, \sigma_1, \sigma_2$ lie outside v^* ’s subtree — o_2, o_3 are top-level occupants, m sits in y ’s hole, and P keeps the profile in the pan — so the tail of v^* counts them and T_{v^*} without repetition. Now (Bo) applies at every node of o_1 ’s subtree (its

hole is not y 's, and $\sigma_1 \leq 1$), giving $X_{v^*} > v^* - s$, so

$$o_2 + 3 + (v^* - s) < \varphi v^* \implies o_2 < (\varphi - 1)v^* - 3 + s, \quad (\text{C1})$$

using $o_3 > 1$ and $\sigma_1 + \sigma_2 > 1$. With $o_2 \geq 3/\varphi$ and $3/\varphi + 3 = 3\varphi$ this forces $v^* > \varphi(3\varphi - s)$ (C2). Moreover v^* has a node child: otherwise $X_{v^*} \leq D < \varphi - 1$ and (Bo) would give $v^* < s + \varphi - 1$, against (C2) whenever $s \leq (2\varphi + 4)/\varphi^2 = 2.7639\dots$; and that child is $\leq o_2$ by minimality of v^* , so $X_{v^*} < o_2 + (\varphi - 1)$ and (Bo) gives $o_2 > v^* - s - (\varphi - 1)$ (C3). Combining (C1) and (C3) with $1/(2 - \varphi) = \varphi^2$,

$$v^*(2 - \varphi) < 2s + \varphi - 4 \implies v^* < \varphi^2(2s + \varphi - 4). \quad (\text{C4})$$

(C2) and (C4) are incompatible exactly when $6\varphi - 1 \geq s(2\varphi + 1)$, that is when

$$s \leq \frac{6\varphi - 1}{2\varphi + 1} = 11 - 4\sqrt{5} = 15 - 8\varphi = 2.05572\dots,$$

and $s < 2$ here. Contradiction; so $\rho > \varphi$. At the extreme $s \rightarrow 2$ the two bounds are $\varphi + 3 = 4.618\dots$ and $\varphi^3 = 2\varphi + 1 = 4.236\dots$, a margin of $2 - \varphi$. The constant is sharp for the argument: for $s > 11 - 4\sqrt{5}$ the pincer no longer closes, which is why it says nothing about a solid pivot. The adversarial reports (three rounds, including two repaired steps and a hostile audit of this pincer) and the verification scripts are in the repository (`code/batalla2.py`, 6/6; `code/microcelda.py`, 5/5). $\square$

For larger profiles the walls depend on how the rest of the profile is stashed; write $W := \sum_{i \geq 3} \sigma_i$, $\Sigma := \sigma_1 + \sigma_2 + W$, $M := \sum \text{children}(m)$ and X_{σ_1} for the prior content of σ_1 's hole.

Theorem 67 (Golden floor, pan exchange, larger profiles). *Let the witness keep $p \geq 3$ rings $\sigma_1 \geq \dots \geq \sigma_p$ (all < 1) in the pan. Blocking implies $\rho > \varphi$ in each of: (i) light, $\sigma_1 + W \leq 1$; (ii) nested, $W \leq \sigma_1 - \omega - X_{\sigma_1}$ (each with the ω domains of Theorem 66); (iii) heavy large, $\sigma_1 + W > 1$ and $\sigma_2 > \varphi - 1$, every j and every $\omega > 0$; (iv) heavy Ψ_B , $\sigma_1 + W > 1$, $\sigma_1 + M > 1$ and some strict leaf exists (in particular whenever $j \geq 2$), $\omega \in (0, 1)$; (v) heavy mirrors: for $p = 3$, $j = 1$, $\sigma_2 \leq \varphi - 1$, blocking is impossible – with (iii), $p = 3$ profiles with one occupant are closed at every $\omega > 0$; (vi) heavy swap, $p = 3$, $j = 2$, $\sigma_1 + M \leq 1$, $\sigma_2 \leq \varphi - 1$, except the sliver $\{\sigma_2, \sigma_3 > 1 - \omega\}$ (which forces $\omega > 2 - \varphi$). The remaining heavy region is stated in Open Problem 24(a).*

Proof. (i) Every placement of the proof of Theorem 66 sends one of σ_1, σ_2 to a target and the other to a row in D_m ; adding W to that row is legal ($\sigma_j + W \leq \sigma_1 + W \leq 1$), so every wall – (D_p): $\Sigma > 1$; (G) with either σ_i ; (Ry); (Bo); the evacuation dichotomy, now $\sigma_2 > 1 - \omega$ or $\sigma_1 + M + W > 1$ – ports verbatim, and every tail only grows by W . In branch B the bookkeeping is $A = \sigma_1 + \sigma_2 + W + M + X = (\sigma_1 + M + W) + (\sigma_2 + X) > 1 + q$ with $q := \sigma_2 + X$: the same Ψ_B program. The pincer is unchanged ($s = \sigma_2 + \omega < 2$). (ii) By Lemma 55, W rows inside σ_1 's hole beside X_{σ_1} , and $\{\sigma_1 \text{ carrying } W\}$ is a single piece of radius $\sigma_1 \leq 1$ wherever σ_1 goes – to D_m in (G_{σ_2}), (Bo) and the evacuation, to y 's hole in (Ry), to the pan in (G_{σ_1}): Theorem 66 ports again. (iii) The tail of m collects the whole profile: $\rho \geq \Sigma = \sigma_2 + (\sigma_1 + W) > (\varphi - 1) + 1 = \varphi$. (iv) The row $\{\sigma_2, \dots, \sigma_p\}$ into the strict leaf's hole ($\sigma_1 \leq 1$ to D_m) gives the wall $L' < (\sigma_2 + W) + \omega + X_{L'}$; with $\tilde{q} := \sigma_2 + W + X_{L'} > 1 - \omega$ (from $L' \geq 1$) and $A := (\sigma_1 + M) + \tilde{q} > 1 + \tilde{q}$, the two tails give $\rho \geq A$ and $\rho > (2 + \tilde{q})/(\tilde{q} + \omega)$: the branch-B program, $\rho > \Psi_B(\omega) > \Psi_B(1) = \varphi$. (v) $\sigma_2, \sigma_3 \leq \varphi - 1 < \frac{2}{3} = b(1) \leq b(o_1)$ fit concentrically in the two mirror pockets of $\{o_1, m\}$ in the

disk $o_1 + 1 \leq R$ (disjoint, Lemma 60), and $\sigma_1 \leq 1$ rows in D_m : the exchange is never blocked there. (vi) The placement “ $\sigma_2 \rightarrow H_m$, $\sigma_1 + M \rightarrow D_m$ (row ≤ 1 by hypothesis), $\sigma_3 \rightarrow \text{pan}$ ” and its $\sigma_2 \leftrightarrow \sigma_3$ mirror force either $\{o_1, o_2, m, \sigma_3\}$ not to repack – by containment to the disk $o_1 + o_2$ and Proposition 31, $b_2(o_1, o_2) < 1$, the mirror-pocket wall of case (ii) of Theorem 66, whose optimization ports with fatter tails ($1 + \Sigma > 2 + \sigma_2$) – or $\sigma_2, \sigma_3 > 1 - \omega$, and then $\rho \geq \Sigma > 1 + \sigma_2 > 2 - \omega$ closes $\omega < 2 - \varphi$. (Only $j = 2$: non-packability is not inherited by subsets, so the containment step fails for $j \geq 3$ – the same reason case (ii) of Theorem 66 is a pair statement.) The case tree, the two coverage cells missed by a first version and repaired, and the verification script are in the repository (`docs/drafts/perfilp.md`; `code/perfilp.py`, 5/5; adversarially verified). $\square$

The remaining heavy region shrinks further. The tool is a *mural-placement criterion* corrected for the pentagram phenomenon: circles $a_0, \dots, a_{k+1}$ tangent to the wall of a disk $\bar{R}$, with a_0, a_{k+1} diametral (so their own separation is exactly π on the far side), can be placed with *all* pairs disjoint if and only if, in the chosen order, the longest path

$$\max_{S \subseteq \{1, \dots, k\}} \sum_{\text{consecutive in } a_0, S, a_{k+1}} \theta(a_i, a_j)$$

is at most π (difference-constraint scheduling over Lemma 27; summing adjacent arcs alone is the criterion that the pentagram refutes).

Theorem 68 (The heavy region closes except one cell). *In the heavy non-nestable region of Theorem 67 ($\sigma_1 + W > 1$, $W + X_{\sigma_1} > \sigma_1 - \omega$, $\sigma_2 \leq \varphi - 1$, $\sigma_1 + M \leq 1$ where applicable), blocking implies $\rho > \varphi$ in each of: (i) $p = 3$, $j \geq 3$: the $j = 3$ case tree of Theorem 66 ports with the row wall $s' := \sigma_2 + \sigma_3 + \omega$ and the profile mass Σ in every tail, and the pincer closes whenever $s' \leq (\varphi - 1)\Sigma + (16 - 9\varphi)$ – which holds on the whole cell: $\sup(s' - (\varphi - 1)\Sigma) = 5\varphi - 7$, a margin of exactly $23 - 14\varphi > 0$, and at $\Sigma = 1$ the frontier degenerates to the $11 - 4\sqrt{5}$ of Theorem 66(iv); (ii) $p = 3$, $j = 2$: three branches – $\sigma_2 + \sigma_3 \leq 1$ resurrects the mirror wall of Theorem 66(ii); $\sigma_2 + \sigma_3 > 1$, $\omega \leq \frac{1}{2}$ runs a Ψ -program with interior crossing $\sqrt{1 + (1 - \omega)^2}$ and $\Psi(\frac{1}{2}) = \varphi$ exactly; and for $\omega > \frac{1}{2}$ the cell is empty of blockings: the tails of $\rho \leq \varphi$ force $o_2 \geq (1 + \Sigma)/\varphi$, $o_1 \geq (o_2 + 1 + \Sigma)/\varphi$, and there the mural placement of $\{o_2, m, \sigma_2, \sigma_3, o_1\}$ in the disk $o_1 + o_2$ always fits (longest path $\leq 2.648 < \pi$, computer-assisted maximization); (iii) $p \geq 4$, $j = 1$: mural coronas with margins ≥ 0.54 (computer-assisted); (iv) $p \geq 4$, $j = 2$: the supremum of the longest path over the closed domain is exactly π , attained only at the excluded boundary corner $\{\sigma_1 = 1, W = 0\}$, where $o_2 = 2/\varphi$, $o_1 = 2$, $\bar{R} = 2\varphi$ and $\sin^2 \frac{\theta(o_2, m)}{2} = \frac{1}{2} - \frac{\sqrt{5}}{10}$, $\sin^2 \frac{\theta(m, o_1)}{2} = \frac{1}{2} + \frac{\sqrt{5}}{10}$ sum to 1 exactly; the open domain stays strictly below π . What remains open of the pan exchange is the single cell*

$$\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$$

(the containment to the disk $o_1 + o_2$ does not re-place o_3 , and the pan-level cyclic corona is future work), together with extra small rings and the solid-pivot regime $\omega \geq 1$ for $j \geq 3$.

Proof sketch and verification. The chains and identities are exact (sympy); the two maximizations are grid-plus-refinement sweeps over compact four-parameter domains with the boundary behaviour controlled by the exact identities above (script `code/rstar.py`, 6/6, and a hostile adversarial round that refuted and repaired two steps: the naive adjacent-arc criterion – the pentagram – and a $j \geq 3$ corona that failed to re-place o_3 , whence the surviving cell; report in the repository). $\square$

E The corona-versus-tails campaign: closure of the residual cells

This appendix records the machinery and the results that close the cells left open by Theorems 66–68 and by the Status paragraph of Section 8. Epistemic labels are stated with each claim: *theorem* means a written proof (with exact identities delegated to `sympy/Lean`); *computational closure* means per-instance certificates over swept parameter ranges, with the certificate soundness proved but the extension to unswept (j, p, k) ranges supported numerically. Every script below is paired with an adversarial verification report in the repository (`docs/drafts/VEREDICTOS.md`).

Small rings adjoin to the profile.

Corollary 69 (Pan exchange, extra small rings). *In the pan-exchange template, every extra ring $e < r_m$ that P keeps in the pan adjoins the reinsertion profile: with $S^+ := S \cup \{\text{extras}\}$, Theorems 66, 67 and 68 hold verbatim for S^+ , and the residual cell of the case assignment over S^+ is again $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$.*

Proof. Reinsertion is existential: any placement that reinserts all of S^+ reinserts S , so blocking for S implies that every witness placement for S^+ fails, and each wall of the cited theorems – derived from exactly such failures, using only the radii of the profile members and the resources D_m, H_m , holes and the pan repack – holds for the enlarged profile a fortiori. Extras inside holes are already counted by the hole tariffs (X -masses). The case assignment of Theorems 67–68 is exhaustive over S^+ (200 000 sampled instances, zero unassigned; script `coronacolas`, block C). $\square$

The duality certificates. The remaining cells are closed by a necessity/sufficiency pair sharing one wall-tangency function: $\sin^2(\theta(a, b)/2) = f(a)f(b)$, $f(x) = x/(R - x)$, as in Lemma 27. *Necessity* (a true lower bound R_{lb} for the radius of any disk packing a given family): for any pair in any packing, the angular separation of centres satisfies $\cos \gamma \leq h(d_a, d_b) = (d_a^2 + d_b^2 - (a + b)^2)/(2d_a d_b)$, whose maximum over the admissible box of centre distances is attained at a corner; for a non-stackable pair ($R < \max + 2\min$) without confinement the binding corner is mural and $h(R - a, R - b) = 1 - 2f(a)f(b)$ exactly, so $\gamma \geq \theta(a, b)$; stackable pairs are handled by subset certificates and by confinement ($|c| \geq 2g + r - R$ when a giant g is present), and the cyclic sum of separations in any packing is 2π . *Sufficiency* (a constructive placement at $R \geq R_{\text{lb}}$): the cyclic corona by longest-path positions with all pairs validated and small rings assigned to Descartes pockets ($k_p = k_a + k_b - 1/R + 2\sqrt{k_a k_b - (k_a + k_b)/R}$) as row bins. Its soundness is a theorem, by maximality of the longest path: every skipped ring s between consecutive spine members a, b satisfies $\theta(a, s) + \theta(s, b) \leq \theta(a, b)$, and this margin vanishes *exactly* at the Descartes pocket radius (the three mutually tangent mural circles have adjacent wall angles summing to the pair’s), so each skipped ring sits murally inside its gap; positions satisfy $\alpha_j - \alpha_i \geq \theta(i, j)$ for all $i < j$ by construction, so only wrap-around pairs need (and receive) explicit validation. At the golden instance the margin is critically zero: $p(\varphi, 1; \varphi + 1) = \varphi/2$ with vanishing Descartes discriminant, and $f(\varphi)f(\varphi/2) + f(1)f(\varphi/2) = 1 = f(\varphi)f(1)$, so the counterexample of Theorem 19 is the exact critical point of the certificate system and is never unblocked by it (script `zigzag`, 5/5; the adversarial round refuted a stronger induction – consecutive triple margins do not imply global pair legality – and the construction therefore validates all pairs explicitly, as it always did).

Closure of the residual cells. With tails in cascade ($\rho \leq \varphi$ forces $o_k \geq (\sum_{i>k} o_i + 1 + \Sigma)/\varphi$, and $o \geq \varphi(1 + \Sigma)$ at near-ties), the constructive corona at $R = R_{lb}$ succeeds with *uniform tangent duality* (deficit 0.0; the boundary placement is tangent, hence legal) across: the heavy cell $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ of Theorem 68 ($j \leq 5, p \leq 6$ swept, 185k instances plus 4284 corners, Euclidean re-validation 1500/1500); the solid-pivot regime $\omega \geq 1, j \geq 3$ (no step uses the width); and, in the nested template of Section 8, the three golden-level gaps declared in the Status paragraph – the $j = 2$ tip $\omega \in [\varphi/2, 1)$, profiles of four or more pieces beyond the reduction branch, and the quantitative gap lemma (extra smalls adjoin as in Corollary 69; the reduction re-routes through the corona where the light-cell inheritance fails, and sweeps directly with the floor $\alpha \geq \max(1 + \omega, \text{true pair of } u + \omega)$ where transport is illegitimate). Scripts `coronacolas` (5/5) and `coronanidada` (5/5), each with an adversarial round that repaired the routing (six repairs in the pan campaign, four in the nested one; reports in the repository). These are computational closures: the per-instance certificate is theorem-backed as above, the swept ranges are stated, and the extension in (j, p, k) rests on the monotonicity evidence recorded by the scripts.

From computational closure to written proof: the insertion lemma. Two later results upgrade the central pan cells from computational closure to written mathematics.

Lemma 70 (Mural compaction). *If a family C packs in the disk R and every pair of C is non-stackable ($R < \max + 2\min$), then C packs murally in R : pushing every circle to the wall at the angle of its true centre is legal.* Proof. Angular separations are unchanged; for any pair in any packing, $\cos \gamma \leq h(d_a, d_b)$ with the box maximum at the mural corner $h(R - a, R - b) = 1 - 2f(a)f(b) = \cos \theta$ (the corners with a vanishing coordinate give $h \rightarrow -\infty$ for non-stackable pairs, whose centres live in an annulus of positive margin $d_a \geq a + 2b - R$), so $\gamma \geq \theta(a, b)$, which is exactly mural disjointness. The hypothesis is tight in two directions (a central ring family, and a family non-stackable only against its largest member, both packable but not murally). $\square$

Lemma 71 (Insertion by shadows). *Let P pack $F = \{x_i\}$ in R and let s satisfy the regime $R > 2s + x_i$ for every i . If $\sum_i 2 \arcsin((s + x_i)/(R - s)) < 2\pi$, then s inserts murally into P without moving any piece.* Proof. The forbidden arc of x_i at any depth d_i has half-width $\arccos h(R - s, d_i)$, and on $v \in (0, \infty)$ the map $v \mapsto h(R - s, v)$ has the single interior minimum $v^* = \sqrt{u^2 - w^2}$ ($u = R - s, w = s + x_i$) with $h(u, v^*) = \sqrt{u^2 - w^2}/u = \cos \arcsin(w/u)$ exactly: the shadow $\arcsin((s + x_i)/(R - s))$ bounds every arc uniformly in depth. If the shadows sum below 2π , a free angle exists. Without the regime the statement fails (a deep piece of a non-stackable pair forbids arcs up to π); the two regimes are mutually exclusive. $\square$

Theorem 72 (The heavy pan cells, written). *In the cell $\{j \geq 3, \sigma_2 \leq \varphi - 1, \sigma_1 + \sigma_2 > 1\}$ of the pan exchange – for every profile size p , every small-ring count and every width, including the solid pivot $\omega \geq 1$ (where the branch $\sigma_2 > \varphi - 1$ is also closed: $\sigma_2 > \varphi/2$ is empty by mass, $2\sigma_2 > \varphi$, and $\sigma_2 \in (\varphi - 1, \varphi/2]$ by the parametric budget with $\Sigma \geq 2\sigma_2$) – $\rho \leq \varphi$ implies the exchange does not block.* Proof sketch. The witness placement: the pan exactly as F certified it when placing m (occupants with their P -subtrees rigid, m with M inside, D_m vacated in the carrier's hole); $\sigma_1 \rightarrow D_m$ (single row); σ_2 inserted by Lemma 71; the remaining small mass $W' \leq \varphi - \sigma_1 - \sigma_2 < 1/\varphi$ (tail of m plus the wall (D)) as one row-disk (Lemma 3) inserted again by Lemma 71. The shadow regime holds across the domain ($o_2 - 2s \geq 4 - 2\varphi > 0$ exactly), and the budget of shadows at $R = o_1 + o_2$ (shadows decrease in R) is bounded by its cascade corner $j = 3, (2\varphi, 2, 2/\varphi, 1)$: 4.7225 and 5.2644 against 2π , margins 1.56 and 1.02;

the o_2 -direction is exactly monotone, the o_1 -direction is an exact bathtub with limit $\pi < 2\pi$, the geometric tail decay $T_k \geq \varphi T_{k+1}$ removes j , and the remaining directions carry a certified directed maximization (script *insercion*, 7/7, with its adversarial report – which refuted and repaired the naive statement of Lemma 71 with mural angles in place of shadows). $\square$

This supersedes the computational closure of $\{p \geq 4, \sigma_1 + M \leq 1, j \geq 3\}$ and of the solid-pivot regime: both cells now carry written proofs, with a single certified maximization in the budget (the same epistemic standard as Theorem 68).

Theorem 73 (Nested exchange, two or more occupants, written). *In the nested template (u the hole of a top-level α , v the pan) with $j \geq 2$ occupants, $\rho \leq \varphi$ implies the exchange does not block – for every width, every σ_2 , every profile size and any extra small rings. Proof sketch. The witness: the pan as P placed it, with only m leaving (its unit disk D_m becomes vacant at top level) into α 's hole, re-packed by F 's certificate (m is the largest disagreement and F processes decreasingly: the hole's occupants at that moment were exactly the shared rings above m); greedy row filling of D_m with the loose mass in decreasing order up to the first non-fitting piece s' ; s' inserted murally by Lemma 71; the rest as one row-disk $w^* < 1/\varphi$ (exact: the placed row plus s' exceeds 1 and the tail of m is at most φ), inserted again. Two exact walls drive the budget: the insertand cap $s' \leq \min(\Sigma/2, \varphi/2)$ (unconditional in ω : the two largest loose pieces sum within both Σ and the tail of m), and the automatic regime – with $j+1 \geq 3$ top-level pieces the cascade forces the second largest to satisfy $t_2 \geq (1 + \Sigma)(1 + \varphi)/\varphi^2 = 1 + \Sigma \geq 2$ ($\varphi^2 = 1 + \varphi$), so $R - x \geq t_2 \geq 2 > \varphi \geq 2s'$ and $2 > 2/\varphi = 2w^*$ for every piece, with margin $2 - \varphi$. The shadow budgets stay below 2π across the domain (certified maximization with deterministic corners, margin ≥ 1.07 ; script *insercionanidada*, 6/6, whose large-slack block sweeps pieces up to 10^4 above the floors – one and two inflated at a time – and widths up to $\omega = 1.35$, with the worst budget attained at the floor and the $t \rightarrow \infty$ limit closing by formula at $\pi < 2\pi$: the certified domain admits arbitrary values above the floors). For $j \leq 1$ the second insertion dies exactly on the golden razor $o_1 \rightarrow 2/\varphi$ against $2w^* = 2/\varphi$ (the shadow ratio is identically 1): that strip remains with its computational closure. $\square$*

Theorem 73 supersedes the computational closures of the golden-level nested gaps with two or more occupants – in particular the $j = 2$ tip $\omega \in [\varphi/2, 1)$ entirely, and profiles of four or more pieces beyond the reduction branch (profile size is mass here).

Theorem 74 (Gap lemma, nested $j \leq 1$, written). *In the nested template with $j \leq 1$ occupants, $\rho \leq \varphi$ implies the exchange does not block, for every width (solid pivot included), every profile and any extra small rings. Proof sketch. The same witness as Theorem 73 except the mural step: after the greedy filling of D_m , the whole pan is re-packed as a corona of at most five circles $\{\alpha, (o_1), \text{the unit disk carrying the } D_m \text{ row, } s', w^*\}$ – a bounded family, so the mural criterion is finite and exact per instance (exhaustive over the ≤ 12 cyclic orders, all pairs validated). The radius carries the written trio bound: any disk packing $\{a, b, c\}$ satisfies $R \geq \max(\text{pair sums}, \min(R_3, M))$ unconditionally, where R_3 solves $\theta(a, b) + \theta(b, c) + \theta(c, a) = 2\pi$ (the angular necessity, a theorem via the corner lemma and the partition of the circle) and M is the stackability threshold $\min(\max + 2\min)$ – a dichotomy on the real pan. On the domain the cascade floor $o_1 \geq (1 + \Sigma)/\varphi > 2/\varphi$ keeps $R_3 \leq M$ (explicitly checked), with equality exactly at the golden point $(\alpha, o_1) = (2, 2/\varphi)$, where pair sums, R_3 and M all collapse to $1 + \sqrt{5}$ and the trio sum equals 2π exactly. The mass ligatures $s' \leq \min(\Sigma/2, \varphi/2)$ and $w^* < \min(1/\varphi, \Sigma - 1, \Sigma - 2s')$ are exact; the $\alpha \rightarrow \infty$ limit closes by formula ($\pi + 2 \arcsin \sqrt{1/\varphi} < 2\pi$, margin 1.33); the cov-*

erage of the compact domain box is a certified maximization (script `gaplemma`, 5/5, adversarial report with two repaired findings). $\square$

With Theorems 73 and 74, the *entire nested template* of case (b) carries written proofs for every j , every width, every profile and extras – at the same epistemic standard as the pan side (one certified maximization per cell). For $j = 0$ the domain sweep has since been upgraded to an *exact algebraic pocket certificate*: at $R = \alpha + 1$ the Descartes discriminant vanishes identically and the wall pocket is $p(u) = u(u + 1)/(u^2 + u + 1)$, increasing; with the coupled mass caps ($s' \leq \Sigma/2$, $w^* \leq \Sigma - 1$) the sub-pocket condition factors as $(\varphi - u)r(u) \geq 0$ with $r(u) = \varphi u^2 + 2(\varphi - 1)u + (\varphi - 1)$ positive-coefficient, so equality holds exactly at $\Sigma = \varphi$ with α on its cascade floor – where the cap $s' = \varphi/2$ is *tangent* to the golden pocket $b_2(\varphi, 1) = \varphi/2$ of Theorem 66’s counterexample; the second cap closes with exact minimum $(3 - \sqrt{5})/4$ (the quartet cell is a theorem, its critical corner the golden pocket itself). The remaining bounded-corona domains – the $j = 1$ quintet and the $k \leq 2$ hole cells – are certified by subdivision as well, on two new exact tools: an *arc-LP characterization* of small mural coronas (feasibility of the per-arc requirement system is equivalent to the corona, with the pairs-fit precondition; its disjoint-family dual serves as pruning and an exact base-enumeration primal as the criterion – closed inequalities, so it certifies *at tangency*), and a closed-form feasibility test for the critical 4-cycle ($-\sigma \geq \max(B_1, B_2)$: the two diagonal NS-2 deficits against the slack). The binding frontier turned out to be an entire *tangent curve*: the trio sum $\theta(a, o) + \theta(o, 1) + \theta(1, a)$ at $R = a + o$ equals 2π exactly along a curve running from the golden point $(2, 2/\varphi)$ through $(3/2, 3/2)$ – where $\theta(3/2, 1; 3) = \pi/2$ exactly – to its mirror image, the boundary of the trio-necessity band; on the curve the corona is handled *by construction* (the trio is exactly tangent and the small pieces insert sub-pocket at zero angular cost), with the golden endpoints – where pair sums, R_3 and the stackability threshold all collapse – excluded into certified neighbourhoods. Behind it all sits one more exact golden identity, kernel-checked in Lean: $\theta(\varphi, 1/\varphi) + \theta(1/\varphi, 1) + \theta(1, \varphi) = \pi$ at $R = 2\varphi$, which makes the dangerous 4-cycle of the $j = 1$ cell exactly tangent and ties its 5-cycle to the shared-root corner constant $\pi + 4\arcsin(1/\sqrt{3})$ (scripts `arcolp` and `bolsillos`, 5/5 each, three adversarial rounds – one of which refuted and repaired the arc-LP statement itself).

The exchange assembly. The exchange step of Theorem 9 is assembled into a case partition (a theorem: containers are the unique pan or holes of rings larger than m , and $u \neq v$): (a) u is the pan – the template of Theorems 66–68 plus the closures above; (b) u is the hole of a top-level α and v is the pan – the nested templates plus the closures above; (c) the *container port*. Assembly facts (theorems): $|S| \leq 1$ never blocks (a single $\sigma < r_m$ rows into D_m ; with $S = \emptyset$, m moves under F ’s own certificate, containers of larger rings being shared); feasibility is per-container and positions are existential (Section 2) – and this is now a proved *realization-and-repack lemma*, not a gloss: any forest whose containers each carry a placement of their children as balls with disjoint interiors is realized globally by composing translations along root-to-leaf paths (inductive invariant: a ring’s subtree stays in its ball, its strict descendants in its hole ball; the annulus material is disjoint from the hole interior; translations preserve distances, and any isometry of a container placement is just another placement), so replacing one container’s placement – subtrees rigid – yields another realization of the same forest, and a witness with a *new* assignment composes the per-container certificates (vacated ball, row, corona, pocket, shadow insertion) through the same lemma (script `repack`, 5/5, adversarial round: exact tangencies, solid rings and subtree isometries all verified, and the checker proved equivalent to

material disjointness). Repacking any container is therefore a legal resource – the pan repack of Theorem 66 is the precedent; and every geometric wall descends by antitone containment to the intrinsic disk of its members, so walls port to any container with the corresponding capacity. Porting *programs* additionally needs their cohabitation hypotheses, which yields the closure split: if α is a direct member of v , case (b) is inherited verbatim with $R \mapsto Y - \omega$ (theorem); an extension lemma transports the (N) -heritage walls with the load travelling inside σ_1 (monolith for placements not using σ_1 's hole, exact absorption of the fattened B'_3 in the golden-line chain). The genuinely new configurations – α outside v – were closed by a dedicated campaign (script *puertocii*, 7/7, two hostile rounds): exact pincers reduce the light profile to a window that is empty for $\omega_{\text{ef}} \leq 3/(2\varphi)$ (the corner degenerates at $\sigma_2 = \frac{1}{2}$ exactly), and inside the window the pan repack places σ_2 in the mirror pocket of the top-level pair, by $\alpha > 2$, $T > \sqrt{5} - 1$ and monotonicity of b_2 against the corner $b_2(2, \sqrt{5} - 1) = 1$; the heavy profile is closed by an exact partition pincer ($b_2(4/\varphi, 2/\varphi) = 12/(7\varphi) > 1$); towers, the $Y \geq \alpha$ branch and the shared-root configurations (where the top-level pair degenerates) are closed by a dedicated mural-trio analysis: the exchange lives in the hole of α with capacity $c = \alpha - \omega \geq \Sigma_S + Y$ (hole tariff), the trio sum decreases in c (exact), and its supremum over the legal window is the certified corner $\pi + 4 \arcsin(1/\sqrt{3}) \approx 5.60 < 2\pi$ (margin 0.68; lightness with $\sigma_1 \geq \sigma_2$ forces $\sigma_2 \leq \frac{1}{2}$, while the analytic danger threshold is $\sigma_2 = \frac{2}{3}$), with the heavy branch by an exact partition and the deeper/mirrored configurations by direct sweeps with derived tariffs.

The port itself is now *written*. When v is the pan and α sits nested in a tower whose root t is top-level, the case is a *corollary* of Theorems 73–74: those theorems use of the configuration exactly four things – a top-level pan family $\{t, o_1, \dots, o_{j'}, m\}$ in P , the vacated unit disk D_m at top level (v the pan makes m top-level in P), the legality of m 's destination by F 's certificate with rigid subtrees (agnostic to *which* hole receives m ; the tower is shared between F and P), and the mass ligatures – and all four hold with the tower root t in the role of “ α ”, the per-rank floors only rising (the tail of t contains the whole tower; $t \geq \alpha + \omega \geq 1 + 2\omega$ by two circles level by level; the hole-mass floor is automatic by an exact two-branch argument), while both certified domains admit arbitrary values above their floors (large-slack sweeps to 10^4 with the worst budget at the floor, and the $t \rightarrow \infty$, $\alpha \rightarrow \infty$ limits closing by formula – a hostile round of this very reduction demanded and obtained the large-slack block). When v is the hole of a top-level Y , the case assembles into a theorem from the exact pincers above together with the two *hole-corona* theorems, which run the nested template *inside* the hole: the cascade chain among a hole's occupants yields the same automatic regime ($x_2 \geq 1 + \Sigma$ for $k \geq 3$ occupants, $\varphi^2 = 1 + \varphi$; tails are global), and for $k \leq 2$ the family is bounded and the armored trio bound applies, on the hole's own low floors (occupants ≥ 1 with no wall relation – a domain the pan boxes do not cover, swept separately with 10^4 slack and the $x_1 \rightarrow \infty$ limit by formula). The breathing branch $Y \geq \alpha$ needs a *strong breathing lemma* since the hole mass includes dust: the full cross-tail pincer plus the wall $\Sigma_S > 1$ force $X_Y + \omega > \varphi(3 - \varphi) = 2\varphi - 1 = \sqrt{5}$, dust is at most $\varphi - \Sigma_S < \varphi - 1$ by the tail of m , and $\sqrt{5} - (\varphi - 1) = \varphi$ exactly: the above- m mass satisfies $X_{>m} + \omega > \varphi$, so $k \geq 1$ – the golden threshold reappears after discounting dust. The α -corona branch is lighter: lightness gives $\Sigma_S - \sigma_2 < 1$ exactly (the rest of S rows into D_m and only σ_2 needs mural insertion), and the capacity window $[\Sigma_S + X_\alpha, 1 + \sigma_2 + X_\alpha]$ is exact with X_α cancelling (script *coronaagujero*, 5/5, adversarial round with zero counterexamples under 55k independent adversarial instances).

A *geometric-tail lemma* then makes every one of these shadow budgets *uniform in the occupant count*. For any cascade family $t_1 \geq t_2 \geq \dots \geq t_n$ (global tails, $\rho \leq \varphi$) with $R \geq t_1 + t_2$,

regime $2s < t_2$ and $t_2 \geq 1 + \Sigma$, the budget is dominated term by term by an explicit finite series: the suffix masses S_i decay geometrically ($S_i \geq \varphi S_{i+1}$, again $1 + 1/\varphi = \varphi$), each tail piece obeys three exact caps ($t_{3+r} \leq \min(t_2, S_3/\varphi^r - (1 + \Sigma), (S_3 - 1 - \Sigma)/(r + 1))$), a piece at rank $3 + r$ drags a suffix of mass at least $1 + \Sigma + p_{\min}$ so the series has finitely many terms, and – the load-bearing link – the cascade of t_1 itself forces $t_1 \geq (t_2 + S_3)/\varphi$: a heavy tail forces a large leader, so the tight pair and the full tail are *jointly infeasible* (allowing both would push the majorant to $6.37 > 2\pi$; with the link its supremum over the whole box is $5.2115 < 2\pi$, attained with *zero* gap at the real family $\{2\varphi, 2, 2/\varphi\}$ plus the unit disk – the very corner every sweep had found). The hypothesis $t_2 \geq 1 + \Sigma$ is forced by three-or-more occupants via $\varphi^2 = 1 + \varphi$ and is exactly the lemma’s frontier: at $n = 2$ the golden razor ($t_2 \rightarrow 2/\varphi$, shadow ratio identically 1) breaks the budget, which is the known boundary where the bounded-family closures take over. Consequently the sweep caps ($j \leq 6, k \leq 14, k \leq 12$) drop out of the written theorems’ statements: shadow coverage holds for *every* occupant count (script `colageometrica`, 5/5, adversarial round: domination unbroken under $\sim 45k$ families including the theorems’ own cascade generators). Finally, a *pocket- φ lemma* removes the profile- and dust-count directions of the scaling question exactly: at $R = o_1 + o_2$ the Descartes discriminant of the tangent pair vanishes identically, the pocket radius $1/(1/o_1 + 1/o_2 - 1/(o_1 + o_2))$ is monotone in o_1, o_2, R and equals φ *exactly* at the cascade minimum $(2\varphi, 2, 2\varphi + 2)$, while the total small mass is $\leq \varphi$ by the tail of m : the entire profile and dust row into a single pocket (p and k free; script `escala`, 5/5, with the nested template verbatim and a geometric non-invasion check). The honest residue of the whole program is therefore: the occupant-count direction j of the *computational* scaling closures only, swept to $j \leq 9$ (pan) and $j \leq 8$ (nested) with uniform tangent duality and the exact geometric growth $T_{k-1} \geq \varphi T_k$ of the cascade ($1 + 1/\varphi = \varphi$) – the written theorems’ occupant counts are closed by the geometric-tail lemma; the computational (as opposed to written-proof) status of the remaining sweeps – the shared-root *multipiece* sweeps, now substantially certified by a pessimistic-corner engine over the arc-LP’s sufficiency direction (valid for every k): a term-by-term majorant matrix (each wall angle bounded at its own corner, participants at their ceilings *including* their contribution to the capacity, the rest at their floors – valid since $\partial\theta/\partial a$ carries the sign of $R - a - b = \Sigma_S + \text{rest} > 0$), a maximum-slack LP whose returned gap vector is re-verified in plain floating point, and an *antipodal* criterion for the $\Sigma_S \rightarrow 1$ edge (Y and m at distance exactly π , strict by algebra, with the remaining pieces as an interval – hence totally unimodular, hence exact – path system on one half-circle, where the circular LP’s margin degenerates even at real points). This certifies the explicit extra occupants (1–3 corona pieces up to size Y , coronas of 4–6 pieces, entire $\sigma_2 < 1$) over the sweep box, and the mirrored branch beyond its $X = 0$ cut in the $X_Y = 0$ direction (shifted windows with $X_\alpha, X_z, X_m > 0$, dual per-box bounds coupling the diametral floor $c' \geq 1 + z$, certified at margins both 0.3 and 0.35). The heavy-profile partition – multipiece A with *unbounded* cardinality – is closed by a *reduction lemma with a golden threshold*, exact in $\mathbb{Q}(\sqrt{5})$: taking B^* the best subset of S of mass ≤ 1 (rowed into D_m) and $A = S \setminus B^*$ with $\beta = \Sigma B^*$, maximality gives $a + \beta > 1$ for every $a \in A$, $\beta \geq \sigma_1 \geq \max A$ (hence $a \leq \varphi/2$), and *at most four* pieces of A exceed $t_0 = (\varphi - 1)/4 = 1/(4\varphi)$ – a dichotomy at $\beta^* = (9 - \sqrt{5})/8$ whose critical identity $5t_0 = \varphi - \beta^*$ is exact, with the twin identity $4t_0 = \varphi - 1$ making four large pieces and dust *incompatible* – while the remaining dust (pieces $\leq t_0$, present only for $\beta > \beta^*$, mass $\mu < 5t_0$) is paid by *mass, not count*: convexity of $\arcsin$ gives $\theta(a, b) \leq \arcsin f(a) + \arcsin f(b)$, so a monotone dust chain costs at most $\pi\mu/(R - t_0)$ regardless of how many pieces carry the mass. The heavy mural is then $\{Y \text{ or } z, m\}$ plus at most four large pieces plus one aggregated dust block, and the two-sided antipodal engine certifies

both heavy branches over their sweep boxes (the direct one entirely; the mirrored one on its $X = 0$ cut). What remains sampled: the $X > 0$ mass of the heavy mirrored sweep, the $X_Y > 0$ mirrored corona – whose exact danger geometry is now settled by a two-round exploration: at the coexistence floor $c' = 1 + z$ the pair (z, m) is exactly diametral, and an extra piece x of X_Y fits beside m precisely when $x \leq p(z, 1; 1 + z) = z(z + 1)/(z^2 + z + 1)$, the *Descartes pocket of the diametral pair* (degenerate form, exact at $R = a + b$; the would-be sliver $x \in (p, 1)$ has width $1/(z^2 + z + 1) - 1/(2\varphi^2)$ at $z = \varphi$, where also $p(\varphi, 1; \varphi + 1) = \varphi/2$ exactly, the golden pocket of Theorem 66’s counterexample); a first adversarial round confirmed this algebra by hand, and a second one then proved the sliver *vacuous*: the global tail of the pivot counts every smaller piece, so $\Sigma_S + X_m + \Sigma X_Y \leq \varphi$, which is incompatible with the light-profile wall $\Sigma_S > 1$ and $x > p \geq 0.91$ (100% of the previously sampled “variety” violates it, minimum 1.96 against φ), and independently the floor itself is *rigid* – the diametral pair is forced tangent and the largest hole left in the container is exactly the pocket, so no legal configuration ever placed such an x murally at the floor. The positive consequence: every *legal* X_Y piece measures at most $\varphi - \Sigma_S < \varphi - 1 < p(z)$ throughout the domain – *always sub-pocket* – so the mirrored $X_Y > 0$ cell closes by vacuity plus sub-pocket insertion, with only the routine k -piece corona certification left declared (scripts `espxy` and `espvals`, 5/5 each, two adversarial rounds; the second refuted a proposed vacated-ball witness – the two modes of the repack lemma, positional and fresh-certificate, must not be mixed in one container – and found the missing global-tail wall itself) – and the sweep caps themselves ($X_Y \leq 3$, $\omega \leq 1.6$ – sampling ceilings, not derived walls) (scripts `r2bmulti` and `areduccion`, 5/5 each, adversarial rounds: no soundness gap – 1 460 directed tests for the engine, hand-redone algebra and adversarial profiles for the lemma; the rounds corrected domain claims to the sweep boxes, replaced the solver-trusting margin by a float re-verification of the returned gap vector, fixed a degenerate half-circle that silently re-introduced the excluded antipodal pair, and contributed the $4t_0$ incompatibility) – plus the near-equal-tops duality gap ($R_{\text{fit}}/R_{\text{lb}} \leq 1.0116$ – now re-delimited by a *conditional duality lemma*: for families whose pairs are all non-stackable at the radius in play, the arc-LP radius of Appendix E is the exact minimal mural-corona radius on both sides, by compactification for necessity and the closed arc-LP for sufficiency, reproducing the classical equal-circle optima $1 + 2/\sqrt{3}$ and $1 + \sqrt{2}$ at tangency; on the synthetic non-stackable domain the true load-versus-tops interval has corner supremum 1.082, certified over a deterministic corner mesh – now an *abstract* arc-LP statement: a subsequent global-tail audit showed that every sampled gap instance of the realistic generator is ρ -illegal (0/30 under the full tail legality), by a one-line *forbidden-triple* fact: three pieces whose second and third are $\geq (\varphi/2)$ -largest already violate $\rho \leq \varphi$ (their tail sum is $\geq 2 \cdot (\varphi/2) = \varphi$ times the largest), while the duality gap needs at least three near-equal tops (ratio $0.9 > \varphi/2$) – and the complementary case, a third piece too small for the forbidden triple, is closed by a *forced sub-pocket* lemma: full tail legality gives $t_3 < (\varphi - r_2)t_1 - 2$, the dominant pair’s pocket at its pairs floor is $q(r_2)t_1$ with $q(r) = r(1 + r)/(1 + r + r^2)$ (the same degenerate-pocket function, scaled by the smaller piece), and $(\varphi - r_2)t_1 - 2 \leq q(r_2)t_1$ holds throughout $t_1 \leq 2/(\varphi - r_2 - q(r_2))$ – at the cell’s edge $r_2 = 0.9$ the ceiling is 23.0, growing to infinity at the real root $r^* = 0.9637$ of the golden cubic $r^3 + (2 - \varphi)r^2 + (2 - \varphi)r = \varphi$, while the generator’s true domain is $t_1 \leq 6.64$ – so the third piece always inserts freely, the trio sits at its pairs floor, and (checked against the confinement strengthening of the lower bound, 100% of 5 300 cell instances) R certificates match: the cell has *no* legal instance of the phenomenon (0 gaps across $\sim 11\,000$ adversarial legal samples, including sub- m third pieces and four-top profiles). The old 1.0116 residue is therefore *withdrawn*: $\rho \leq \varphi$ itself forbids the configurations that carried it, and only the abstract arc-LP

statements (the conditional duality lemma and the synthetic 1.082 corner) remain, with the empirical cell boundary 0.9 and the converse “gap $\Rightarrow$ cell” declared open (evidence in favour: outside the cell the confinement raises the lower bound exactly where a third piece exceeds the pocket, and the constructive corona fits at the raised bound); scripts `f3cierre`, `auditcolas` and `f3vacio`, 5/5 each, adversarial rounds – the first refuted the original “closure” claim and its sampled 1.030 bound, the second refuted a partial-tail audit and supplied the forbidden-triple mechanism, the third corrected the declared generator domain (6.64, not 5.1) and verified the confinement attack) and the extras coronas on their swept ranges; the shared-root *trio* itself is now certified by subdivision over its entire legal domain in the direct orientation – light interior, both frontiers and every depth at once, via the free-leader reduction – and over the $X = 0$ cut of the mirrored one, where the same branch-and-bound also proves the mass wall $\Sigma_S \leq \varphi$ *necessary*: without it the corner $\sigma_2 \rightarrow 1$ is legal and its corona genuinely fails, the (z, m) pair going exactly diametral (script `r2bcert`, 5/5, adversarial round; the pan-repack legality behind the mirror-pocket pincers is likewise no longer on this list: it is the realization-and-repack lemma above); and the optimization asterisk of the *bounded-family* domain sweeps. For the shadow budgets themselves that asterisk is now closed: since every term of a shadow budget is monotone in each piece radius and in s , and antitone in R , corner evaluation gives a valid upper bound on any box, and an exhaustive branch-and-bound over the majorant’s hypothesis box (with the exact prunings $t_2 \geq 1 + \Sigma$, $u \leq \varphi t_2$ and the t_1 -link, the π -cap for $t_1 \rightarrow \infty$, and a normalized form with an analytic geometric remainder for $t_2 > 10^3$) *certifies* $\sup G \leq 5.25$ on the principal box and $< 2\pi - 0.05$ globally – margins 0.98 and 0.19 radians, twelve orders of magnitude above IEEE rounding (script `optimizacion`, 5/5, adversarial round with a repaired normalized-link bound and a certified-honesty control: tightening the target below the true supremum 5.2115 makes the branch-and-bound stall at exactly that value).

F Verification details for Theorem 14

All feasibilities in the twin instances are two-circle-exact except three facts, proved here; every numerical claim is also reproduced by the scripts in the repository (`code/gemelas.py`).

(V1) $\{10, 4.99, 4.50\}$ does not pack in a disk of radius 15. As in Lemma 12: $|c_A| \leq 5$, $|c_X| \leq 10.01$, $|c_Y| \leq 10.5$, $|c_A - c_X| \geq 14.99$, $|c_A - c_Y| \geq 14.5$, $|c_X - c_Y| \geq 9.49$; the first two give $a := |c_A| \geq 4.98$. With $u = -c_A/|c_A|$, $c_X \cdot u \geq (124.5 - a^2)/(2a) \geq 9.95$ and $c_Y \cdot u \geq (100 - a^2)/(2a) \geq 7.5$ (both decreasing in a , evaluated at $a = 5$), with transverse norms at most $\sqrt{10.01^2 - 9.95^2} \leq 1.095$ and $\sqrt{10.5^2 - 7.5^2} \leq 7.349$. Hence $|c_X - c_Y|^2 \leq 10.01^2 + 10.5^2 - 2(9.95 \cdot 7.5 - 1.095 \cdot 7.349) \leq 77.30$, so $|c_X - c_Y| \leq 8.80 < 9.49$.

(V2) With the pan pair $\{10, 5\}$ placed, no third circle of radius $z \in \{4.74, 4.76\}$ fits. From $|c_A| \leq 5$, $|c_B| \leq 10$ and $|c_A - c_B| \geq 15$ the triangle inequality forces equalities: the pair is diametrically rigid, $c_B = 10u$ for a unit vector u and $c_A = -5u$. For a third circle of radius z : $c_z \cdot u \geq ((10+z)^2 - (15-z)^2 - 25)/10$, which equals 8.8 for $z = 4.76$ and 8.7 for $z = 4.74$; then $|c_z - c_B|^2 = |c_z|^2 - 20 c_z \cdot u + 100 \leq (15 - z)^2 - 20 c_z \cdot u + 100$, giving $|c_z - c_B| \leq 5.38 < 9.76$ and $\leq 5.60 < 9.74$ respectively.

(V3) $\{10, 4.76, 4.74\}$ packs in a disk of radius 15. Explicit boundary configuration: $c_A = (-5, 0)$,

$$\begin{aligned} c_X &= 10.24(\cos 30.77^\circ, \sin 30.77^\circ) \approx (8.799, 5.238), \\ c_Y &= 10.26(\cos 32.02^\circ, -\sin 32.02^\circ) \approx (8.700, -5.440). \end{aligned}$$

Then $|c_A| = 5$, $|c_X| = 10.24 = 15 - 4.76$, $|c_Y| = 10.26 = 15 - 4.74$ (all inside); $|c_A - c_X| = 14.76$ and $|c_A - c_Y| = 14.74$ (exact tangencies, chosen via the boundary angles); and $|c_X - c_Y| \approx 10.68 \geq 9.50$ with ample margin.

(V4) Hole facts. The hole of the 10 has radius 9.495; $4.99 + 4.50 = 9.49 \leq 9.495$ (the pair fits, two-circle exact) while $4.76 + 4.74 = 9.50 > 9.495$ and $5 + 4.50 = 9.50 > 9.495$ (blocked, two-circle exact).

References

- [1] J. P. Pedroso, S. Cunha, J. N. Tavares. Recursive circle packing problems. *International Transactions in Operational Research* 23(1–2):355–368, 2016.
- [2] A. Gleixner, S. J. Maher, B. Müller, J. P. Pedroso. Price-and-verify: a new algorithm for recursive circle packing using Dantzig–Wolfe decomposition. *Annals of Operations Research* 284(2):527–555, 2020.
- [3] E. G. Coffman Jr., M. R. Garey, D. S. Johnson. Bin packing with divisible item sizes. *Journal of Complexity* 3(4):406–428, 1987.
- [4] E. D. Demaine, S. P. Fekete, R. J. Lang. Circle packing for origami design is hard. In *Origami⁵: Proc. 5th International Conference on Origami in Science, Mathematics and Education*, A K Peters, 2011, pp. 609–626.
- [5] S. P. Fekete, P. Keldenich, C. Scheffer. Packing disks into disks with optimal worst-case density. In *Proc. 35th International Symposium on Computational Geometry (SoCG)*, LIPIcs 129, 35:1–35:19, 2019. Journal version: *Discrete & Computational Geometry* 69:51–90, 2023.
- [6] M. Abrahamsen, T. Miltzow, N. Seiferth. Framework for $\exists\mathbb{R}$ -completeness of two-dimensional packing problems. In *Proc. 61st IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 1014–1021, 2020.
- [7] S. P. Fekete, S. Morr, C. Scheffer. Split packing: algorithms for packing circles with optimal worst-case density. *Discrete & Computational Geometry* 61(3):562–594, 2019.
- [8] R. L. Graham, B. D. Lubachevsky, K. J. Nurmela, P. R. J. Östergård. Dense packings of congruent circles in a circle. *Discrete Mathematics* 181:139–154, 1998.
- [9] U. Pirl. Der Mindestabstand von n in der Einheitskreisscheibe gelegenen Punkten. *Mathematische Nachrichten* 40:111–124, 1969.
- [10] H. Melissen. Densest packings of eleven congruent circles in a circle. *Geometriae Dedicata* 50:15–25, 1994.

- [11] F. Fodor. The densest packing of 19 congruent circles in a circle. *Geometriae Dedicata* 74:139–145, 1999.
- [12] D. B. Ekanayake, D. J. LaFountain. Tight partitions for packing circles in a circle. *Italian Journal of Pure and Applied Mathematics* 51:115–136, 2024.
- [13] J. Edmonds. Matroids and the greedy algorithm. *Mathematical Programming* 1:127–136, 1971.
- [14] B. Korte, D. Hausmann. An analysis of the greedy heuristic for independence systems. *Annals of Discrete Mathematics* 2:65–74, 1978.
- [15] A. Gupte. Convex hulls of superincreasing knapsacks and lexicographic orderings. *Discrete Applied Mathematics* 201:150–163, 2016.
- [16] I. Litvinchev, L. Infante, E. L. Ozuna Espinosa. Approximate circle packing in a rectangular container: integer programming formulations and valid inequalities. In *Computational Logistics (ICCL 2014)*, LNCS 8760, Springer, 2014, pp. 47–60.